CLD100

Cloud for SAP

PARTICIPANT HANDBOOK
INSTRUCTOR-LED TRAINING

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Typographic Conventions

American English is the standard used in this handbook.

The following typographic conventions are also used.

This information is displayed in the instructor's presentation	
Demonstration	
Procedure	
Warning or Caution	
Hint	
Related or Additional Information	
Facilitated Discussion	
User interface control	Example text
Window title	Example text

Contents

Course Overview

TARGET AUDIENCE

This course is intended for the following audiences:

- Application Consultant
- Business Analyst
- Business Process Architect
- Business Process Owner/Team Lead/Power User
- Developer
- Development Consultant
- Enterprise Architect
- Executive
- Program/Project Manager
- Solution Architect
- System Architect
- Technology Consultant
- User

UNIT 1 SAP Cloud Overview

UNIT OBJECTIVES

- Get an introduction to SAP Cloud
- Explain landscape architecture-relevant terms
- Explain cloud security
- Explain SAP Leonardo

Unit 1
Lesson 1

Introducing SAP Cloud

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Get an introduction to SAP Cloud

SAP Cloud, Introduction

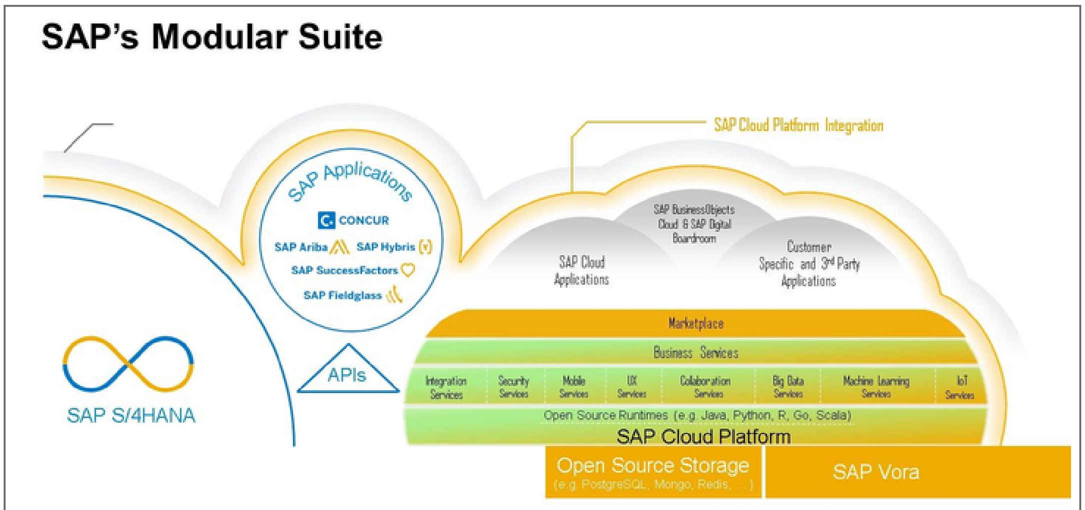


Figure 1: SAP Cloud, Introduction

Introduction to Cloud Computing at SAP

SAP is a cloud company driven by SAP HANA in-memory technology. We have 110 million cloud subscribers and 41 state-of-the-art data centers around the world. We offer cloud apps (SaaS) for all lines of business, a market-leading cloud platform (PaaS), and flexible on-demand infrastructure (IaaS).

You can rely on proven enterprise cloud security and hosting services - and choose from a public, private, or hybrid cloud environment.

Introduction: What is cloud computing?

Simply put, cloud computing makes it possible for users to access data, applications, and services over the Internet. The cloud eliminates the need for costly hardware, such as hard drives and servers - and gives users the ability to work from anywhere. Over 90% of businesses are already using cloud technology in a public, private, or hybrid cloud environment.

The major benefits of cloud include:

- Elasticity: Scale up or down quickly to meet computing demand
- Affordability: Only pay for what you use, and minimize hardware and IT costs
- Availability: Get 24/7 cloud system access from anywhere, on any device
- Simplicity: Free IT from managing servers and updating software

Cloud software offers a number of distinct advantages that help sharpen your competitive edge.

Introduction Introducing SAP Cloud Trust Center

Trust, security, and compliance in the cloud - that's our commitment to you.

Your business is built on trust, and you expect the same from your software provider. As a leading software provider and a cloud company, we're dedicated to building - and keeping - our customers' trust.

That's why we're constantly improving our cloud solutions by staying ahead of the latest threats, building security into every layer of our cloud offerings, and adapting to the latest compliance standards.

See: **https://www.sap.com/about/cloud-trust-center.html**

LESSON SUMMARY
You should now be able to:

- Get an introduction to SAP Cloud

Unit 1
Lesson 2

Explaining Landscape Architecture-Relevant Terms

LESSON OBJECTIVES

After completing this lesson, you will be able to:

Explain landscape architecture-relevant terms

Landscape Relevant Terms

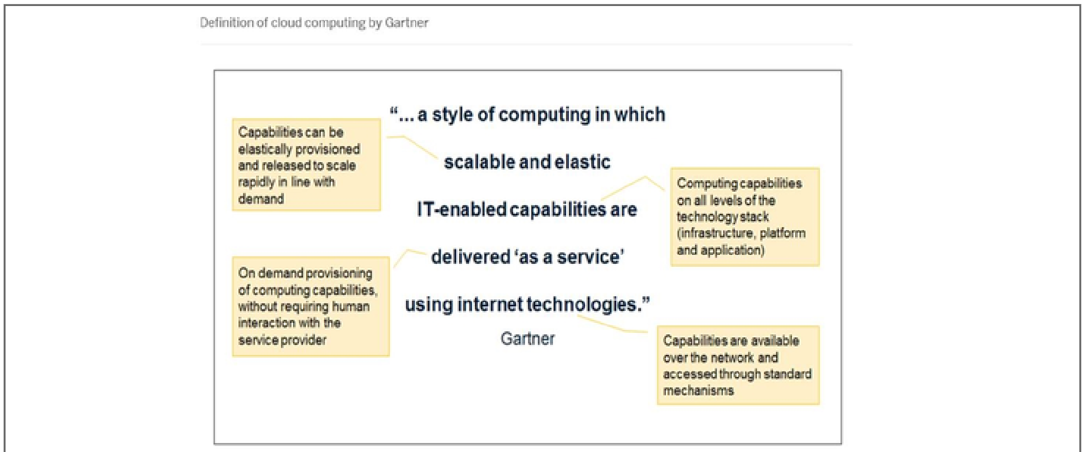


Figure 2: Key Characteristics of Cloud Computing

According to the National Institute of Standards and Technology [NIST], key Characteristics of Cloud Computing are defined by:

On-demand self-service

A consumer can unilaterally provision computing capabilities, without requiring human interaction with each service provider.

Broad network access

Capabilities are available over the network and accessed through standard mechanisms.

Resource pooling

Resources are pooled to serve multiple consumers using a multi-tenant model.

Rapid elasticity

Capabilities can be elastically provisioned and released to scale rapidly in line with demand.

Measured service

Cloud systems automatically control and optimize resource use by leveraging a metering capability

Cloud computing includes various service models defining which layers of the software stack are provided by the cloud provider. They are the 'as a service' offerings, where cloud services are offered either as software as a service (SaaS), platform as a service (PaaS) or infrastructure as a service (IaaS).

There are multiple cloud deployment models which describe various approaches to provisioning software in the cloud. They are characterized by different infrastructure, software lifecycle management, and licensing approaches. We differentiate two main cloud deployment models: public, including private editions, and private, including managed cloud offerings. These deployment models address customers need for flexibility to support a large variety of use cases.

Cloud Service Models: Overview

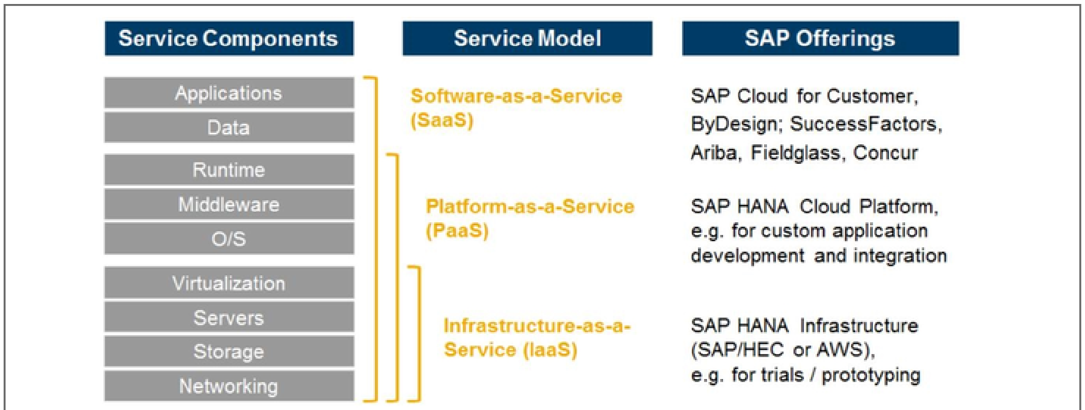


Figure 3: Overview of Cloud Service Models

The figure gives an overview of different Cloud Service Model offerings.

Cloud Service Models: 'Infrastructure as a Service' (IaaS)

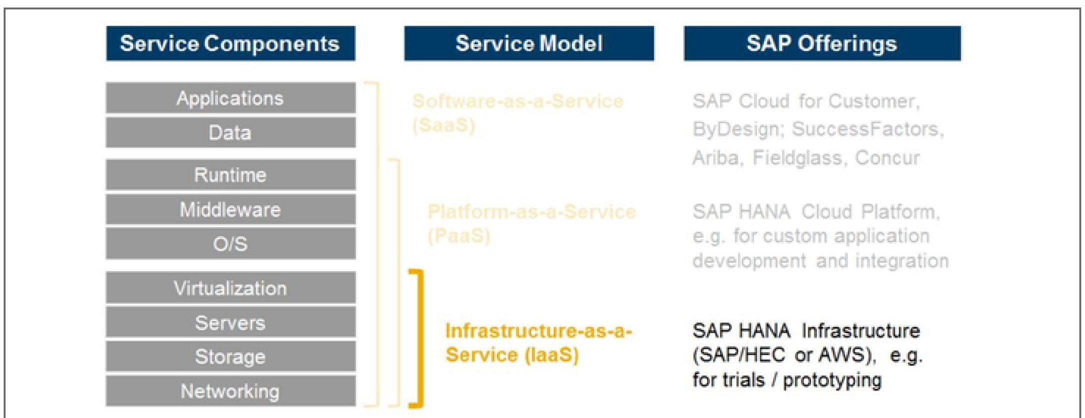


Figure 4: Cloud Service Models of 'Infrastructure as a Service' (IaaS)

Infrastructure as a Service (IaaS) is a cloud service model in which an organization outsources the equipment used to support operations, including storage, hardware, servers, and networking components. The IaaS provider owns the equipment and is responsible for housing, running and maintaining it. In other words, IaaS generally offers an infrastructure, i.e. a virtual machine with a given amount of processing power, availability and storage. The client typically pays on a per-use basis.

The goal of IaaS is to provide a flexible, standard, and virtually operating environment that becomes the foundation for higher-level cloud offerings, such as Platform as a Service (PaaS) and Software as a Service (SaaS).

This service model is suitable for companies which occasionally have temporary infrastructural needs or are trying to make the shift from capital to operating expenditure.

SAP has a number of services in the IaaS layer and also has partnerships with companies like Amazon in order to make products like SAP HANA available in the cloud.

Cloud Service Models: 'Infrastructure as a Service' (PaaS)

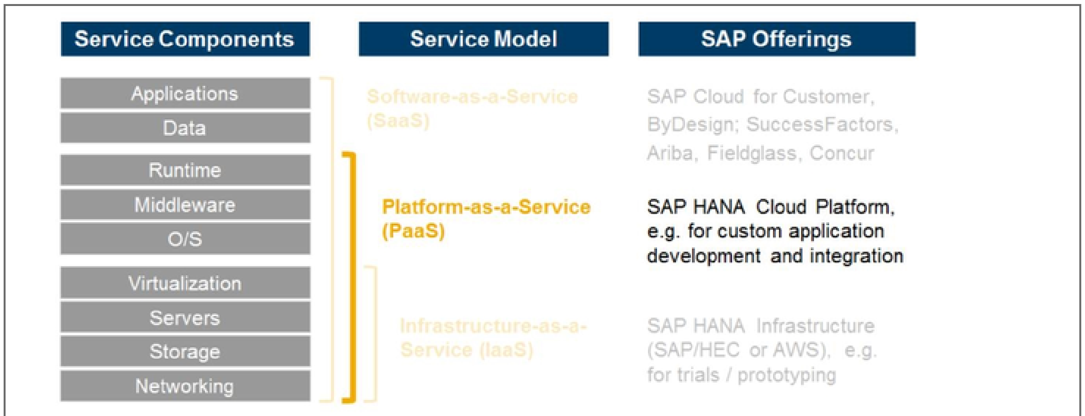


Figure 5: Cloud Service Models of 'Infrastructure as a Service' (PaaS)

Platform as a Service (PaaS) delivers a whole platform with related services such as persistence, authentication, and connectivity. PaaS provides a development platform designed to help customers, independent software vendors, and partners to rapidly create innovative software applications in the cloud.

SAP's key offering in this context is the SAP Cloud Platform. This allows SAP partners, SAP customers and SAP developers to build, deploy and operate non-ABAP applications in an open and standards-based cloud environment which offers a number of shared application services.

Cloud Service Models: 'Infrastructure as a Service' (SaaS)

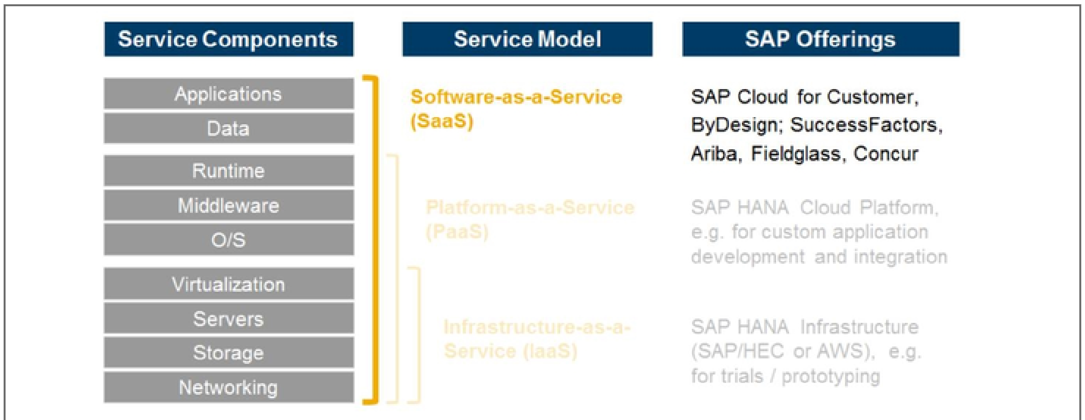


Figure 6: Cloud Service Models of 'Infrastructure as a Service' (SaaS)

Software as a Service (SaaS), sometimes referred to as **on-demand software**, is a software delivery model in which software and associated data are centrally hosted on the cloud. The software is thus literally provided "as a service." Software applications are delivered and managed remotely over a secure Internet connection and a standard Web browser. Access is charged on a subscription basis at a fixed price per user, usually on a monthly basis, and users only pay for applications they actually use.

By providing standardized capabilities for a usage based cost, SaaS provides significant cost efficiency.

Today, the SaaS is mainly used for non-core and non-differentiating applications, such as customer relationship management, human resources or procurement applications.

SAP has complete suites for SME markets, such as SAP Business ByDesign and SAP Business One. For large enterprises, it provides a range of business solutions like Ariba, Concur, Fieldglass, and Cloud for Customer, and SuccessFactors.

Deployment Models: Overview

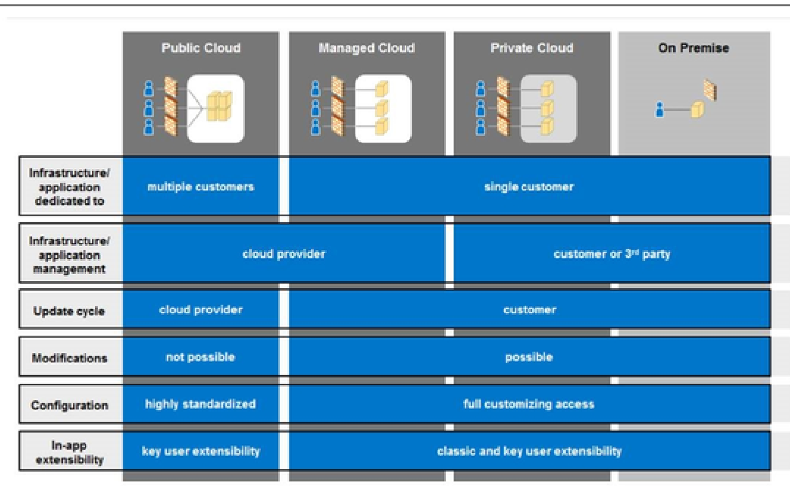


Figure 7: Overview of Deployment Models

A deployment model defines how a company deploys software. This could be onPremise, in the private cloud, in the public cloud, or a hybrid approach. It describes various software deployment approaches in terms of infrastructure, software lifecycle management, and licensing.

Each deployment model can be differentiated using the following characteristics:

- Exclusiveness of the application (single vs. multiple customers)
- Access type (VPN vs. web access only)
- Location of the hardware (customer, 3rd party or cloud provider)
- Infrastructure and application management provider (customer, 3rd party or cloud provider)
- Upgrade flexibility/automation (prescribed by customer or cloud provider)
- Possibility for modifications
- Type of configuration (standardized, guided or full access)
- In-app extensibility (tool managed, governed or full access)
- Pricing model (perpetual vs. subscription)

Deployment Models: Hybrid Cloud

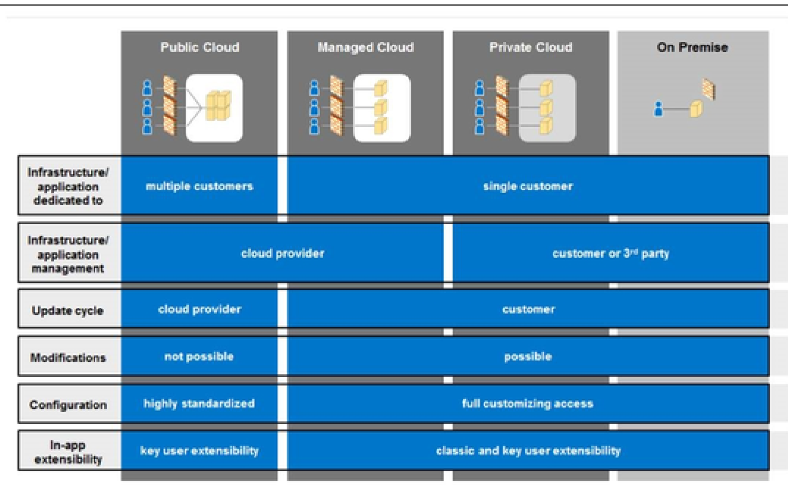


Figure 8: Deployment Models 'Hybrid Cloud'

A hybrid cloud deployment uses resources comprised of a mix of two or more distinct deployment models (typically onPremise or private and public cloud deployments).

There are no hybrid cloud solutions per se. It is rather how companies orchestrate dedicated and cloud resources that create a hybrid cloud. This holistic approach to the consumption of IT requires a careful review of each application and data resource matching the right option to each application.

In an hybrid cloud deployment public cloud, private cloud and onPremise applications are combined and seamlessly integrated. By joining dedicated and cloud resources, businesses can establish an IT landscape that provides the best balance of convenience and security for the company.

Deployment Models: Public Cloud

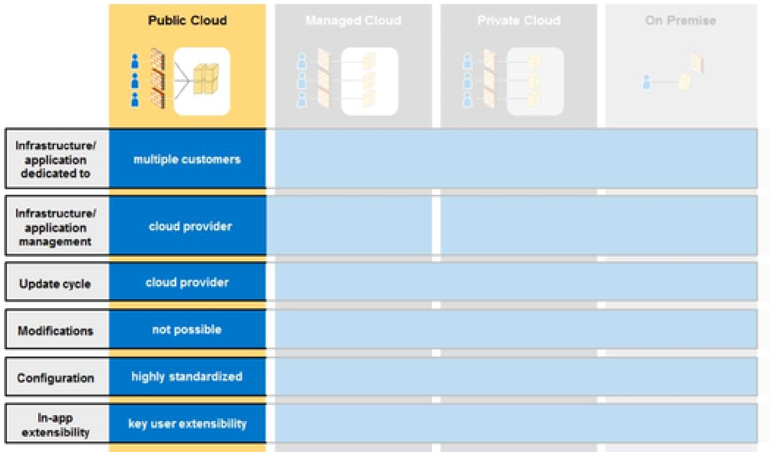


Figure 9: Deployment Models 'Public Cloud'

A public cloud deployment provides customers with access to the provider's applications running on a cloud infrastructure, while the resources are located on the cloud provider's premises, and are shared by multiple customers accessing them via the Internet.

One of the important factors driving public cloud adoption is the need to reduce overall expenditure on IT infrastructure. Since public cloud is a computing model that relies to a great extent on standardization, it offers less flexibility for customization, downtimes or maintenance windows. This model is suitable for companies that have few regulatory hurdles to overcome and are looking to outsource parts of their IT environment in order to reduce capital expenditure and bring down operational IT costs.

The public cloud often hosts non-differentiated and non-critical but computing intensive workload.

Service and Deployment model of SAP Cloud Platform

The SAP Cloud Platform belongs to Paas as service model and serves a public deployment model.

Deployment Models: Managed Cloud: HEC

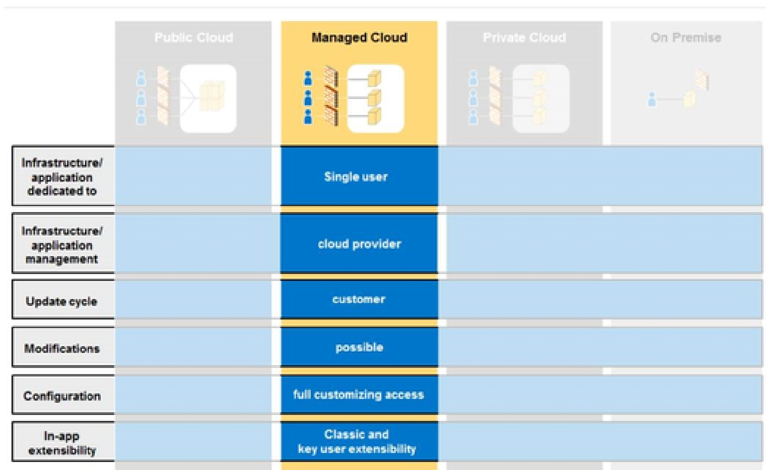


Figure 10: Deployment Models 'Managed Cloud' HEC

A managed cloud deployment implies resources dedicated to one customer and accessed via a VPN while the infrastructure is owned, managed and operated by the cloud provider in the cloud provider's data center.

The managed cloud model is broadly comparable to a private cloud model, while the party responsible for operations and maintenance of the cloud environment is the cloud provider. The managed cloud is usually built on a private cloud platform with dedicated hardware owned by the cloud provider, although certain non-location specific cloud services can be adapted to the requirements of the customer.

This approach allows companies to tap the power of cloud computing without the pain of becoming an expert in everything. This is of particular importance if companies want to leverage the benefits of cutting edge infrastructures from cloud providers like SAP HANA platform without the immediate need to allocate their own resources or develop skills in that area.

Companies who are thinking of implementing this deployment model should make sure that the managed cloud provider in question meets their particular requirements in terms of infrastructure location, data control, service engagement flexibility and so on.

In the SAP context, the private managed cloud offering is the SAP HANA Enterprise Cloud

Deployment Models: Private Cloud

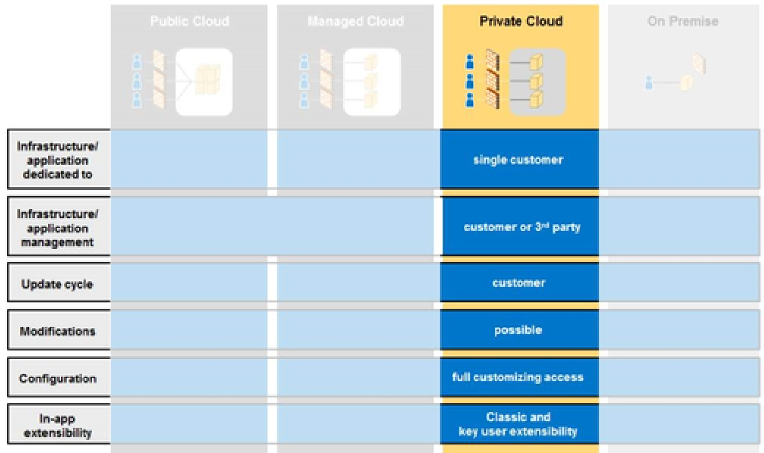


Figure 11: Deployment Models 'Private Cloud'

Private cloud deployment implies resources dedicated exclusively to one specific customer while the infrastructure is owned, managed and operated by the customer or a third-party and is located on the premises of the customer or a third party.

The private cloud model is largely comparable to when a company buys, builds and manages its own infrastructure, but it can help mitigate challenges and concerns related to data security. These concerns are taken care of, as the private cloud environment is built and maintained exclusively for one specific customer. Internet security concerns can be addressed by means of a secure-access VPN or by having the physical location of the computing infrastructure within the client's firewall.

This model is also suitable in cases where data or applications have to conform to regulatory or local standards which prescribe that the company has full control over the data.

The private cloud model is also suitable for applications which require heavy customization, as they deliver competitive edge functionality, but the company wants to leverage cloud qualities such as scalability, managing load spikes and so on.

This model can also apply to mission-critical applications where companies have to take account of customer-specific downtimes and maintenance windows.

Deployment Models: onPremise

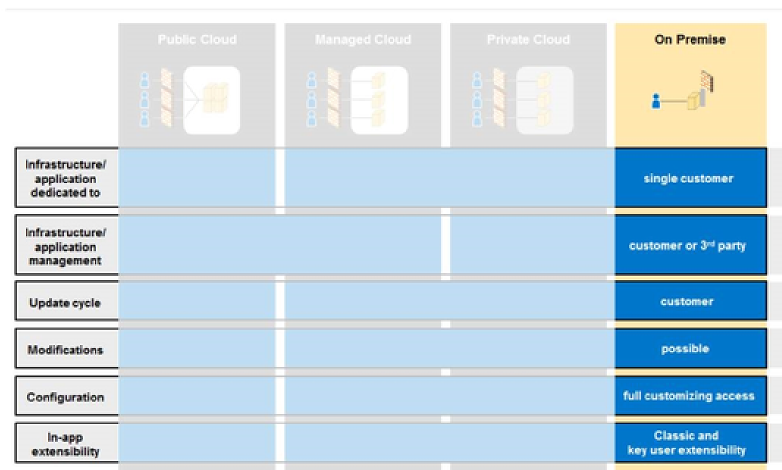


Figure 12: Deployment Models 'onPremise'

In the onPremise deployment model, the software is deployed on servers at the customer's premises, where the customer manages and controls the infrastructure, software and data.

OnPremise deployment is the classic approach, with no use made of the cloud. This approach is suitable when companies require full control over all aspects of the IT environment, while accepting the need to finance, operate and maintain the environment themselves.

LESSON SUMMARY

You should now be able to:

- Explain landscape architecture-relevant terms

Unit 1
Lesson 3

Explaining Cloud Security

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain cloud security

SAP Cloud Platform, Security Architecture

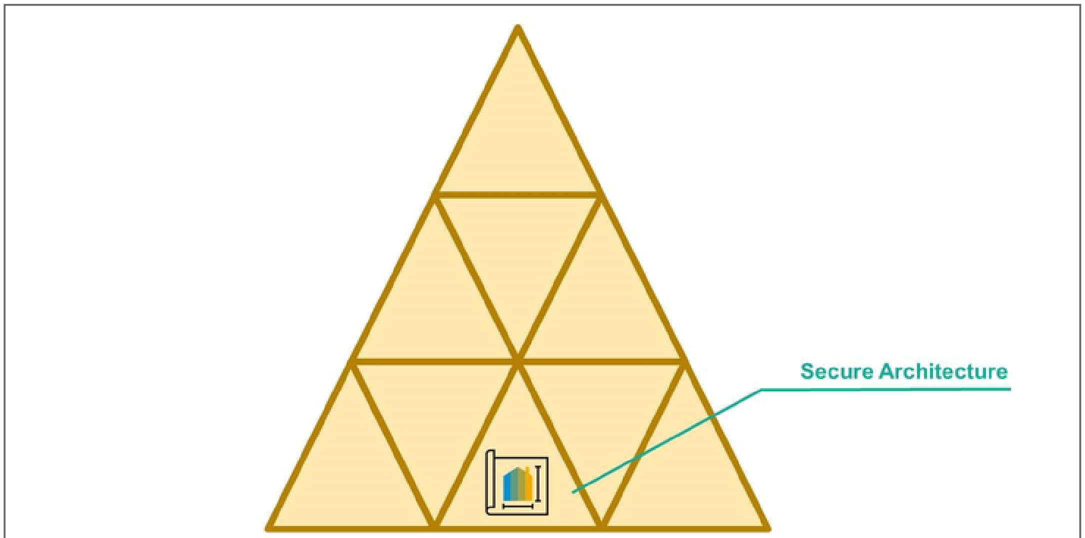


Figure 13: SAP Cloud Platform Security

We start with the SAP Cloud Architecture. The figure will show the different aspects, we will discuss in this lesson, building the security architecture in the whole SAP Cloud Platform solution.

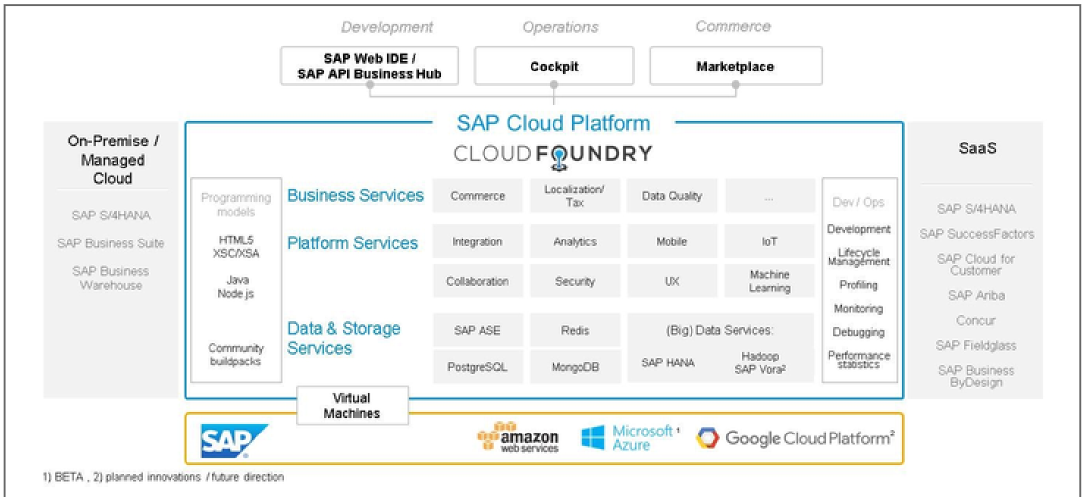


Figure 14: SAP Cloud Platform, High-level overview

The figure shows the overall architecture including both Neo and Cloud Foundry environment.

The security architecture of SAP Cloud Platform aims to establish security measures that are among the highest in the industry:

A multitenant environment, which allows execution of custom code with two lines of defense:

- Application sandboxing - restricting and managing the capabilities of an application within the container in which it run
- Network sandboxing - restricting and managing the capabilities of an application to access other systems in the landscape

Customer and network segregation:

- SAP Cloud Platform is set up in a fenced network, separated from the SAP internal network.
- The internal traffic is controlled by firewalls.
- Administrative access requires strong authentication.

Secure Communication:

- Secure communication in accordance with the protection requirement of the transmitted information.
- Measures for securing the exchange of information
- SAP uses at least a 128-bit symmetric key or a 2,048-bit asymmetric key as encryption methods, as well as strong and internationally recognized cryptographic algorithms

Secure Application Containers - provide state-of-the-art capabilities to application providers to implement secure applications.

System hardening - all nonessential services are deactivated in the system; user accounts that are not required are deleted.

Client media encryption - device encryption is established for storage of data at rest on all type of devices if used for the data classified as "confidential," including customer data. By default, all SAP-controlled laptops and PCs are supplied with an encrypted hard drive.

Deletion of data - backup data is retained for a period of 14 days. Logs from customer applications are retained as follows:

- Maximum of 14 days for development logs
- Maximum of 18 months for audit logs, unless an extended retention period is required by the customer
- Paper data is shredded accordingly, based on applicable local laws and industry-specific regulations

Data Centers and Physical Security

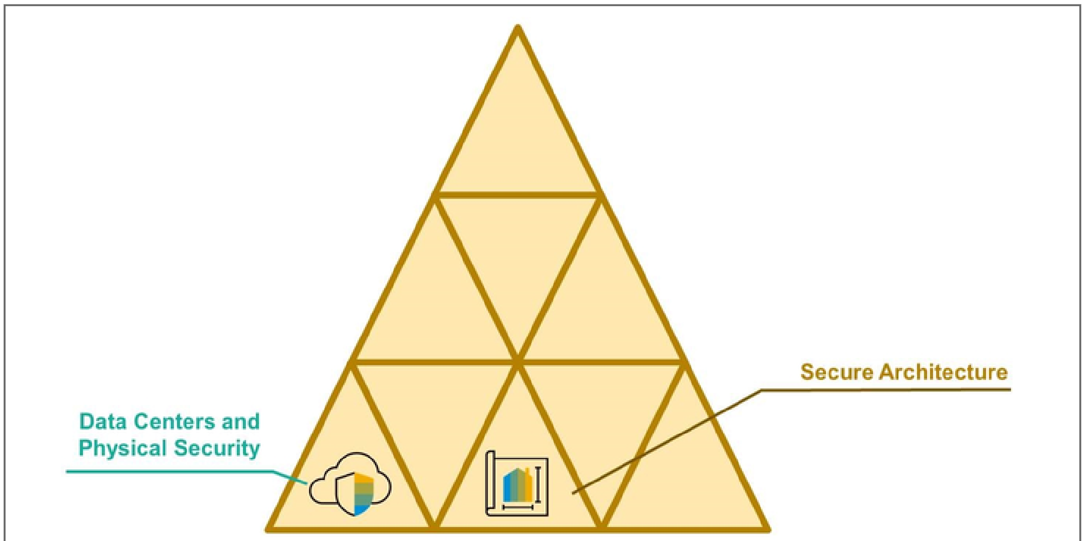


Figure 15: Data Centers and Physical Security

The next topic to discuss, are the Data Centers and Physical Security.

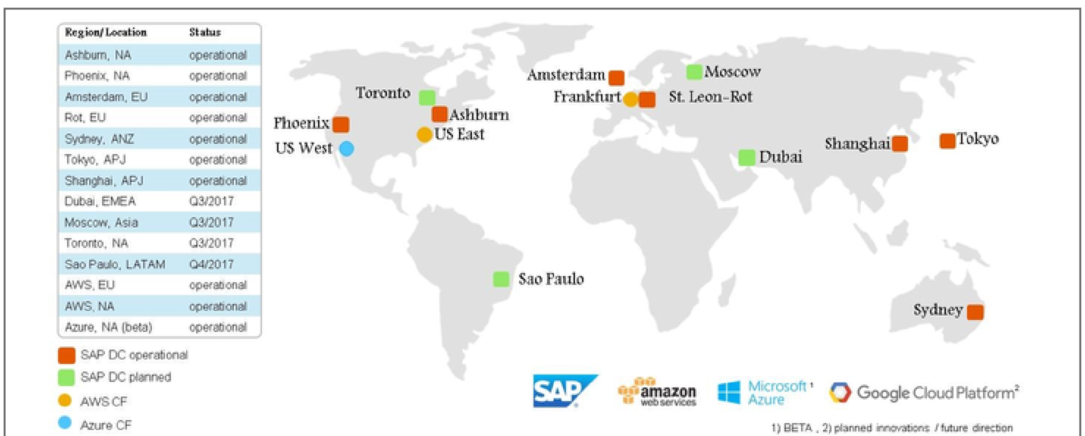


Figure 16: Data centers build-out (2017)

Disclaimer: Dates are subject to change and considerable delays and may be introduced due to unforeseen circumstances.

Physical Access

24/7 monitoring

Access only to authorized individuals.

Technicians enter special rooms using custom-configured ID cards.

Biometric scans for high-sensitivity areas.

Access to Data

Intrusion detection system for identifying suspicious activities

Firewalls made by different manufacturers protect the data in the data center

Data and backup files are exchanged with customers in an encrypted format or transmitted via secure fiber-optic cables.

Power Supply

Multiple-redundancy power supply system

Batteries automatically and immediately actuated in case of a power sully system failure - they cover the power outage until the power diesel generators are started

Emergency power diesel generators can then supply power to the data center for an extended period.

Hardware

All virtual and physical servers, HANA databases, storage units and networks in use access a pool of physical hardware.

Load can be directly re-allocated to other components without impairing system stability, if individual components fail.

Data can be recovered from the backup system, if hardware fails due to a fire.

Data Privacy

SAP ensures compliance with data protection provisions.

Data from cloud customers falls under the jurisdiction selected by the customer and is not forwarded to third parties.

SAP's support services ensure that data protection is also maintained during required maintenance operations.

Fire Protection

Data center is subdivided into many fire compartments.

Thousands of fire detectors and aspirating smoke detectors (ASD) monitor all rooms including control for overheating electronic components, etc.

Building

100,000 metric tons of reinforced concrete

480 concrete pillars, each extending 16 meters into the ground. The are

30 centimeters thick exterior walls

Server rooms surrounded by 3 concrete walls.

Backup

Backups are carried out in the form of disk-to-disk copies for rapid data creation and recovery.

Full backups done on a daily basis

Interim versions are created several times per day and are then archived, like all backups, at a second location for security purposes.

Data Governance and Legal Compliance

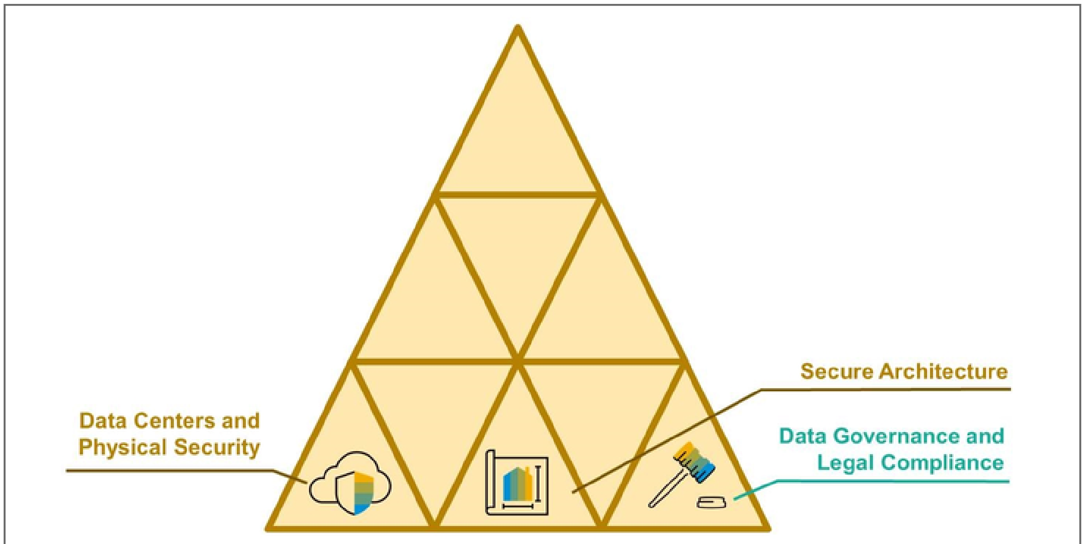


Figure 17: Data Governance and Legal Compliance

The next topic is Data Governance and Legal Compliance.

All customer data processed in SAP Cloud Platform belongs to the customer and is classified as "Confidential" according to SAP's data classification standard, unless the customer chooses to make it visible to the public by means of tools and services within SAP Cloud Platform:

Access authorization is provided following the need-to-know principle

Disclosure to third parties happens only after these parties become subcontractors by signing agreements

Encryption is applied for transferring data outside the SAP network.

Device encryption is required for storage of the data

Storage infrastructure is provided only by SAP or subcontractors

Certified (annual audit cycle) data protection management system (DPMS), which is based on the British Standard "BS 10012:2009 Data Protection."

Employees' awareness via the code of business conduct

Further details:

Confidentiality and privacy statement (CPS) and other contractual confidentiality provisions are mandatory for all of SAP's vendors and third-party service providers that perform services on behalf of SAP and that have access to confidential information.

Mandatory security awareness training sessions are conducted on a regular basis. The training is designed not only for vendors and third-party providers but also for SAP employees.

European data protection service - SAP Cloud Platform can, by customer request, optionally be operated and supported in European Union (EU) access mode. Only SAP employees who are located in European Economic Area member countries or Switzerland can operate and support customer data.

SAP Cloud Platform is certified according to ISO 27001:2013, SSAE 16-SOC 1/ISAE 3402* Type 2, and SOC 2 Type 2 security standards, that also ensure platform stability, security, and performance at a platform operations and process level

Requests from law enforcement authorities are checked - SAP legal department evaluates every inquiry in detail and customer data is shared with authorities only if the request is legally valid.

Operational Security Management and Business Continuity

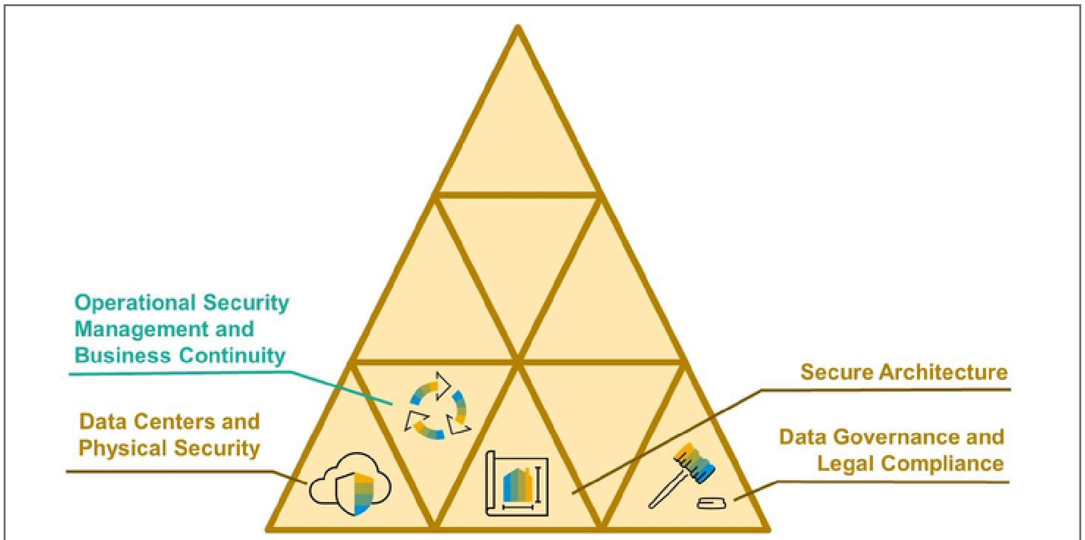


Figure 18: Operational Security Management and Business Continuity

The next topic is Operational Security Management and Business Continuity.

Reliability, integrity, availability, and authenticity of customer data ensured with operating the SAP Cloud Platform securely:

Change management follows a formal process that is reviewed and approved regularly - change request documentation, impact analysis, planning, testing, implementation, tracking and maintenance.

SAP's security-patch management process mitigates threats and vulnerabilities. Security patches are rated based on the Common Vulnerability Scoring System standard for OS, DB, and virtualization in cloud services. Critical security patches applied on a priority basis.

Malware management process ensures secure service delivery free of viruses, spam, spyware, and other malicious software - anti-malware agent deployment, regular scans, and malware reporting processes.

SAP's comprehensive administrative user access management follows the principles of minimal authorization (the need-to-know principle) and segregation of duties.

Incident, Thread and Vulnerability Management

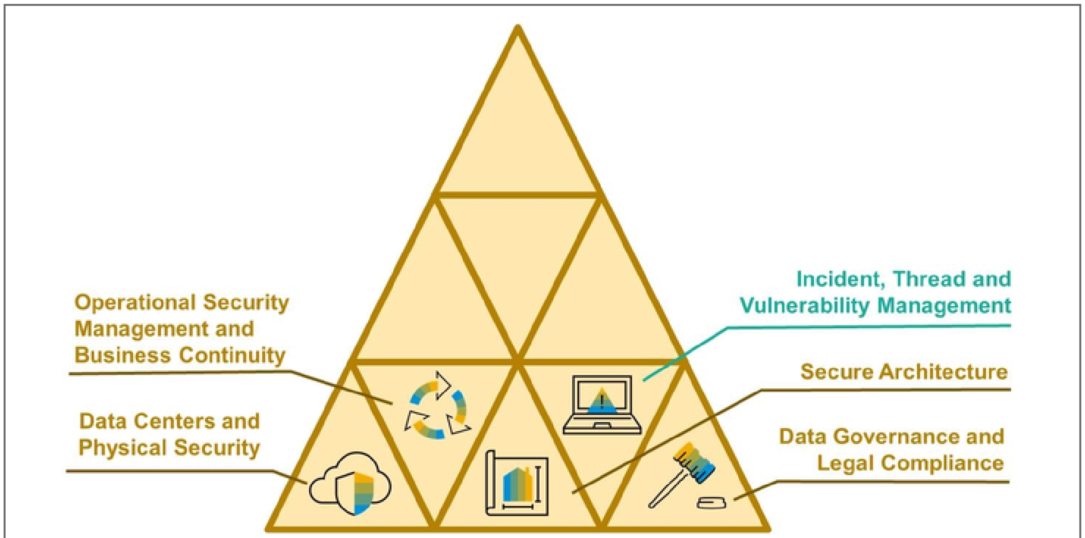


Figure 19: Incident, Thread and Vulnerability Management

The next topics are Incident, Thread and Vulnerability Management.

Reliability, integrity, availability, and authenticity of customer data ensured with operating the SAP Cloud Platform securely:

Security incident management process that is following the trusted security principals, regulated by the:

- International Organization for Standardization (ISO)
- International Electrotechnical Commission (IEC) 27035:2011

Early identification, assessment, and mitigation:

- Vulnerability scans

Performed at least four times a year to ensure the highest possible level of security.

- External penetration testing:

  All Internet-facing systems are scanned on a weekly basis. In addition,

  Manual verification and penetration testing is performed to validate risk and priority.

  Biannual assessment and penetration tests are performed by independent security researchers to verify the security posture of the external and Internet-facing cloud infrastructure.

Transparency is one foundation for trust:

Security information and event management systems for analysis, reporting, and alerting.

- All critical systems and infrastructure components within SAP Cloud Platform log relevant data, which is stored for a minimum of six months.
- Data security is ensured through security configuration compliance checks and event monitoring.
- General security monitoring is performed 24x7 for all activities. Once a warning or an alert comes up, it is processed through our ticketing system, and critical events are handled according to the incident management process

Notification is communicated through the defined communication channel and contains the following information:

- Details relating to the security incident that has occurred, known at the time of notification and the IT infrastructure or application affected by the security incident
- An overview of the mitigation actions performed to restore security
- All further applicable notifications required by country-specific regulations "on obligation to notify"

SAML Identity Provider, SSO and Strong authentication

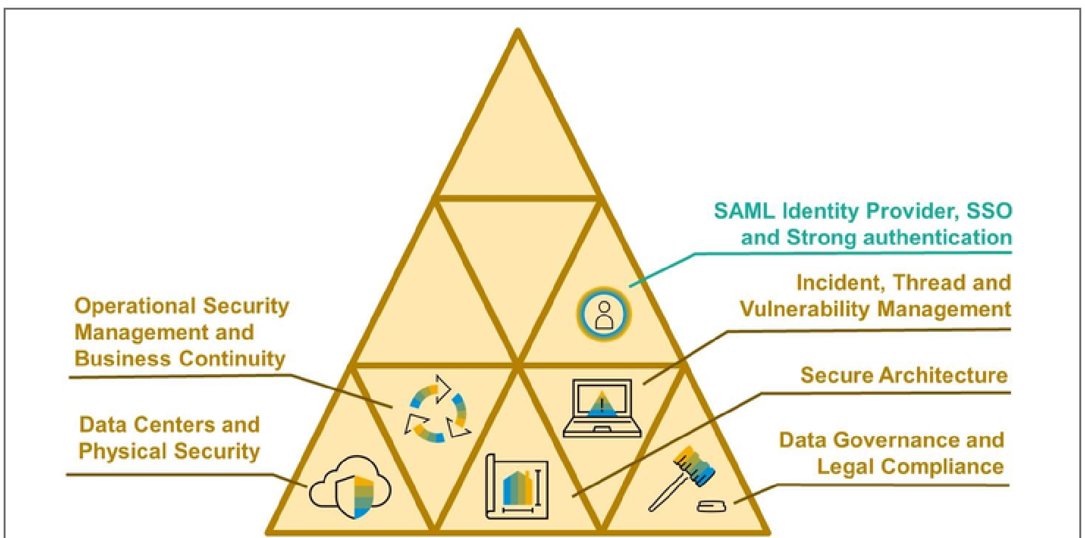


Figure 20: SAML Identity Provider, SSO and Strong authentication

The next topic is the SAML Identity Provider, SSO and Strong authentication.

Central user authentication, web single sign-on and user management as Software-as-a-Service, based on open industry-standards - SAML 2.0, SCIM:

Authentication against cloud or on-premise user store for SAP and non-SAP cloud applications

A variety of authentication methods:

- Username and password
- Kerberos/SPNEGO
- Two-factor-authentication (TOTP based on RFC 6238)
- Risk-based authentication rules (app/group level)
- Social login (LinkedIn, Twitter)

User self-services (e.g. registration, password reset, user profile)

Look & feel and policies customization

Programmatic integration via SCIM standard

User and Access Provisioning

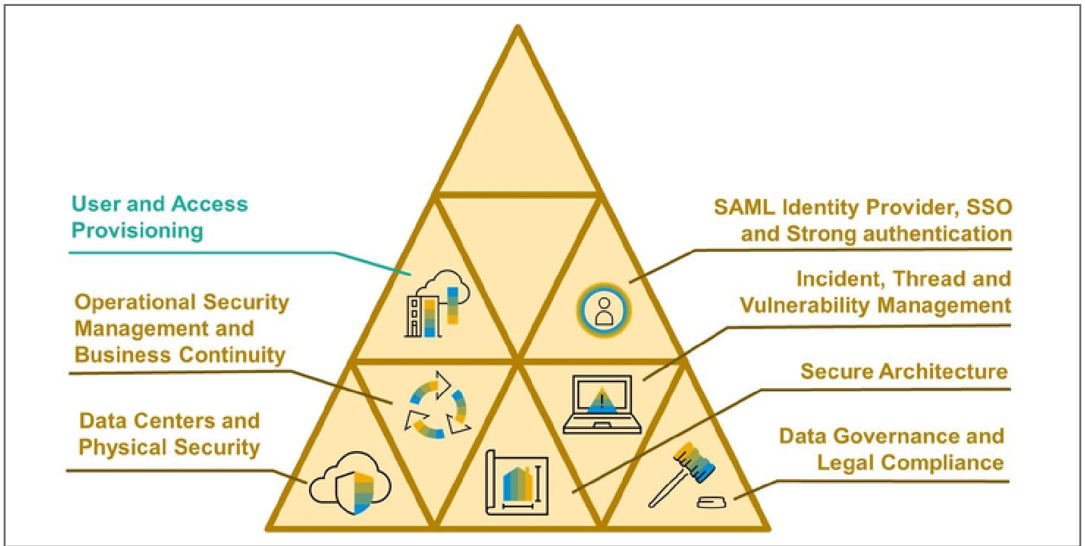


Figure 21: User and Access Provisioning

The next topic is the User and Access Provisioning.

SAP Cloud Platform, Identity Provisioning Service is a Comprehensive, low cost approach to identity lifecycle management in the cloud:

Automatically create and manage user accounts

Policy based authorizations management

Re-using existing corporate user stores (SAP AS ABAP, SAP Cloud Platform Identity Authentication (IAS), SAP SuccessFactors, MS Active Directory, LDAP Server, SCIM compliant systems)

Optimized for SAP cloud applications and SCIM-enabled solutions (SAP Cloud Platform apps, IAS, SAP Hybris C4C, CloudFoundry UAA Server, Google G Suite, SCIM Systems, SAP Jam, Concur)

Integrated with SAP Cloud Platform, integration service

Integration option with SAP Identity Management

Repository for Cryptographic Keys and Certificates

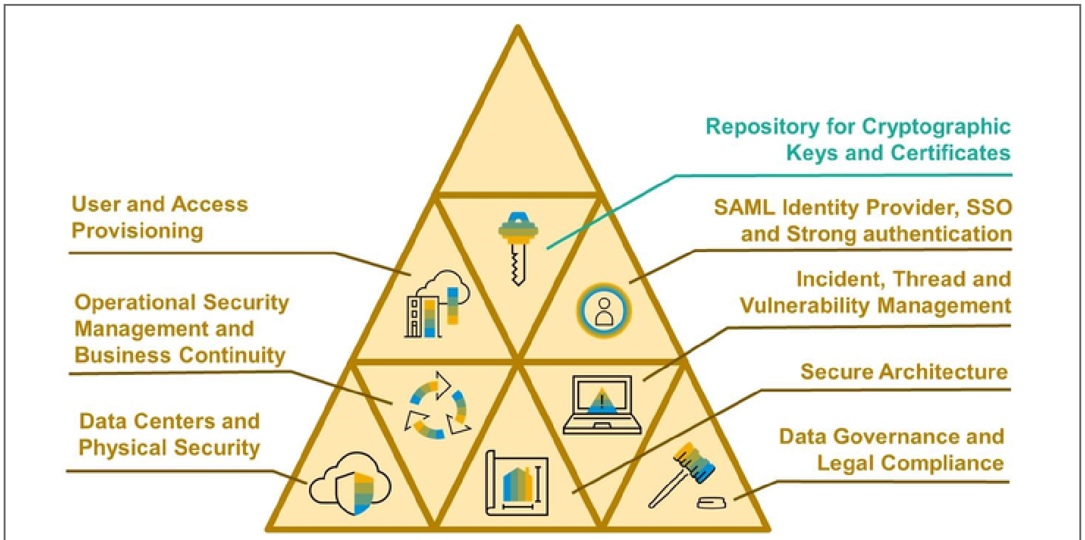


Figure 22: Repository for Cryptographic Keys and Certificates

The next topic is the Repository for Cryptographic Keys and Certificates.

The Keystore Service provides a repository for cryptographic keys and certificates to the applications hosted on SAP Cloud Platform:

Using the Keystore Service, the applications could easily retrieve keystores and use them in various cryptographic operations:

- Signing and verifying of digital signatures
- Encrypting and decrypting messages
- SSL communication

Exposed to applications via API

Administrators can list, upload, download, and delete the keystores using the Platform's console client

The Keystore Service stores and provides keystores encoded in the following formats:

Java Keystore (JKS) - It supports password-based integrity validation for its content. Key entries are protected with password-based encryption. Password has to be specified in order to retrieve a key entry.

Extended Java Keystore (JCEKS) - It provides the same functionality as the JKS format with stronger protection for private keys.

PKCS #12 file (P12) - This format supports password-based integrity validation for its content. Key entries are protected with password-based symmetric encryption. A password has to be specified in order to retrieve a key entry.

Privacy Enhanced Mail Certificate (PEM) - It does not support integrity validation. Key entries are not protected with password.

Password Storage

Using the password storage API, you can securely persist passwords and key phrases such as passwords for keystore files.

Once persisted in the password storage, passwords and key phrases:

- Exposed to applications via API
- Survive application restarts and updates
- Are a subject of automatic backup
- Stay persisted unless you explicitly delete them via the API, or you undeploy your application.

Before transportation and persistence, passwords are encrypted with an encryption key which is specific for the application that owns the password.

Application Security- authentication, permissions

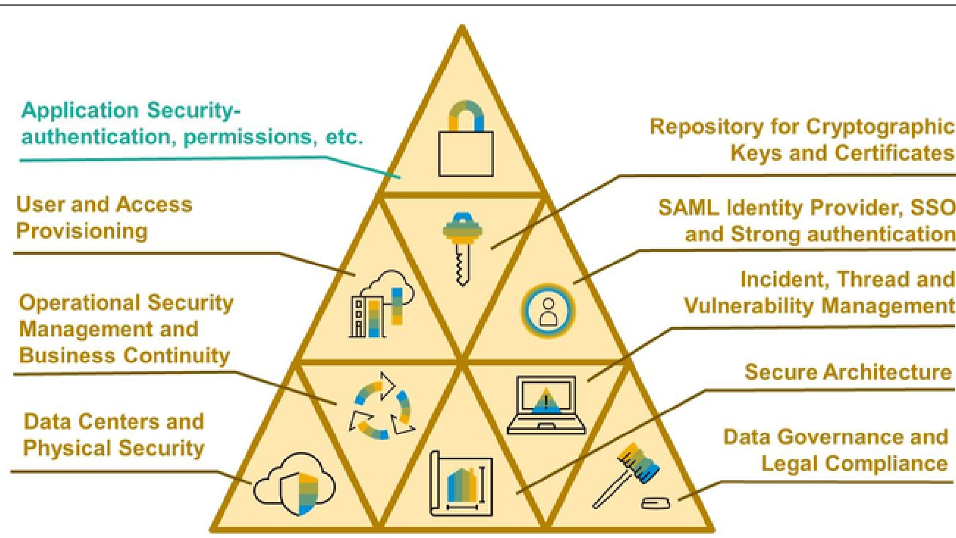


Figure 23: Application Security- authentication, permissions

The last concept in this lesson are Application Security- authentication, permissions, etc. Discussing, how to develop applications with high level of security.

SAP Cloud Platform's rich security services make it easy for you to create secure and reliable cloud applications, enabling you to focus on what you do best - delivering amazing apps:

SAP Cloud Platform protects against unauthorized access by integrating with a wide range of authentication systems - The Platform offers support for different authentication and single sign-on scenarios based on technologies such as the Open Authorization Framework (OAuth), Security Assertion Markup Language (SAML), and X.509 certificates.

A role-based authorization concept ensures that only authorized users are able to access your system resources.

Delegate authentication to an on-premise user store* (SAP Single Sign-On and Microsoft Active Directory).

Secure Software Development Lifecycle

Security capabilities and configurations available to you for developing business applications with highest level of security on the SAP Cloud Platform:

Enable Authentication: SAML 2.0 Authentication or OAuth 2.0

Enforce Authorizations: Use the isUserInRole method to manage authorizations for users.

Enable Logout: You can provide a logout operation for your application by adding a logout button or logout link. When logout is triggered the user is redirected to the identity provider to be logged out there

Test security configurations - Local test identity provider (IdP) is available with the Platform to test single sign on (SSO) and identity federation of your applications end-to-end.

Principal Propagation to OAuth-protected applications:

Propagate users from external applications with SAML identity federation to OAuth-protected applications running on SAP Cloud Platform. Exchange the user ID and attributes from a SAML assertion for an OAuth access token, and use the access token to access the OAuth-protected application.

Protection Against Common Web Attacks:

Prevent a cross-site request forgery (XSRF) attack - SAP Cloud Platform uses a randomly generated unique value ("nonce") per request, and stores it in the user's session.

Prevent a cross-site request forgery (CSRF) attack

Cross-site Scripting (XSS) - one of the most common types of malicious attacks to Web applications. In order to help protecting against this type of attacks, SAP Cloud Platform provides a common output encoding library

LESSON SUMMARY
You should now be able to:

Explain cloud security

Unit 1
Lesson 4

Explaining SAP Leonardo

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain SAP Leonardo

SAP Leonardo, Introduction

What is SAP Leonardo?

SAP Leonardo, SAP's approach to innovation, helps companies navigate the new digital revolution and guide their transformation into a digital business. SAP Leonardo is made up of four elements:

- proven methodology,
- technology expertise,
- industry accelerators, and
- co-innovation with our customers.

By leveraging SAP's expertise in advanced technologies, including Machine Learning, Blockchain, Data Intelligence, Big Data, IoT, and Analytics, you can continuously innovate your business processes, faster than you could before, and with less risk of failing or wasting resources.

Through the use of SAP Leonardo technologies, you can foster digital transformation by optimizing and digitizing your business processes, maximizing your data for increased revenue and growth opportunities, and building leading-edge innovations that deliver rich, new user experiences.

SAP Leonardo seamlessly integrates future-facing technologies and capabilities into the SAP Cloud Platform. This powerful portfolio enables you to rapidly innovate, scale new models, and continually redefine your business.

SAP Leonardo Digital Innovation System

SAP Leonardo is open, extendable, and ready to be woven into every facet of your business. SAP Leonardo allows for rapid implementation and seamless scaling through the existing SAP platform and application portfolio. Automate and augment existing processes, develop new applications, make existing SAP and third-party applications more intelligent, and power new business models.

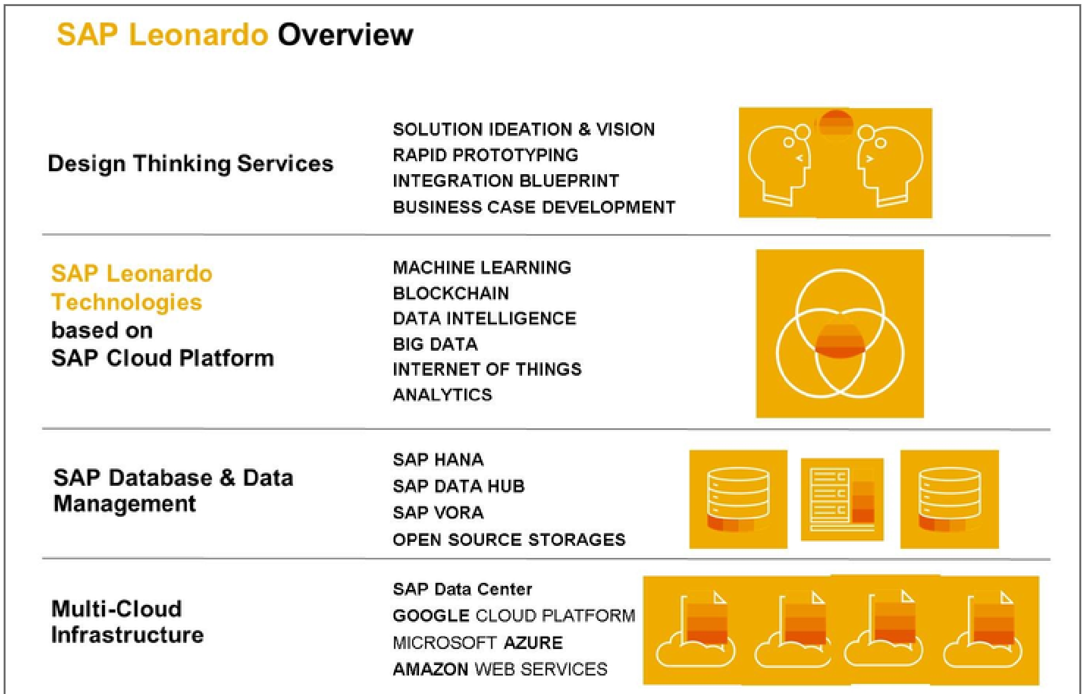


Figure 24: SAP Leonardo

As discussed earlier, SAP Leonardo is a design-led approach to innovation. Through the use of next-gen technologies, such as Big Data, Machine Learning and IoT, you can foster digital transformation by optimizing and digitizing your business processes, maximizing your data for increased revenue and growth opportunities, and building leading-edge innovations that deliver rich, new user experiences.

With SAP Leonardo, you can quickly create a connected world where, for example, machines provide automatic alerts when they need maintenance or replenishment; fleets are routed based on real-time traffic updates; cities predict their own needs for water, power, and waste management; and retailers can maximize margins and revenue, in real-time.

SAP Leonardo builds on SAP’s digital transformation framework, seamlessly connects with the overall SAP landscape, and supports an open-source and multi-cloud infrastructure – all with the goal of enabling faster innovation with less risk.

With dozens of industry awards and recognition, SAP is the largest Analytics company in the world (by market share), we have been recognized by Forrester as a leader in Machine Learning and Predictive Analytics, and acknowledged by customers like Trenitalia as an innovative partner in IoT. However, Leonardo isn't just about technology - it's about business, and the real power is when these technologies all work seamlessly together to solve a business problem. THAT is the power of Leonardo.

SAP Leonardo Technologies

The following are the technologies of SAP Leonardo:

Machine Learning

Using the wealth, power and speed of data in your SAP systems to classify and predict future outcomes.

Blockchain

A powerful and secure way of building trusted transactions between parties.

Data Intelligence

Using SAP Leonardo Machine Learning, Big Data and Analytics to generate your profit driven by SAP experts.

Big Data

Easily create, monitor and manage powerful data pipelines.

Internet of Things

Collect and analyze sensor data.

Analytics

The set of tools and techniques that enable the generation of powerful descriptive insights into your data.

LESSON SUMMARY

You should now be able to:

- Explain SAP Leonardo

Learning Assessment

1. Which of the following is not a main benefit of cloud?

   Choose the correct answer.

   - [ ] **A** Elasticity
   - [ ] **B** Creativity
   - [ ] **C** Affordability
   - [ ] **D** Availability
   - [ ] **E** Simplicity

2. What are the three cloud service models?

   Choose the correct answers.

   - [ ] **A** Software-as-a-Service
   - [ ] **B** HANA as-a-Service
   - [ ] **C** Storage as-a-Service
   - [ ] **D** Platform as-a-Service
   - [ ] **E** Infrastructure as-a-Service

3. Infrastructure as-a-service delivers a whole platform with related services such as persistence, authentication, and connectivity.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

4. What are the four deployment models?

   Choose the correct answers.

   - [ ] **A** Public Cloud
   - [ ] **B** On Premise
   - [ ] **C** Private Cloud
   - [ ] **D** Managed Cloud
   - [ ] **E** Partner Cloud

5. Private cloud deployment implies resources dedicated exclusively to one customer.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

6. SAP Leonardo builds on SAP's digital transformation framework with the goal of enabling faster innovation with less risk.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

7. What are the key functionalities in SAP Cloud Platform?

   Choose the correct answers.

   - [ ] **A** Open APIs
   - [ ] **B** Integration
   - [ ] **C** Edge Computing
   - [ ] **D** Microservices

Learning Assessment - Answers

1. Which of the following is not a main benefit of cloud?

   Choose the correct answer.

   - [ ] **A** Elasticity
   - [x] **B** Creativity
   - [ ] **C** Affordability
   - [ ] **D** Availability
   - [ ] **E** Simplicity

   This is correct. See Unit 1, Lesson 1 of CLD100, Col 18.

2. What are the three cloud service models?

   Choose the correct answers.

   - [x] **A** Software-as-a-Service
   - [ ] **B** HANA as-a-Service
   - [ ] **C** Storage as-a-Service
   - [x] **D** Platform as-a-Service
   - [x] **E** Infrastructure as-a-Service

   This is correct. See Unit 1, Lesson 2 of CLD100, Col 18.

3. Infrastructure as-a-service delivers a whole platform with related services such as persistence, authentication, and connectivity.

   Determine whether this statement is true or false.

   - [ ] True
   - [x] False

   This is correct. See Unit 1, Lesson 2 of CLD100, Col 18.

4. What are the four deployment models?

   Choose the correct answers.

   [X] **A** Public Cloud

   [X] **B** On Premise

   [X] **C** Private Cloud

   [X] **D** Managed Cloud

   [ ] **E** Partner Cloud

   This is correct. See Unit 1, Lesson 2 of CLD100, Col 18.

5. Private cloud deployment implies resources dedicated exclusively to one customer.

   Determine whether this statement is true or false.

   [X] True

   [ ] False

   This is correct. See Unit 1, Lesson 3 of CLD100, Col 18.

6. SAP Leonardo builds on SAP's digital transformation framework with the goal of enabling faster innovation with less risk.

   Determine whether this statement is true or false.

   [X] True

   [ ] False

   This is correct. One of the main goals of SAP Leonardo is the goal of enabling faster innovation with less risk. See unit 1, Lesson 4 of CLD100, Col18.

7. What are the key functionalities in SAP Cloud Platform?

   Choose the correct answers.

   [X] **A** Open APIs

   [X] **B** Integration

   [ ] **C** Edge Computing

   [X] **D** Microservices

   This is correct. Edge Computing is not a key functionality of SAP Cloud Platform. See unit 1, Lesson 4 of CLD100, Col18.

UNIT 2

Infrastructure and Database Service in Platform-as-a-Service (PaaS)

UNIT OBJECTIVES

- Get an introduction to Platform-as-a-Service
- Perform a deep dive into PaaS
- Explore the PaaS infrastructure
- Explain data and storage in PaaS

Explore development and IT operations services

Explore the internet of things (IoT)

Explain mobile services

Use runtime and containers

Unit 2
Lesson 1

Introducing Platform as a Service (PaaS)

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Get an introduction to Platform-as-a-Service

PaaS, Introduction

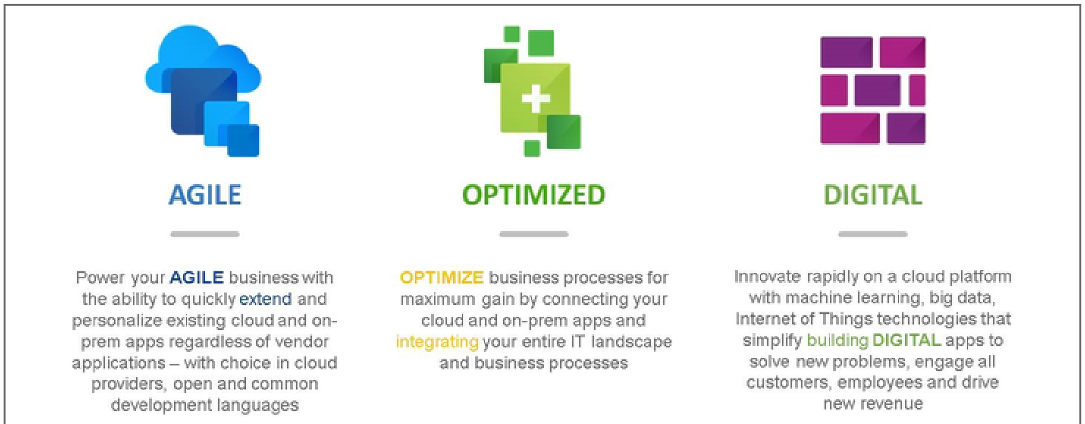


Figure 25: SAP Cloud Platform, the enterprise platform-as-a-Service for leading digital innovation

SAP Cloud Platform is positioned as the **agility platform** for SAP and signifies its importance as the solution for **Mode 2** IT environments. It helps companies in their digital transformation journey become agile, optimized and digital.

SAP Cloud Platform, Become AGILE by extending Cloud and On-Premises Apps

Achieve business agility by leveraging and extending your current applications:

- **Extend current cloud apps** to add new features and personalization unique to the business, its people and goals
- **Transform your current on-premises applications** into beautiful, mobile-ready apps for your employees, customers and partners
- **Achieve** measurable results with **pre-built app extensions** created by SAP and our partners
- **Example:** ZS Associates uses SAP Cloud Platform to extend the functionality of SAP SuccessFactors Employee Central to have greater confidence in company data and establish a strong foundation for analytics.

SAP Cloud Platform, OPTIMIZE business processes by integrating your apps and data

Optimize business processes by reducing data silos and operational inefficiencies

Connect your apps, data, and business processes regardless of who built them or where they are deployed

Deliver the right information and business processes into the right hands on the right device at the right time

Integrate on-premises and **cloud applications to deliver hybrid clouds** with real-time data and event streams

Example: Owens-Illinois, the world's leading glass packing manufacturer, uses SAP Cloud Platform to achieve cost effective and secure exchange of business-to-government (B2G) e-invoices for legal and tax compliance in Peru.

SAP Cloud Platform, Become DIGITAL with innovative technologies and apps

Innovate and build digital cloud apps using open standards

Build innovative apps to meet new challenges, attract new customers and drive new revenue opportunities

Develop using open standards such as Java, Python, Node.js as we incorporate Cloud Foundry and OpenStack technology into SAP Cloud Platform

Create native apps and deploy to any smart phone, tablet, and computer

Example: B. Braun opted for the SAP Cloud Platform and SAP Fiori for iOS to develop an iOS app that scans, tracks a digital twin of each surgical instrument ensuring its availability when needed but also replacing a labor and paper intensive process.

PaaS, Introduction, Modular Suite

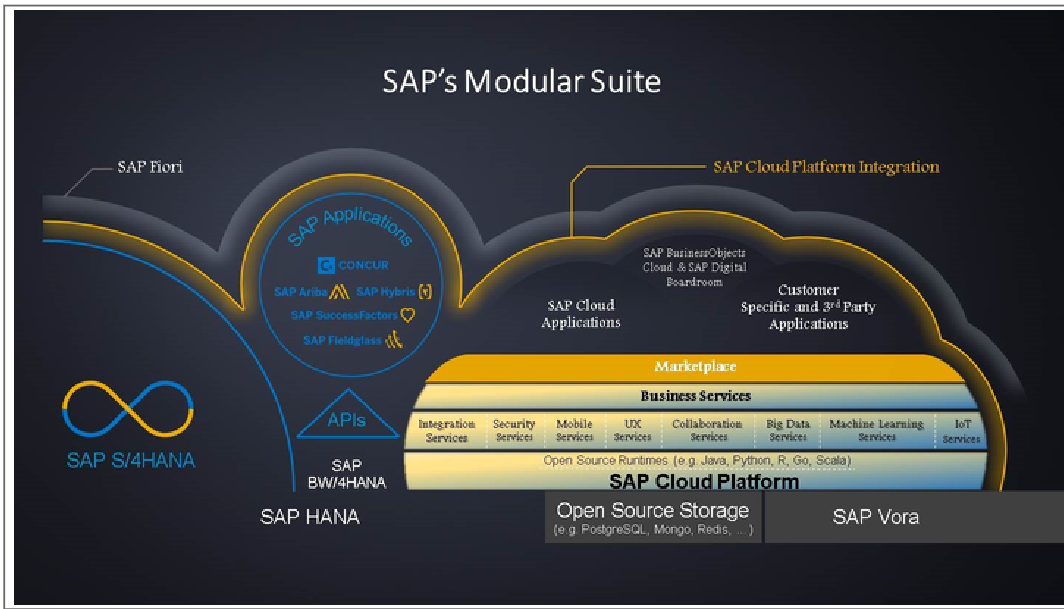


Figure 26: PaaS, Introduction

The figure gives an overview of the modularity of the suite.

Capabilities of the SAP Cloud Platform

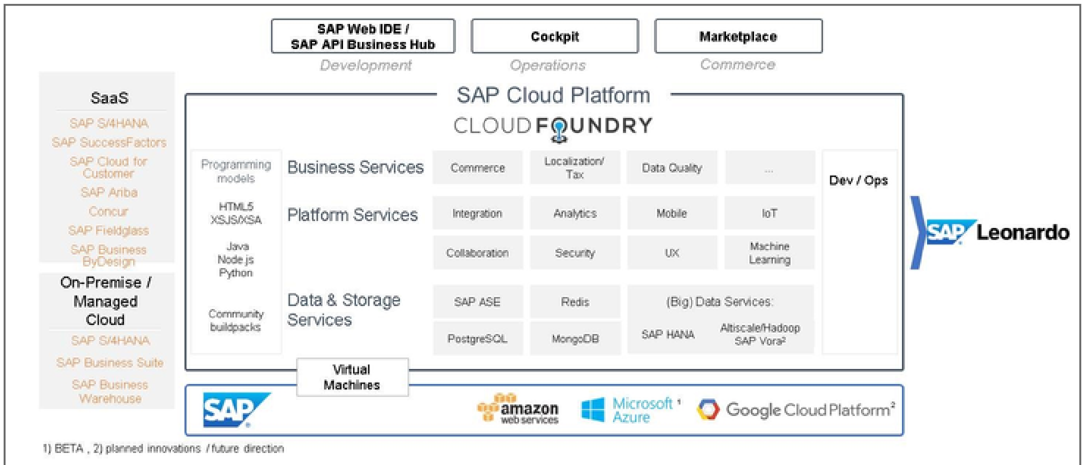


Figure 27: Capabilities and built of the SAP Cloud Platform

The figure gives an overview of all capabilities of the SAP Cloud Platform.

LESSON SUMMARY

You should now be able to:

Get an introduction to Platform-as-a-Service

Unit 2
Lesson 2

Performing a Deep Dive into PaaS

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Perform a deep dive into PaaS

PaaS, Deep Dive

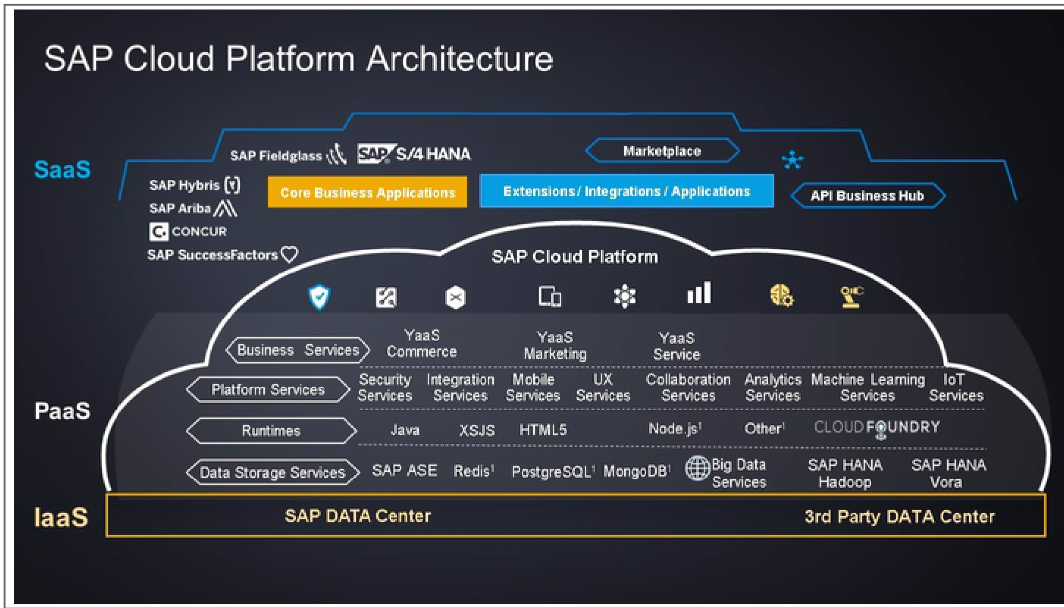


Figure 28: PaaS, Deep Dive

The figure gives a closer look at the architecture of the SAP Cloud Platform. The architecture consists of 3 layers:

- the IaaS (Infrastructure as a Service)
- the PaaS (Platform as a Service)
- the SaaS (Software as a Service)

PaaS requires an infrastructure. It additionally shows the hierarchy of IaaS, PaaS and SaaS. PaaS requires a IaaS.

SAP Cloud Platform: a platform for SAP Leonardo applications

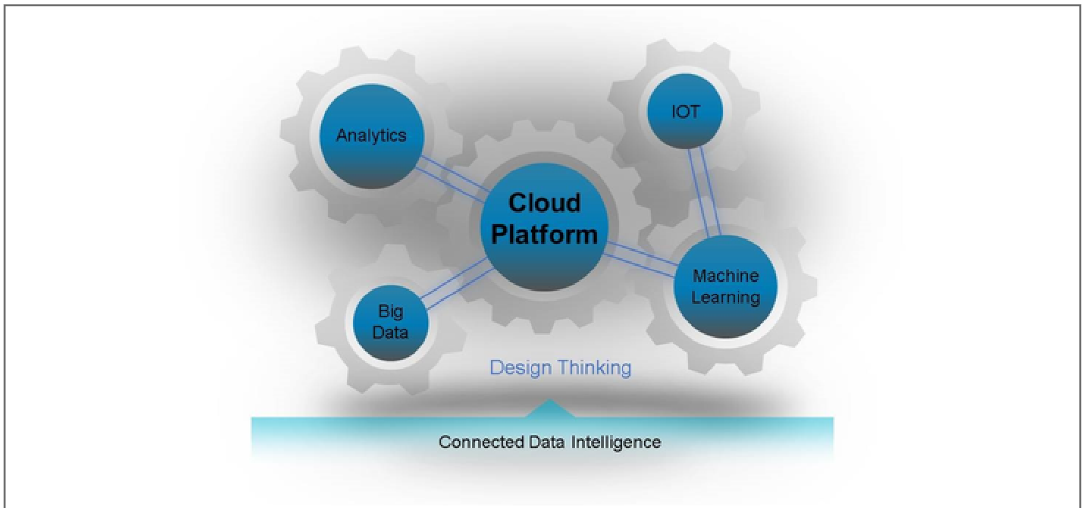


Figure 29: SAP Cloud Platform -Platform for SAP Leonardo applications

Leonardo is based on the PaaS-platform. Leonardo apps will be implemented on the SAP Cloud Platform.

PaaS, Deep Dive

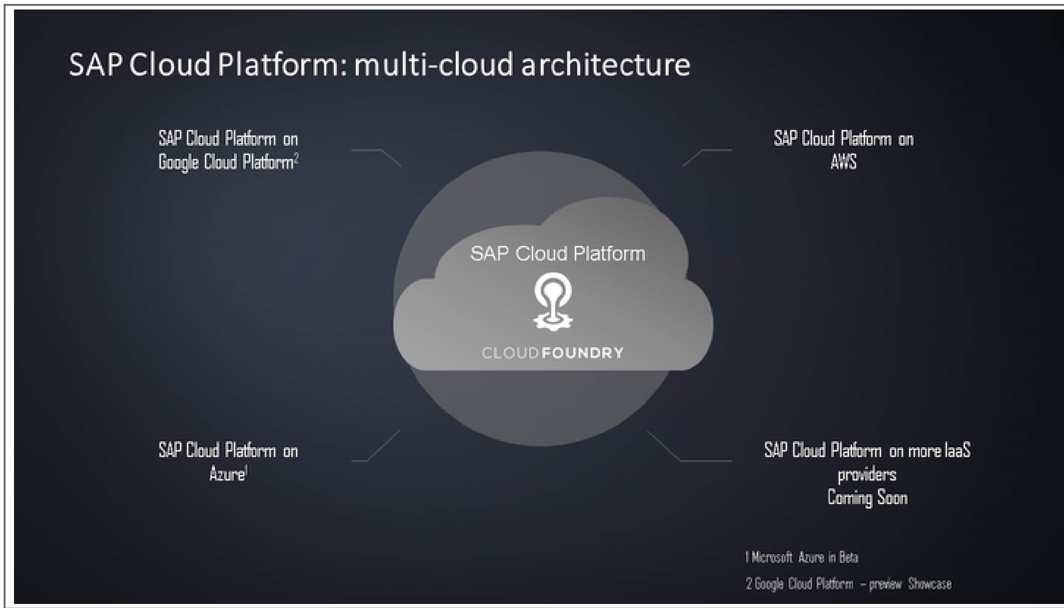


Figure 30: SAP Cloud Platform: multi-cloud architecture

The figure shows a future trend: the PaaS will be able to run on other platforms, like AWS, Google or Azure. Note: IaaS = HEC.

SAP Cloud Platform, Enterprise capabilities to be agile, optimized and digital

Figure 31: Enterprise capabilities to be agile, optimized and digital

The figure shows the different categories, for which Services are available. It is a big picture over all capabilities of the IaaS platform.

LESSON SUMMARY
You should now be able to:

- Perform a deep dive into PaaS

Unit 2
Lesson 3

Exploring the PaaS Infrastructure

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explore the PaaS infrastructure

PaaS, Infrastructure

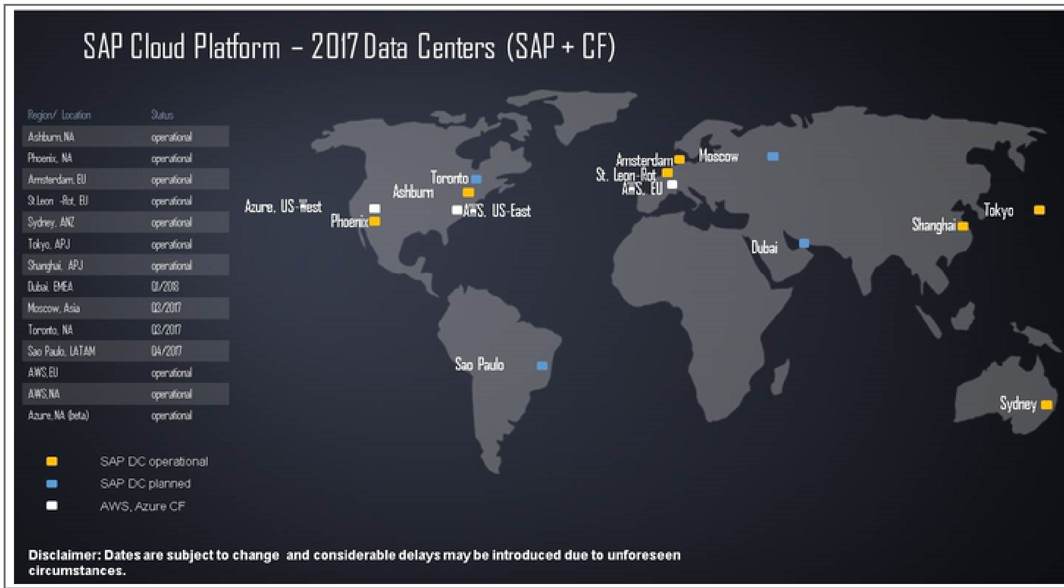


Figure 32: PaaS Infrastructure

The figure shows the data centers, providing the SAP Cloud Platform.

Infrastructure: SAP Cloud Platform, Data Centers

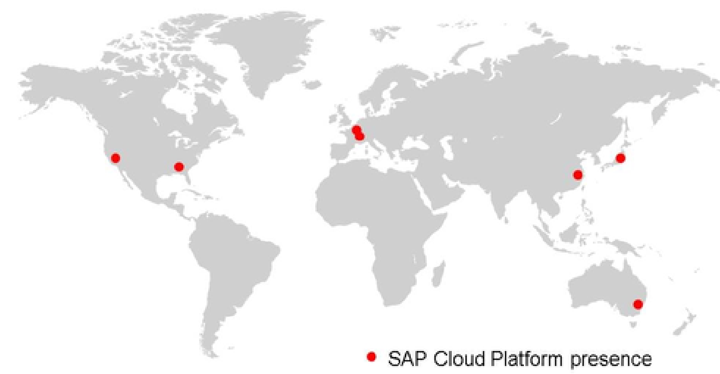


Figure 33: SAP Cloud Platform, Data Centers

SAP Cloud Platform runs in SAP's data centers around the globe. It offers SAP tier-3 and 4 data centers with 24/7 global support and 99.9% availability.

Data centers

SAP managed data centers & selected partners operating according to SAP standards

Customers / partners choose the data center which they want to use

Currently SAP Cloud Platform is available in data centers in:

- St. Leon-Rot, Germany
- Amsterdam, Netherlands (only for disaster recovery)
- Ashburn (Virginia), United States
- Phoenix (Arizona), United States
- Sydney, Australia
- Tokyo, Japan
- Shanghai China

The data centers are world-class tier-3 and 4 data centers with advanced network security and reliable data backups. For more information about SAP data centers, see http://www.sapdatacenter.com.

Infrastructure: Data Center Certifications & Attestations

SAP Cloud Platform runs only in secure and certified environments

Certified operations

World-class data centers

Advanced network security

Reliable data backup

Built-in compliance, integrity, and confidentiality

Secure and compliant infrastructure with 99.9% availability

- Certifications, e.g. ISO 27001, SOC 1 Type 1, SOC 1 Type 2, SOC 2 Type 1
- SAP warrants 99.9% system availability over any calendar month

The following certifications and controls are kept in each of the data centers:

ISO 27001
: Certification for Information Security Management Systems

SOC SSAE 16
: Statement on Standards for Attestation Engagements No. 16

SOC 1 Type 2
: Statement of effectiveness of Type 1 examination

SOC 2 Type 1
: Service Organization Controls Report (Attestation report)

SAP Cloud Platform runs in secure and certified data centers: SAP has certified quality management for the data centers and certifications for IT operations, high availability, and energy efficiency as well as testified compliance to international accounting regulations.

LESSON SUMMARY
You should now be able to:

- Explore the PaaS infrastructure

Unit 2
Lesson 4

Explaining Data and Storage in PaaS

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain data and storage in PaaS

Data and Storage

SAP Cloud Platform, Data and Storage

SAP Cloud Platform offers a variety persistence options allowing flexibility when developing solutions

These are the persistence options:

SAP Cloud Platform HANA service

- Massive real-time speed
- Breakthrough analytics
- Dramatic simplification
- Amazing new applications

SAP Cloud Platform ASE service

- Superior performance, with ability to easily meet growing business needs
- 24x7 availability with proven track record in Wall Street and Telco environments
- Flexible - can be used for custom app development, OEM environments, and SAP Business Suite

SAP Cloud Platform Document service

- Content repository for unstructured or semi-structured content
- Based on OASIS content management interoperability service (CMIS)

ObjectStore on SAP Cloud Platform

- supports the storage and access of unstructured data (e.g. Files, BLOBs) as 'Objects'
- helps users to create blob storage space

MongoDB on SAP Cloud Platform (beta)

- Flexible - great for processing semi-structured data
- Easy scaling, especially horizontally

PostgreSQL on SAP Cloud Platform (beta)

Open source ORDBMS

Users can extend SQL by adding new data types, functions, operators and aggregate functions

Supports various authentication mechanisms, such as Kerberos, SSPI and LDAP

ACID (atomicity, consistency, isolation, and durability) in database transaction is guaranteed

Redis on SAP Cloud Platform (beta)

Open source advanced key-value cache and store

High performance in-memory data structure store with on-disk persistence

Lower barrier of entry and faster time to value through big open-source community

SAP Cloud Platform Big Data Services

Provides a performance-driven and robust Big Data framework that uniquely includes Spark, Hadoop, Hive, Vora and more.

SAP Cloud Platform, SAP HANA Service

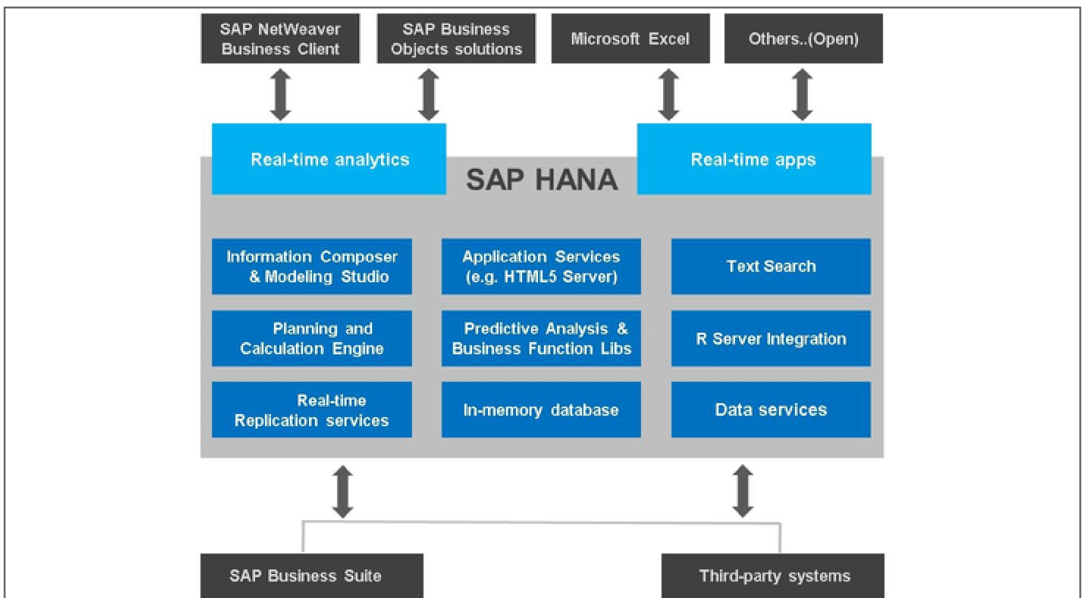


Figure 34: SAP HANA Services

SAP HANA is a modern in-memory platform for real-time analytics and applications. It enables organizations to analyze business operations based on large volume and variety of detailed data in real-time, as it happens.

Key capabilities and Benefits

Key capabilities

SAP HANA base and platform editions supported

Supported sizes (RAM): 32 GB, 64 GB, 128 GB, 256 GB, 512 GB, 1 TB (bigger sizes up to 3 TB upon request)

Spatial/GIS provides function to analyze and process geospatial information

Predictive Analysis Library (PAL) is a set of predictive algorithms in the SAP HANA Application Function Library (AFL)

Text search and analysis enable SAP HANA in-database analysis on unstructured and structured related content

Multitenant Database Containers (MDC) tenants

Available on Cloud Foundry and Neo environments

Benefits

Massive real-time speed

Breakthrough analytics

Dramatic simplification

Amazing new applications

SAP Cloud Platform SAP ASE Service

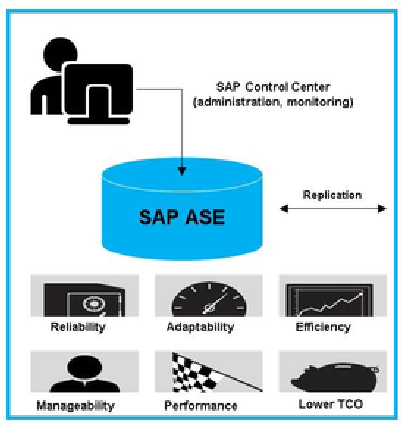


Figure 35: SAP ASE Service

SAP Adaptive Server Enterprise (SAP ASE) is a high-performance transactional database. It is a cost-effective database solution that can handle large numbers of transactions and concurrent users with superior performance, reliability, efficiency and is used by customers worldwide.

Key capabilities and Benefits

Key capabilities

Supported sizes (HDD): 32 GB, 80 GB,160 GB, 320 GB, 640 GB

Scales to over a million transactions per minute for large concurrent workloads and data volumes

Provides high availability and disaster recovery for continuous operations

Provides lowest cost/best TCO for large number of users, transactions and data volumes with efficient resource utilization and easy administration

Integrated with the SAP HANA for application development and report acceleration

Benefits

Superior performance, with ability to easily meet growing business needs

24x7 availability with proven track record in Wall Street and Telco environments

Flexible – can be used for custom app development, OEM environments, and SAP Business Suite

SAP Cloud Platform Document service

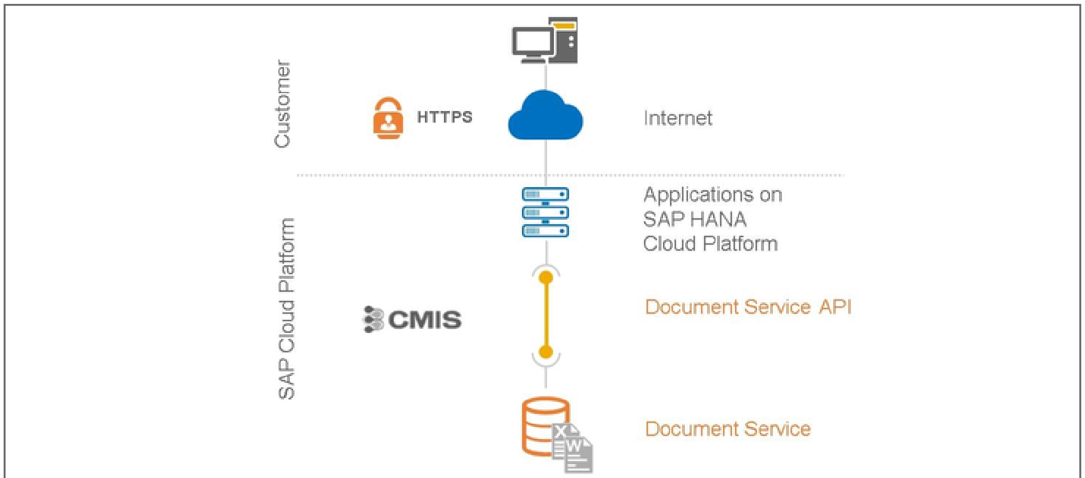


Figure 36: Document Service

SAP Cloud Platform Document service provides an enterprise cloud content repository for unstructured or semi-structured content.

Key capabilities and Benefits

Key capabilities

Content repository for unstructured or semi-structured content

Define hierarchies and metadata

Support for access controls, checkout and versioning

Based on OASIS content management interoperability services (CMIS)

Benefits

Multi-tenancy and fully managed

Files are always encrypted (AES) before storage

MongoDB on SAP Cloud Platform

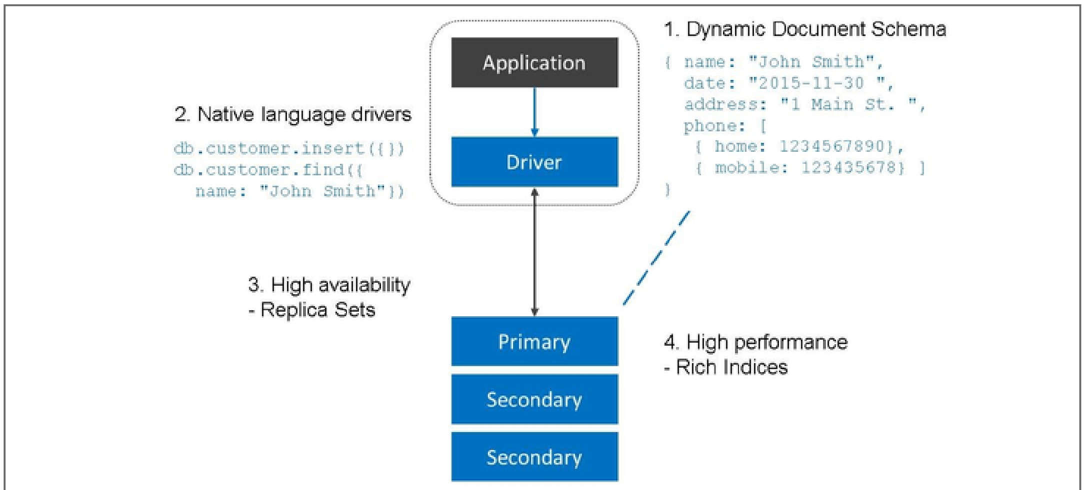


Figure 37: MongoDB on SAP Cloud Platform

SAP Cloud Platform provides support of MongoDB, the open-source document-oriented NoSQL database. Instead of utilizing tables to organize data as in traditional relational databases, MongoDB favors JSON-like documents with dynamic schemas.

Key Capabilities and Benefits

Key capabilities

Supports dynamic schema; unlike RDBMS where table content has to conform with a rigid column structure

Supports Replica Sets

Any field can be indexed; secondary indices are also supported

Benefits

Flexible - great for processing semi-structured data

High availability

High performance

PostgreSQL on SAP Cloud Platform

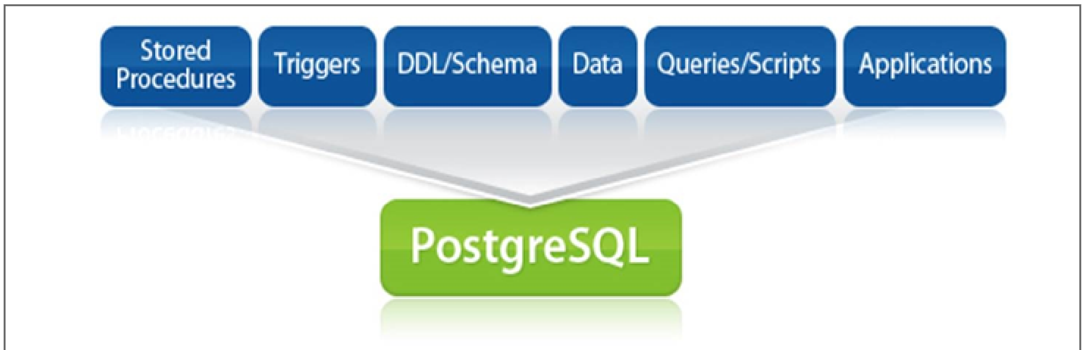


Figure 38: PostgreSQL on SAP Cloud Platform

SAP Cloud Platform provides support of PostgreSQL, the open source object-relational database management system (ORDBMS).

Key Capabilities and Benefits

Key capabilities

- Supports majority of standard SQL
- Offers inheritance
- Supports updatable views
- Manages concurrency through MVCC (multiversion concurrency control)

Benefits

- High Availability
- Brings together the concept of object oriented design and relational database
- ACID compliance

Redis on SAP Cloud Platform

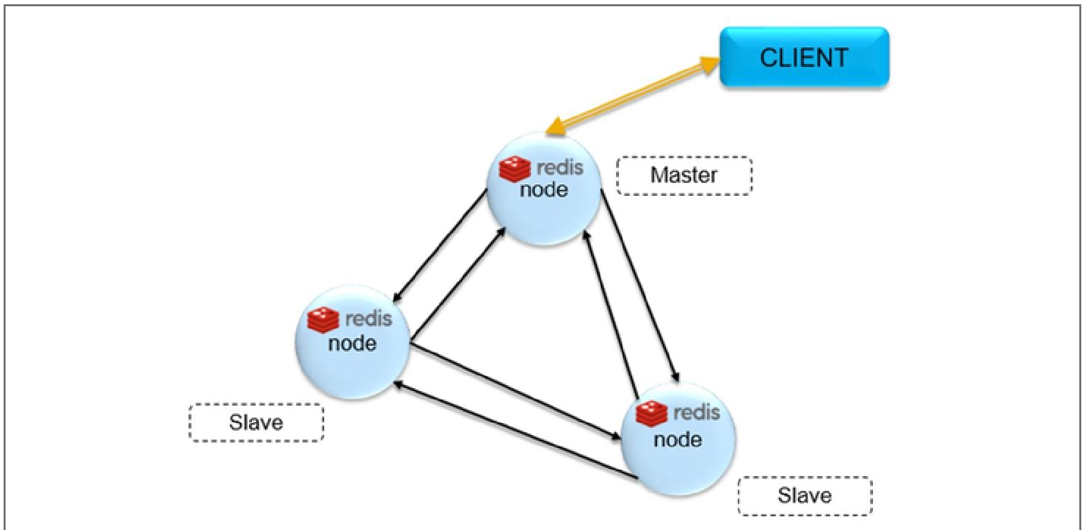


Figure 39: Redis on SAP Cloud Platform

SAP Cloud Platform provides support of Redis, the open-source advanced key-value cache and store.

Key capabilities and Benefits

Key Capabilities

Advanced key-value cache and store

High performance in-memory data structure store with on-disk persistence

Supports not only basic strings but also data types like lists, sets or hash tables of strings

Benefits

Extremely Fast

High availability

Multi-faceted tool

Data & Storage (CF) ObjectStore on SAP Cloud Platform

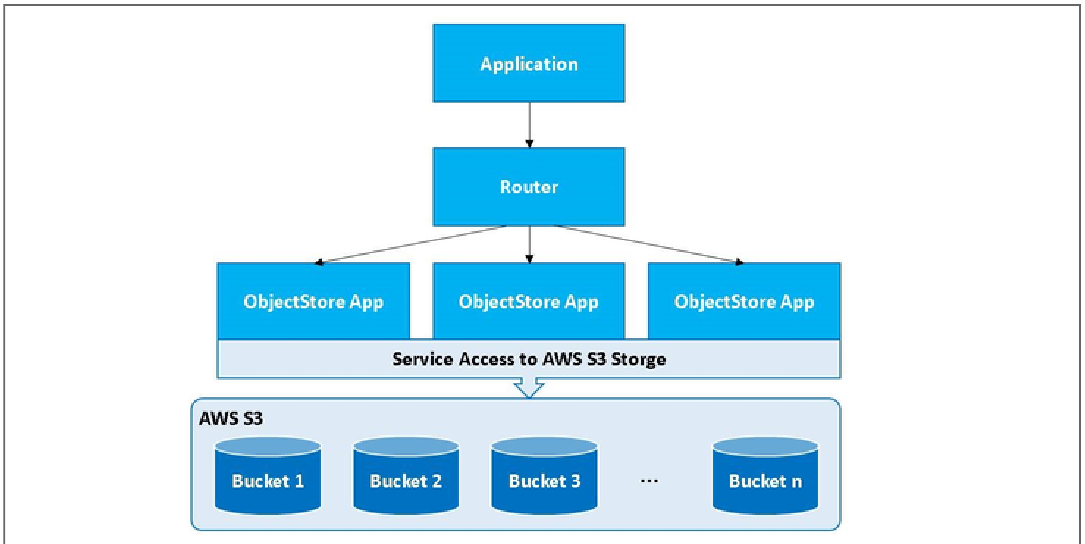


Figure 40: ObjectStore

ObjectStore on SAP Cloud Platform supports the storage and access of unstructured data (e.g. Files, BLOBs) as 'Objects' in Amazon Simple Storage Service (Amazon S3) Buckets. This service will help users to create blob storage space and use it from their respective applications.

Key Capabilities and Benefits

Key capabilities

Create/Delete an AWS S3 bucket

Provides secure credentials to access and control the Objects in the bucket

Possibility to interact with the objects in the bucket using any SDK (supported by AWS)

Benefits

High availability

High Performance

Ease of access

Proven industry usage

Data & Storage: SAP Cloud Platform Big Data Services

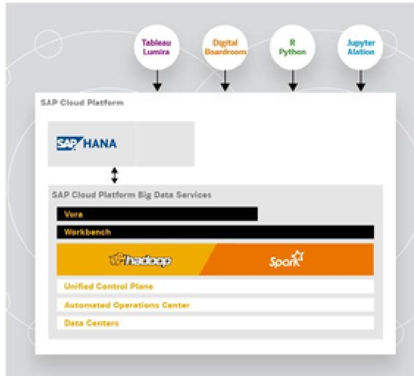


Figure 41: SAP Cloud Platform Big Data Services

SAP Cloud Platform Big Data Services provides a performance-driven and robust Big Data framework that uniquely includes Spark, Hadoop, Hive, Vora and more.

Key Capabilities and Benefits

Key Capabilities

Cloud-optimized, high performance Hadoop and Spark platform

Support for business analyst and data science applications

Fully managed service

Secure high bandwidth connectivity options

Benefits

More time to focus on analytics instead of cluster operations

Scale easily to meet short-term demand and long-term business growth

Industry-leading performance and reliability for high job success rate and low TCO

SAP Cloud Platform Big Data Services are comprehensive services, including deployment, automated operations management, and proactive support. Data scaling, compute bursting, and job monitoring are all automatically included. Other cloud providers just offer Hadoop or Spark on a generic cloud server - everything else is up to the customer. SAP offers a complete, optimized Big Data environment, including security, scalability, operational services, and support, all in one offering. Customer's data science and analytics team can focus on business priorities, not Hadoop or Spark management and operations.

The services feature the key capabilities for Big Data - Apache Hadoop, Apache Spark, Apache Hive, Apache Pig and SAP Vora. It also includes support for business analyst applications as well as data science applications.

LESSON SUMMARY
You should now be able to:

Explain data and storage in PaaS

Unit 2
Lesson 5

Exploring Development and IT Operations (DevOps) Services

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explore development and IT operations services

Development and IT Operations Services

Development and IT Operations (DevOps) allow developers to develop and manage applications - including complete life cycle management. These services not only increase developer productivity by simplifying develop but also improve team productivity with the ability to code and collaborate anywhere

The following are facts about Development and Operations Services:

SAP Cloud Platform cockpit for all operational needs

- Configuration
- Deployment
- Monitoring

Standards-based development environment

- Use of the popular open-source Eclipse IDE on the local computer together with the SAP Cloud Platform SDK
- SAP HANA web-based Development Workbench browser-based IDE for development and debugging of XSJS for SAP HANA
- SAP Web IDE to enable developers to build new business applications in the cloud with zero footprint
- Git service to handle your version control system needs
- Debugging service to inspect runtime behavior and state of web application
- Monitoring service to receive the metrics of Java applications running on SAP Cloud Platform
- Performance Statistics Service(Beta) to to monitor the resources used by the applications and to investigate the causes of performance issues.

SAP App Center

Single destination to evaluate, consume and market SAP and partner applications built on SAP Cloud Platform.

SAP Translation Hub

Access SAP's translation experience across multiple products and languages.

SAP Cloud Platform App Autoscaling service (beta)

Enables applications to automatically increase or decrease the compute capacity to meet changing demands.

SAP Cloud Platform Job Scheduler service (beta)

Allows applications to define and manage jobs that run on either one-time or recurring schedules.

DevOps, SAP Cloud Platform cockpit

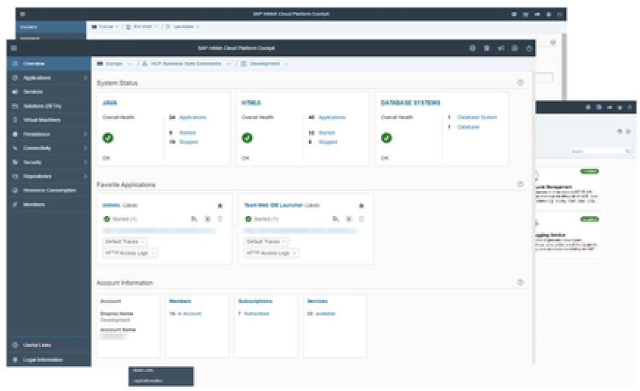


Figure 42: SAP Cloud Platform cockpit

The SAP Cloud Platform cockpit is the consolidated destination for all operational needs from configuration to deployment and monitoring on SAP Cloud Platform.

Key capabilities and Benefits

Key capabilities

- Entry point to the SAP Cloud Platform on the web
- SAP Cloud Platform and application configuration
- Deployment of Java, XSJS and HTML5 applications
- Monitoring of resources, databases and applications

Benefits

- Complete overview and access point of the SAP Cloud Platform
- Easy web-based operations administration

DevOps Services, Development Overview

Figure 43: DevOps Services, Development Overview

Descriptions:

Eclipse

Use of the popular open-source Eclipse IDE on the local computer

Full fledged Integrated Development Environment with the ability to add your favorite plugins and profit from the huge ecosystem

SAP HANA Web-based Development Workbench

Browser-based IDE for lightweight creation and editing of development objects and debugging for SAP HANA

Four sub-tools: Editor, Catalog, Security, Traces

SAP Web IDE

Enable developers to build new business applications in the cloud with zero footprint

Simplify SAP Fiori and SAP UI5 development and support end-to-end application lifecycle with one tool

DevOps Services, Eclipse

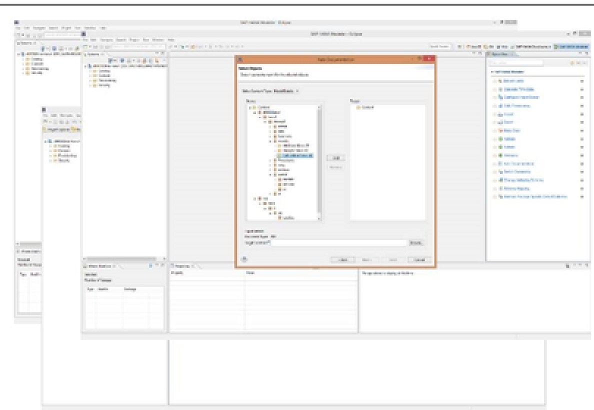


Figure 44: Eclipse

The open-source Eclipse IDE provides a unified IDE for developing cloud applications on SAP Cloud Platform.

Key capabilities and Benefits

Key capabilities

Develop Java, SAP HANA native and HTML5/SAPUI5 applications

Supports offline development

Full access to custom plugins provided by the Eclipse ecosystem

Benefits

Use of the de facto standard IDE

Leveraging existing knowledge and experience in working with the Eclipse IDE

Integrates with the SAP Cloud Platform, debugging service and SAP Cloud Platform, profiling service

SAP HANA Web-Based Development Workbench

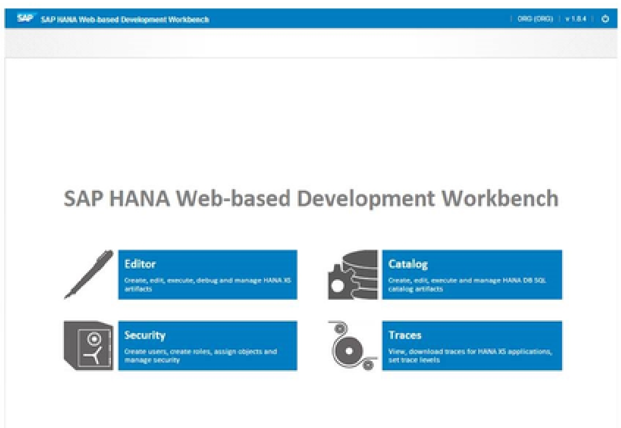


Figure 45: SAP HANA Web-Based Development Workbench

The SAP HANA Web-based Development Workbench is a completely browser-based IDE for lightweight creation and editing of development objects and debugging for SAP HANA without the need for SAP HANA Studio.

Key capabilities and Benefits

Key capabilities

Four sub-tools: Editor, Catalog, Security, Traces

Access to the full SAP HANA Repository and Catalog in a browser, also access User and Role details

Develop XSJS applications for SAP HANA in the cloud, with code editors with syntax coloring, code folding and client and server side checks

Application creation wizards which can create the package as well as the starting artifacts

Benefits

Works completely in the cloud

Zero footprint and low TCO

DevOps Services, SAP Web IDE

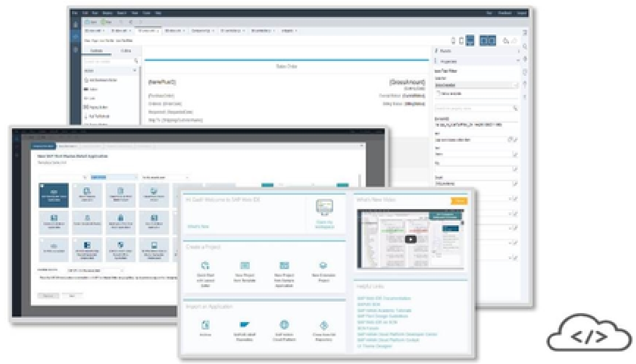


Figure 46: SAP Web IDE

SAP Web IDE is SAP's powerful, web-based integrated development environment that simplifies end-to-end application development for SAP Fiori and SAPUI5 apps including development, build, deployment, and customer extensions.

Key capabilities and Benefits

Key capabilities

Develop & extend SAP Fiori, IoT and "hybrid-mobile" apps

Drag & drop tools, wizards and templates

Code and layout editor optimized for SAPUI5

Integrated Git source control, build and deployment

Integration with various data sources

SDK, extend SAP Web IDE functionality

Benefits

Zero installation and administration

Reduces cost, complexity and effort

Increases developer and team productivity

Develop once, deploy to cloud and on-premise SAP platforms: SAP Cloud Platform, ABAP, SAP Mobile Platform and Enterprise Portal

Feature videos: https://www.youtube.com/channel/UCgScBj_AEGo_mI5accM4XhA

DevOps, SAP Cloud Platform Git service

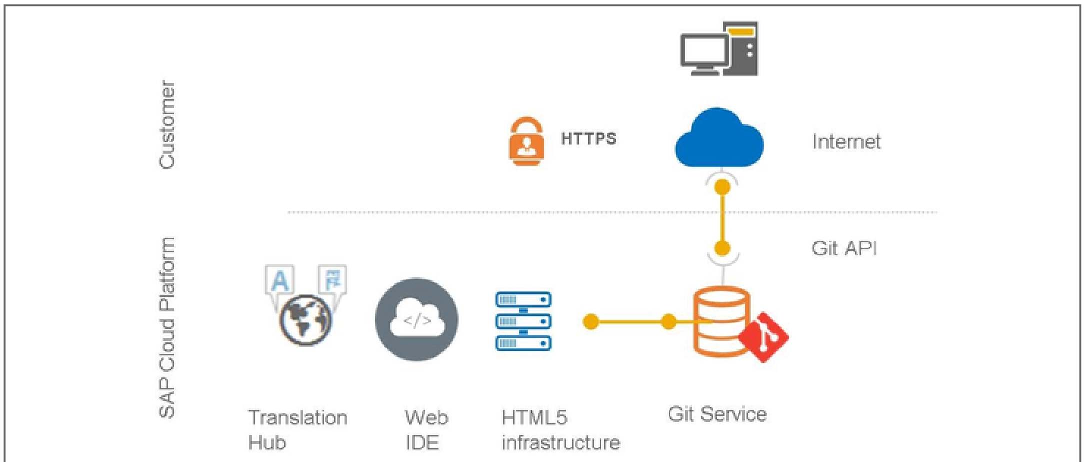


Figure 47: SAP Cloud Platform Git service

SAP Cloud Platform Git service provides hosting of Git repositories in the cloud to handle your version control system needs.

Key capabilities and Benefits

Key capabilities

Git repositories for versioning source code

Create and delete Git repositories

Manage permissions

Use any Git client to fetch / push changes

Browse content of Git repositories online

Integrated with HTML5 infrastructure, SAP Web IDE and SAP Translation Hub

Based on Eclipse JGit and Gerrit Code Review

Benefits

Fully managed version control system

Empowers distributed development

DevOps, Debugging Service

The Debugging Service can be used to inspect the runtime behavior and state of a Web application after it is created and running in SAP Cloud Platform.

Key capabilities and Benefits

Key capabilities

Enables detecting and diagnosing errors in an application

Controls the execution of a program by setting breakpoints, suspending threads, stepping through the code and examining the content of variables

Allows to debug an application on SAP Cloud Platform local runtime and in the cloud, in Eclipse IDE

Benefits

Provides debugging capabilities for Java applications in the cloud

Much better performance than commodity debugging solutions due to optimized protocol

DevOps, Monitoring Service

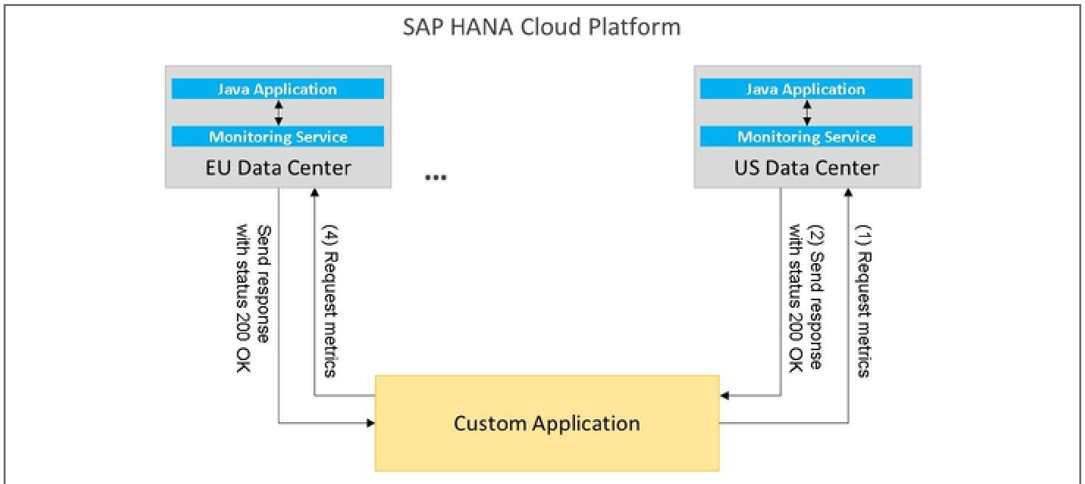


Figure 48: Monitoring Service

SAP Cloud Platform, Monitoring Service is a service to receive the metrics of Java applications running on SAP Cloud Platform.

Key capabilities and Benefits

Key capabilities

Provides the functionality to monitor the state and metric details of a Java application and its processes

Uses REST API requests to get the state or metrics of an application or an application process

Benefits

Enables elastic scaling for Java applications

Retrieves and shows the metrics of many Java applications located on different accounts and in different data centers

Notification of all critical metrics of many Java applications vie e-mail, SMS, or another channel

Enables taking actions for application self-healing when critical metrics are received

DevOps, Performance Statistics Service (beta)

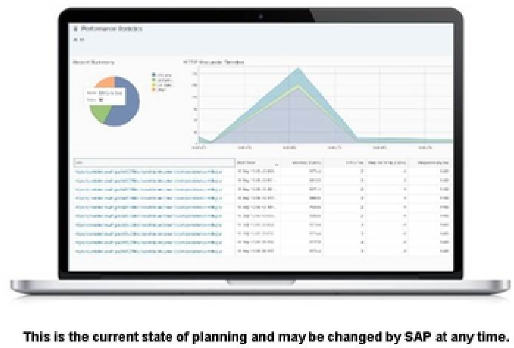


Figure 49: Performance Statistics Service (beta)

Performance statistics is a SAP Cloud Platform service that enables to monitor the resources used by the applications and to investigate the causes of performance issues.

Key capabilities and Benefits

Key capabilities

Breakdown of the time and resources such as CPU, memory, response time, and DB calls for each HTTP request to the application

Collection of performance statistics data

Allows to get insight on specific requests and the respective behavior of the application

Benefits

Central collection of important application statistics

Provides insights into critical performance bottlenecks and therefore helps reducing operational costs

DevOps, Profiling Service

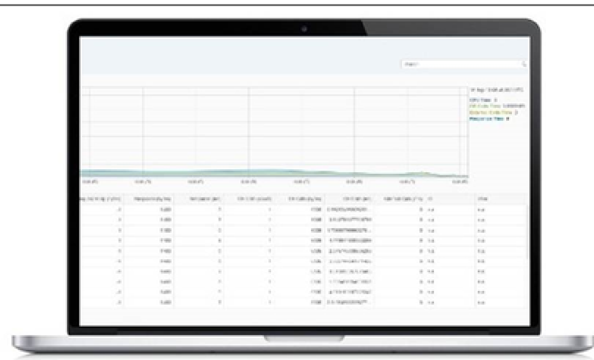


Figure 50: Profiling Service

Using the SAP JVM Profiler helps to analyze resource-related problems in Java applications whether the JVM is running locally or in the cloud.

Key capabilities and Benefits

Key capabilities

Retrieve and analyze profiling data

Display profiling data as statistics and graphs

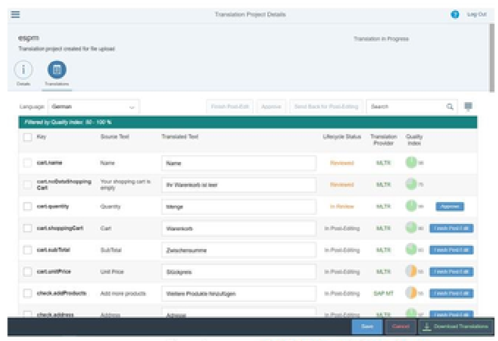
Check and optimize memory usage

Identify frequently called operations and slow performance

Benefits

Monitor and improve stability, security and scalability of your Java applications on SAP Cloud Platform

DevOps, SAP Translation Hub



Read more: SAP Translation Hub

This is the current state of planning and may be changed by SAP at any time.

Figure 51: SAP Translation Hub

SAP Translation Hub is an SAP Cloud Platform service that opens the door to the wealth of SAP's translation experience across multiple products and languages. With this service, a developer, whether customer or partner, can quickly pre-translate the field labels and buttons on their application UIs into various languages (currently, SAP Translation Hub supports 44 languages).

Key capabilities and Benefits

Benefits

SAP Machine Translation engines are trained with SAP data

Post-editing activities can be planned based on the quality index of translations

Addresses needs of cloud and on premise customers

Can be integrated in existing environments (e.g. SAP Web IDE or SE63 in ABAP systems) and serves independent file upload scenarios

Status Management on text and translation project level.

You can now check and adjust the translations that SAP Translation Hub provides by using different lifecycle statuses to manage the post-editing and review steps for each translation in a translation project. For more information, see Managing Post-Editing and Review Steps.

DevOps, SAP App Center

Keypoints:

For Customers:

- **Discover, Try, Buy, & Manage** 3rd party apps in one place
- Centralized **User & License Management**, across all vendors
- **Enterprise-Class** Purchasing (two-tier system: requester & procurement/IT)
- **PO-based** ordering, invoice settlement **via SAP Ariba payment solutions** or other payment gateways
- **Central Communication** platform with all 3rd party vendors

For Partners:

- **List Solutions** on the SAP App Center
- Manage **Price Books/Editions**, settings for **SAP Ariba payment solutions** or other payment gateways, **API** end-points configuration
- Manage **Trials, Leads, and Sales** in one place, for all customers
- **Central Communication** platform with all customers
- Manage Payments & Analytics

The scope of the App Center is extended, from a marketing listing only, to an end-to-end marketplace where customers can discover, try, buy and deploy applications. SAPs goal is to consolidate all Partner marketplaces into one. This will happen over time, but already SuccessFactors, SAP Analytics, Hybris and Concur have jumped onboard.

It is still possible for partners to list their solution for Marketing purposes only; in that case, the listing on the App Center remains free. On transactions, SAP acts as an "agent" (not a reseller of the partner solution), and collects a fee from the transaction in exchange for the Cloud Service offered on the marketplace.

DevOps (CF), SAP Cloud Platform App Autoscaling service (beta)

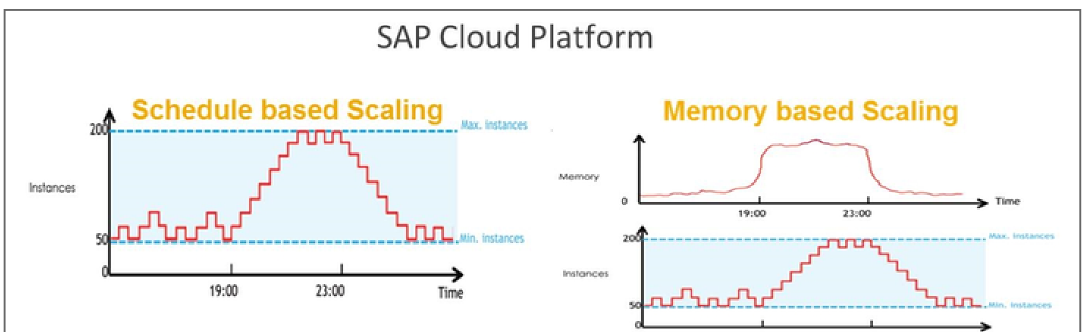


Figure 52: SAP Cloud Platform App Autoscaling service (beta)

SAP Cloud Platform App Autoscaling service is a service that enables applications to automatically increase or decrease the compute capacity to meet changing demands.

Key capabilities and Benefits

Key capabilities

Enables applications to define scaling policies with upper and lower thresholds that dynamically adjusts the number of instances of the application based on its memory consumption

Supports memory based scaling and scheduled based scaling

Benefits

Enables automatic scaling of applications hosted on SAP Cloud Platform

Enables applications to define the required compute resources at a scheduled time

https://wiki.wdf.sap.corp/wiki/display/HCPIN/Auto+Scaling

DevOps (CF), SAP Cloud Platform Job Scheduler service (beta)

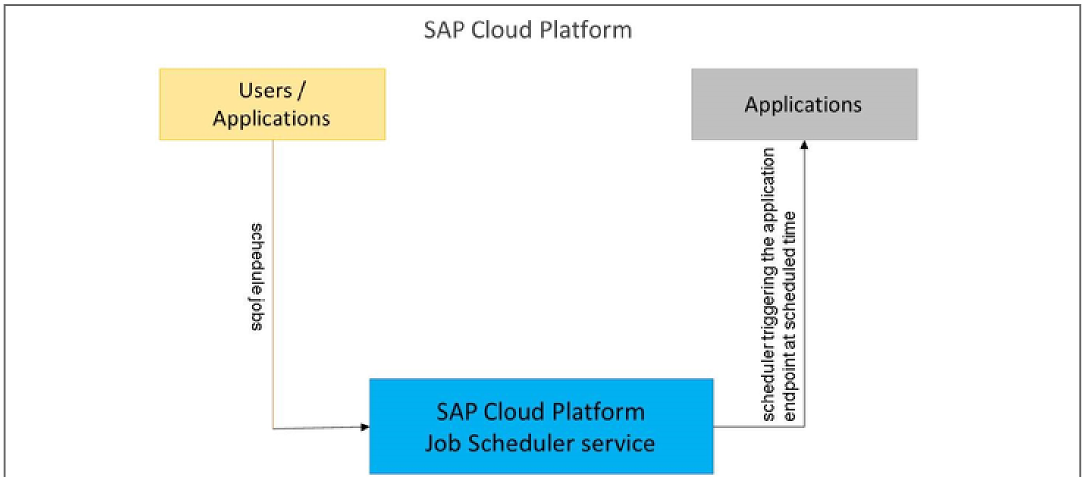


Figure 53: SAP Cloud Platform Job Scheduler service (beta)

SAP Cloud Platform Job Scheduler service is a service that allows applications to define and manage jobs that run on either one-time or recurring schedules.

Key capabilities and Benefits

Key capabilities

Runtime agnostic service to easily schedule any REST endpoint actions in applications

Easy to use REST APIs to schedule jobs, e.g. to schedule long running jobs asynchronously

Multiple schedule formats far beyond cron – to define simple and complex recurring schedules

Service Dashboard to monitor the schedules and their execution

Benefits

Enables applications to schedule jobs

Allows monitoring of jobs and schedules, the success/failure of the executions using the Service Dashboard

https://wiki.wdf.sap.corp/wiki/display/HCPIN/Job+Scheduler

LESSON SUMMARY

You should now be able to:

Explore development and IT operations services

Unit 2
Lesson 6

Exploring the internet of Things (IoT)

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explore the internet of things (IoT)

SAP Cloud Platform, Internet of Things

The SAP Cloud Platform enables you to go from sensor to action with SAP Cloud Platform Internet of Things

The following solutions are the building blocks of the IoT:

SAP Cloud Platform Internet of Things

- Connect to remote devices to manage life-cycle from onboarding till decommissioning
- Receive device data and send commands to remote devices
- Use IoT message management capabilities to collect sensor data and store it in the SAP Cloud Platform persistence layer (SAP HANA or SAP ASE)
- Message processing, integration with Smart Data Streaming

SAP Cloud Platform Streaming Analytics

- Analyzes and transforms high-velocity, high-volume event streams in real time
- Generates alerts or notify applications to respond
- Can be used with SAP Cloud Platform IoT Services to analyze incoming data in real-time

SAP Cloud Platform Remote Data Sync

- Synchronizing huge numbers of remote databases into a consolidated SAP HANA database in the cloud
- Always-available data at IoT devices, making economical use of network connectivity
- Remote data sync service is a synchronization technology for exchanging data between relational databases

SAP Cloud Platform: Internet of Things, Capabilities

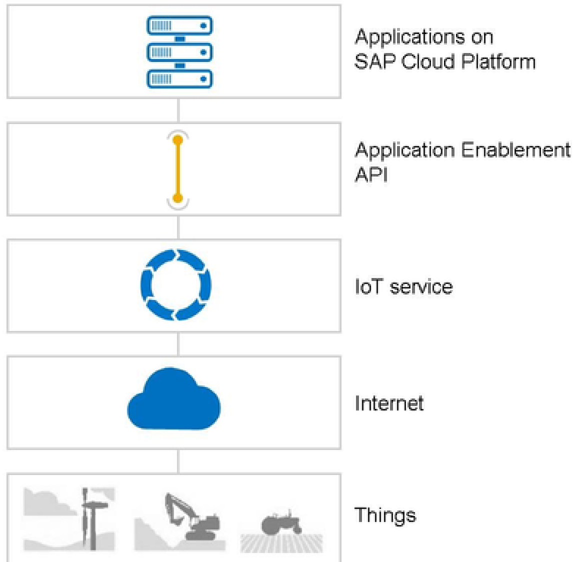


Figure 54: Internet of Things, Capabilities

The SAP Cloud Platform Internet of Things can be used to connect devices to the SAP Cloud Platform in order to use data from these devices in applications.

Key Capabilities:

IoT Cockpit as the central UI

Fine-tuned access control for IoT service users on the application level

Receive device data and send commands to remote devices using the Message Management Service

REST APIs for device modelling and data consumption

Notification management (alerts / rules / events)

Certificate-based on-boarding and authentication of devices to IoT Gateway Edge and Cloud

Device to IoT Gateway Cloud Protocol support

- HTTP (REST), MQTT

Device to IoT Gateway Edge Protocol support

- HTTP (REST), MQTT, CoAP, SNMP, ModBus, Zigbee, XMPP

Provisioning of a software development kit (SDK) for development of custom protocol adapters or agents and custom filters (interceptors) on IoT Gateway Edge

Support of various SAP Cloud Platform services

- PostgresSQL, KAFKA

Integration into the SAP IoT Application Enablement toolkit in support of rapid development of IoT applications through the IoT app builder

Get started quickly using guidelines, tutorials, and code examples coming with the Starter Kit for the SAP Cloud Platform Internet of Things publicly available on GitHub

Further aspects about IoT

IoT Cockpit as the central UI:

Is the user interface of the Internet of Things service

Allows the customer to manage users and tenants

Allows the customer to create and manage devices

It is a fully fledged user interface providing high flexibility in managing projects, tasks, scenarios and setups during design and runtime of the SAP Cloud Platform Internet of Things

Fine-tuned access control for IoT service users on the application level:

Fine-tuned access control for different users and user roles to manage visibility of tenants, devices, and relevant data

User access that requires each user to assume a unique digital identity across applications and the Internet of Things service, which enables access controls to be assigned and evaluated against this identity

Use of a unique identity across all system components that simplifies the monitoring and verification of potential unauthorized access and allows the system to keep tabs of excessive privileges granted to any individual within the Internet of Things service and other applications

Ability to track user access from the user lifecycle perspective, as user access can be tracked from new subscription, to suspension, to termination of contract

Receive device data and send commands to remote devices using the Message Management Service:

SAP Cloud Platform Internet of Things - Message Management Service facilitates sending messages between two or more clients (producers and consumers)

Allows application components to create, send, receive, and read messages; allows communication between different components of a distributed application to be loosely coupled, reliable, and asynchronous

Highly scalable and reliable messaging as a service

Scalability to support integration for millions of devices

REST APIs for device modelling and data consumption:

Provides a rich feature set and flexibility to compile complex and customized, specific device-management setups

Notification management (Alarms / Rules / Events):

Notification-management capabilities for setting rules, events, and alarms in the SAP Cloud Platform Internet of Things Cockpit

Customers enabled to manage and monitor their device behavior using a single user interface

Certificate-based on-boarding and authentication:

A device can register itself using the device type certificate. This certificate must be included in the device key store for the HTTPS connection. It will then be used during the initial SSL handshake as a client certificate

IoT Gateway (Edge and Cloud) for the SAP Cloud Platform Internet of Things uses of a certificate-based device onboarding mechanism, which is the standard procedure used by the majority of IoT device-connectivity platform vendors

Device to IoT Gateway Cloud Protocol support:

Internet of Things Gateway Cloud:

- Support of device connectivity via HTTP Rest to SAP Cloud Platform Internet of Things
- Support of device connectivity via native MQTT to SAP Cloud Platform Internet of Things

Provides customers with flexibility to use various protocols to communicate with SAP Cloud Platform Internet of Things

Device to IoT Gateway Edge Protocol support:

Internet of Things Gateway Edge:

- Provides smart plug-ins to support major IoT protocols Acts as a virtual gateway that brings devices and assets online
- Is responsible for collecting data from the device and sending data to the device (bi-directional communication)

Provides customers with flexibility to use the SAP Cloud Platform Internet of Things Gateway Edge running in the field with the support of major IoT protocols

Provisioning of a software development kit (SDK) for development of custom protocol adapters or agents and custom filters (interceptors) on IoT Gateway Edge:

Toolset to build connections to devices that speak proprietary protocols in addition to the edge protocols that SAP Cloud Platform Internet of Things supports out of the box

User ability to configure and set the device data payload already at the edge by using the SDK to set filters and their behavior

Support of various SAP Cloud Platform backing services:

PostgresSQL, KAFKA

Support for a variety of commercial and open source-based databases from SAP Cloud Platform, which benefits the customers and partners of SAP Cloud Platform Internet of Things - and their IoT projects (more to come see planned innovations)

Integration into the SAP IoT Application Enablement toolkit in support of rapid development of IoT applications through the IoT app builder:

Integration into SAP IoT Application Enablement that makes SAP Cloud Platform IoT an integral part of the SAP Leonardo portfolio, which is a highly innovative IoT portfolio that extends the digital core with adaptive applications, Big Data management, and connectivity to enable new business processes

Get started quickly using guidelines, tutorials, and code examples coming with the Starter Kit for the SAP Cloud Platform Internet of Things publicly available on GitHub:

- Understand the methodology of SAP Cloud Platform Internet of Things more readily
- Access basic and advanced sample scenarios

SAP Cloud Platform: Streaming Analytics

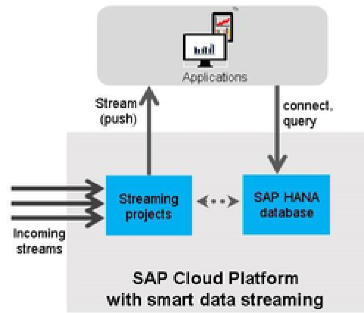


Figure 55: Streaming Analytics

The SAP Cloud Platform Streaming Analytics analyzes and transforms high-velocity, high-volume event streams in real time, allowing applications to detect and respond to changing conditions in real-time, and with the ability to compute and capture useful information in the SAP HANA database.

Key capabilities and Benefits

Key capabilities

- Monitor and analyze incoming data to watch for trends, correlation, patterns, missing data
- Generate alerts or notify applications to respond
- Filter, sample, aggregate and correlate incoming data
- Combine raw events into actionable information
- Capture only the data you want, in the form you need it
- Stream live updates to operational dashboards
- Can be used with SAP Cloud Platform IoT Services to analyze incoming data in real-time

Benefits

- Respond immediately to new information
- Optimize the data being stored in SAP HANA, storing only useful information in optimized data models

Smart data streaming

We have it listed as a planned innovation, but its on the cusp of being generally available.

Combined with our IoT Services, you'll be able to fully leverage the connectivity that you have into devices being able to monitor the data as it comes in.

Some of the things that goes beyond just being an interface for putting data into SAP Cloud Platform.

You'll be able to filter the data, allowing you to capture only the data that you want.

Or perhaps you want to set up alerts when there are anomalies in the data stream

For example, generally speaking, sometimes the data that you collect from things is not too critical where you have to have up to the second (not up to the minute) updates, but in cases where you may be applying IoT in Healthcare, it's absolutely critical that alerts are provided before even the application that depends upon that data can react.

Possible renaming in the pipeline to SAP Cloud Platform, streaming service.

SAP Cloud Platform: Remote Data Sync

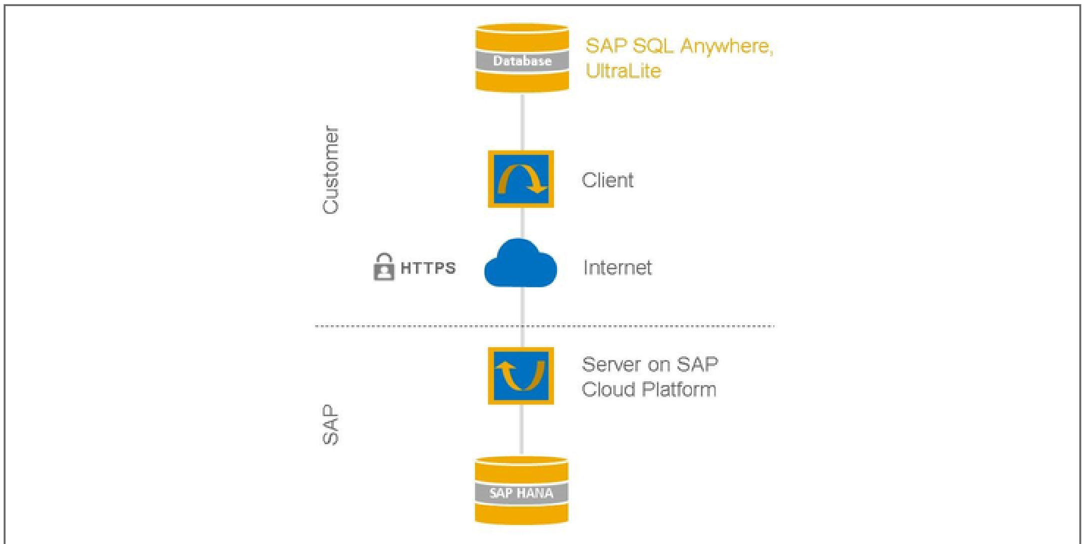


Figure 56: Remote Data Sync

SAP Cloud Platform Remote Data Sync service provides capabilities for synchronizing huge numbers of remote databases into a consolidated SAP HANA database in the cloud.

Key capabilities and Benefits

Key capabilities

Remote data sync service is a synchronization technology for exchanging data between relational databases

Remote database (in IoT gateway) needs to be an SAP SQL Anywhere database

Example: store sensor data in remote databases, consolidate in SAP Cloud Platform, e.g. for analytics and monitoring

Benefits

Always-available data at IoT devices, making economical use of network connectivity

Ensures transactional integrity over unstable networks

Sophisticated strategies for resolution of data change conflicts

LESSON SUMMARY

You should now be able to:

- Explore the internet of things (IoT)

Unit 2
Lesson 7

Explaining Mobile Services

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain mobile services

SAP Cloud Platform, Mobile Services

SAP Cloud Platform mobile portfolio delivers key capabilities such as multiple authentication methods, secure access to on-premises and cloud-based systems.

The following are the build blocks of mobile services in the SAP Cloud Platform:

Mobile service for development and operations

- Build and deploy mobile apps in the cloud.
- Supports hybrid and full cloud solutions.
- Server counterpart of the new SAP Cloud Platform SDK for iOS.
- Support of iOS, Android and Windows mobile platforms.

Mobile service for SAP Fiori

- transforms your SAP Fiori web app into a native/hybrid app.
- enables app lifecycle management and an enterprise app store app for deployment, ratings and reviews.
- provides infrastructure support for advanced mobile features such as push and offline.

Mobile service for app and device management

- Branded, multi-channel, self-service enterprise app store.
- Serve employees, partners and contractors.
- Support managed and unmanaged devices.
- Streamline publishing, analysis and management of apps and services for iOS, Android and Windows mobile platforms.

Mobile Services, mobile service for app and device management

Figure 57: Mobile service for app and device management

An integrated Enterprise Mobile Management SaaS solution running on SAP Cloud Platform.

Key capabilities and Benefits

Key capabilities

Branded, multi-channel, self-service enterprise app store.

Mobile app and device lifecycle management serving employees, partners and contractors.

Streamlined publishing, analysis and management of apps/services.

Support for iOS, Android and Windows mobile platforms.

Benefits

Increase mobile app adoption.

Lower the overall cost of supporting enterprise mobility.

Improve enterprise compliance and security.

Mobile app and mobile device management running as a service on SAP Cloud Platform.

Differentiation: Runs on Cloud Platform, enterprise mobile app store with end user self service (enroll, lock device, wipe device, locate device)

A single service can serve:

Multiple device ownership models - BYOD, Corporate-owned devices, Specialized Devices used in retail & hospitality, manufacturing, transportation & logistics, medical, etc.

Multiple lines of business (field sales/service, executive, HR, etc).

Many geographies (country legal/privacy requirements).

Management styles supported:

MDM app on the device

- Complete capabilities to manage devices, apps, and users

No MDM app on the device:

- Manage apps on the device without regard for the device

Mobile Services, mobile service for development and operations

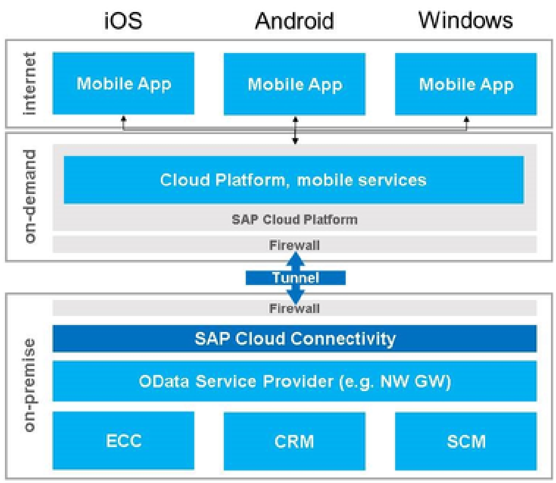


Figure 58: Mobile service for development and operations

The SAP Cloud Platform, mobile service for development and operations enables developers build and run mobile apps for iOS, Android and Windows with a scalable cloud backend.

Key capabilities and Benefits

Key capabilities

Build and deploy mobile apps in the cloud.

Content to go: all time and easy access to business information.

Supports hybrid and full cloud solutions.

Server counterpart of the new SAP Cloud Platform SDK for iOS.

Support of iOS, Android and Windows mobile platforms.

Benefits

Flexible and scalable runtime for mobile apps.

Supports SAML as standard way to authorize users on their device.

Enables offline applications and push notifications.

Easy to integrate back-end systems via REST API.

Integrate to SAP & non-SAP systems.

Further features:

Mobile services are available for developers who are quickly develop an application and enabling to support a mobile device.

Regardless of which mobile platform they are developing for, the SAP Cloud Platform ms will assure connectivity to the data and applications in a secure manner.

For those familiar with SMP, SAP Mobile Platform, those features are now available through SAP Cloud Platform.

So with for example the off-line capabilities, we recognize the need to sometimes be able to work with data locally (on the device and then be able to synchronize that with the back-end).

Support of hybrid app development is done via HTML5 with Apache Cordova.

Apache Cordova is essentially a container for web applications written in HTML, CSS, Java. Typically these don't have access a phone's camera, GPS, accelerometer, etc., but with Apace Cordova, it can.

Support for native app development is provided via the SAP Mobile Platform SDK, included in the SAP Cloud Platform ms license, for Android, iOS and Windows.

Mobile Services, mobile service for SAP Fiori

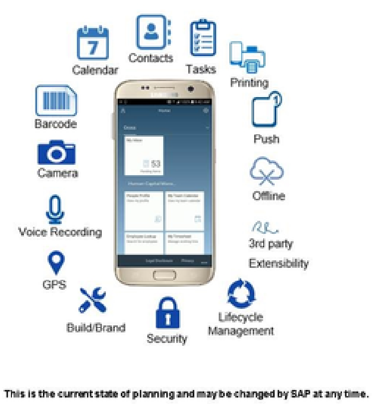


Figure 59: Mobile service for SAP Fiori

Optimize your SAP Business Suite UX renewal on mobile devices by deploying a consumer grade experience, complete with enterprise grade security.

Key capabilities and Benefits

Key capabilities

Cloud build service transforms your SAP Fiori web app into a native/hybrid app, necessary to leverage native device features.

App deployment service enables app lifecycle management and an enterprise app store app for deployment, ratings and reviews.

Data service provides infrastructure support for advanced mobile features such as push and offline.

Development plugin for Web IDE provides access to Cloud Build Service, Plugin management and the Developer Companion.

Benefits

Mobilizes SAP Fiori Cloud apps and SAP Fiori on-premise apps with a single solution.

Integrates with SAP Mobile Secure for mobile device and app management.

Simplifies support for basic and advanced SAP Fiori use cases.

Completely cloud-based developer infrastructure simplifies creating and testing delightful mobile apps.

SAP customers are continuing to renew their user experience through the implementation of SAP Fiori. And in any conversation that includes user experience, the mobile experience is a critical piece to the overall success.

With the mobile service for SAP Fiori (currently available only through SAP Fiori Cloud premium), SAP has introduced an end to end service designed to optimize and enrich the mobile user experience. In order to deliver mobile success, several key capabilities are included as part of the solution:

Build/Packaging service transforms your SAP Fiori web app into a native/hybrid app, necessary to leverage native device features. Benefit: Native, packaged apps perform better than their web-only equivalent and can quickly and easily interact with native device features using relatively low cost development skills. SAP Fiori mobile provides these benefits regardless of where your SAP Fiori content is installed - on-premise or in the cloud.

App deployment service enables app lifecycle management and an enterprise app store app for deployment, ratings and reviews. Benefit: When building native apps, app deployment is always a concern. How do I get this app to my end users? SAP Fiori mobile includes Mobile Place, an enterprise app store, simplifying the app management process. The app store is fully integrated with the build service eliminating the need to transport your app manually from the place where it was created to the place where it will be deployed. This increases transparency and minimizes errors.

Data service provides infrastructure support for advanced mobile features such as push and offline. Benefit: Advanced features such as push and offline require two things - first they need the app to be built to support it, second they need to have infrastructure to support things like data synchronization and device specific push technology. SAP Fiori mobile provides the infrastructure necessary to support these features. Apps must still be built to support them, this can be done through Web IDE and is also included on some SAP Fiori apps out of the box.

Development plugin for Web IDE provides access to Cloud Build Service, Plugin management and the Developer Companion. Benefit: SAP Fiori mobile enables developers to focus on solving business problems and uses cloud technology to provide build, packaging and testing services. This enables the developer to be more productive.

Each feature in the SAP Fiori mobile roadmap is aligned around the persona that is deriving the benefit. SAP Fiori mobile general drives benefits for 3 personas:

Administrator/Operations - In general this persona is responsible for operating the landscape. He is also capable (using a simple, easy to use UX) to package SAP Fiori apps/the FLP for distribution. In this persona he can't WRITE SAP Fiori apps themselves, but he controls what the SAP Fiori app connects to and monitors runtime execution. Ultimately, the goal is to have as many SAP Fiori apps with mobile qualities out of the box as possible, so that unlocking mobile value will be a simple configuration exercise and not require a developer.

Developer - The developer is responsible for creating new SAP Fiori apps and extending existing SAP Fiori apps. The ultimate vision for the developer persona is to making Web IDE a tool for amplifying and/or creating mobile qualities in SAP Fiori apps, to simplify app testing, and to integrate his output seamlessly into an app store that the administrator can then manage.

End User - ultimately, everything is for the end user. He's the person that is executing the SAP Fiori apps, experiencing the benefits of mobility and solving business problems.

Recent Innovations

Operations and Administration

Deepened SAP Fiori Cloud integration support - Application list in drag and drop interface will be accurate (it is currently hard coded), will show friendly name in addition to the SAP Fiori client ID (along with a sort by friendly name) and will reflect not only standard SAP Fiori apps but also those that meet system requirements for mobility.

Usage Information Capture - allows admin to gather basic information on how an app is used, how long it's run, what screens, etc. Admin is responsible for taking the data and creating their own visualizations in this phase.

SAP Fiori client customization support - There are several items in the SAP Fiori client that can be branded, but that require modification to the source code in order to implement. In SAP Fiori mobile, customers only get access to source code for their app not the container (SAP Fiori client) itself. With this feature we will make more customizations available to the administrator over time. In the first release we will support the ability for the administrator to remove the app password (in addition to customizing it), and we will also allow the admin to disable the app privacy plugin. The app privacy plugin affects how the app shows up when "task switching" (switching from one app to another) and when using screen sharing software. If the plugin is included screen sharing is disabled and task switching hides app details. If removed, those features are permitted. This is is very useful particularly when creating enablement material (like screen recordings using screen cast software)

Development

Developer Companion - A complete app rebuild (including hybrid components) can take anywhere from 7-10 minutes to complete. When a developer is making a web app change, he should be able to see this on a mobile device faster than that! With the Developer Companion this is possible. Developers can create a Developer Companion hybrid app which is functionally the same as the productive SAP Fiori mobile app they are creating, with the exception that the companion app has a "Refresh" button. If an HTML change is made, rather than doing a complete rebuild the developer can simply tap the "Refresh" button and the new content will be downloaded.

Simplified Push testing - With minimal coding, developers can add and test basic push support using the Developer Companion (or production app). GCM Sender and/or Push Certificate information can be included in the build process, and a push message UX has been added to the SAP Fiori Mobile DevX to allow a developer to test the end to end push process from Cloud to device.

Web IDE Device Test Cloud integration - integration of Cloud device testing providers into Web IDE will allow for more robust developer testing. This is currently supported for Perfecto Mobile only.

End User

CDN support for App Distribution - This feature enables SAP Fiori applications to be deployed via a content distribution network (CDN). When this feature is enabled, apps will be deployed from a location much closer to where the user is physically located if it were deployed from one of the SAP Cloud Platform data centers. Anecdotal evidence shows anywhere from 60-80% reduction in download speed.

Custom Themes - SAP Fiori packaged apps now support the delivery of a custom theme along with the rest of the UX that is embedded into the app.

Additional Resources

SAP Cloud Platform, mobile service for SAP Fiori GTM site

- https://jam4.sapjam.com/groups/about_page/1XdKZ5JkvrL75SHNWkga3y

Product Video

- https://www.youtube.com/watch?v=TrVAZoMKly8

LESSON SUMMARY
You should now be able to:

Explain mobile services

Unit 2
Lesson 8

Using Runtime and Containers

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Use runtime and containers

Runtime and Containers

SAP Cloud Platform, Runtimes and Containers

SAP Cloud Platform supports different runtimes/programming models (Java, XSJS, HTML) and offers standards-based development.

The following are basics about Runtimes and Containers of SAP Cloud Platform:

Java, XSJS and HTML5 programming models

- Application runtime containers for Java development
- Development of "native" SAP HANA applications by means of SAP HANA Extended Application Services (XSJS)
- HTML5 infrastructure to build and run HTML5 applications in the cloud
- Cloud Foundry-based runtimes with BYOL option

(CF) SAP HANA extended application services, advanced model

Support SAP HANA XS Advanced that will enable developers to deploy, run and scale applications developed for SAP HANA XS Advanced on-premise also on SAP Cloud Platform.

SAP Cloud Platform Virtual Machines

- Install software side by side with the runtimes and services of SAP Cloud Platform
- Full control over virtualized hardware resources (compute power, memory, storage)
- Full operating system (Linux) access (connection via SSH)

(CF) buildpacks

- enables the usage of additional programming languages such as Node.js, python etc., more runtimes and frameworks with SAP Enterprise Support for Java

Runtimes and Containers, Runtimes/Programming Models Overview

Java, HTML5 and XSJS

Java

Develop, deploy and use Java applications in a cloud environment

Java SE 6/7 and Java EE 6 Web Profile certified

E.g. complex integration projects

HTML5

Develop and run lightweight HTML5/SAPUI5 applications in a cloud environment

Benefit from SAPUI5 code templates and SAP best practices to rapidly build applications

E.g. mash-up connections to existing data sources or backend systems

XSJS

Develop and scale XSJS applications in a cloud environment

Reduce the footprint of the solution by not having a separate application server in the solution

E.g. data-centric analytical apps and dashboards

> Note:
> Application stereotypes are for example purpose and not exclusive:
>
> Support of the various programming models
>
> JAVA XSJS HTML5

Java Runtime

The Java Runtime enables developers to leverage existing knowledge and use that to bring applications on SAP HANA and the cloud.

Key capabilities and Benefits

Key capabilities

Develop, deploy and run Java applications in the cloud

Applications run on a runtime container where they can use the platform services APIs and Java APIs according to standard patterns

Benefits

Standards-based Java development

Java 6/7/8 supported

Supports the wide-spread Apache Tomcat Web container

Support for Spring framework and family

Integrated with the SAP Cloud Platform, debugging service and the SAP Cloud Platform, profiling service

XSJS Runtime

The XSJS Runtime enables developers to build native applications on SAP HANA in the cloud. The runtime uniquely leverages the application performance capabilities of SAP HANA.

Key capabilities and Benefits

Key capabilities

Develop and scale XSJS applications in a cloud environment

Easily put modern SAPUI5 Fiori frontends and XSOData services right on top of your database content

Use of Eclipse-based tools (SAP HANA Studio) for administration, development and connecting to your SAP HANA instance on SAP Cloud Platform

Alternatively use of the SAP HANA Cockpit and the SAP HANA Web-based Development Workbench for zero footprint administration and development

Benefits

Native SAP HANA development

Reduce the footprint of the solution by not having a separate application server in the solution

HTML5 Programming Model

The HTML5 programming model enables developers to leverage existing knowledge and use that to build modern applications on SAP HANA in the cloud.

Key capabilities and Benefits

Key capabilities

Develop and run lightweight HTML5 applications in a cloud environment

Leverage the SAPUI5 JavaScript library

Connect to OData services residing on-premise or in the cloud

Benefits

Standards-based UI development

Develop once, deploy everywhere: Run on any device - mobile, tablet, and desktop

Benefit from SAPUI5 code templates and SAP best practices to rapidly build applications

SAP Cloud Platform Virtual Machine

SAP Cloud Platform will support virtual machines that will give full control over virtualized hardware resources and thus are blurring the line between PaaS and IaaS on SAP Cloud Platform.

Key capabilities and Benefits

Key capabilities

Install software side by side with the runtimes and services of SAP Cloud Platform

Full control over virtualized hardware resources (compute power, memory, storage)

Full operating system (Linux) access (connection via SSH)

Benefits

Can be exposed to the internet (via DMZ)

Can be accessed by Java applications and from SAP HANA native applications running on SAP Cloud Platform

(CF) SAP HANA Extended Application Services, Advanced Model

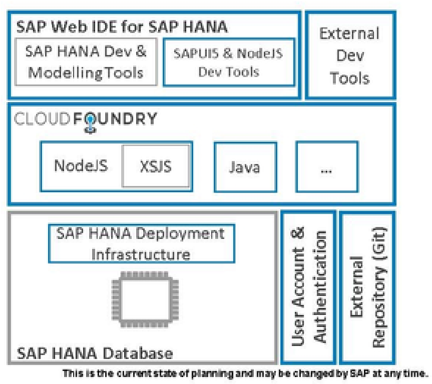


Figure 60: (CF) SAP HANA Extended Application Services, Advanced Model

SAP Cloud Platform will support SAP HANA XS Advanced that will enable developers to deploy, run and scale applications developed for SAP HANA XS Advanced on-premise also on SAP Cloud Platform.

Key capabilities and Benefits

Key capabilities

Support polyglot runtime containers of Cloud Foundry for SAP HANA application development (e.g. NodeJS, Java)

New SAP HANA Deployment Infrastructure (HDI) to enable the deployment of isolated native SAP HANA content multiple times

New User Account Authentication (UAA) service to handle user management and support external SAML2 and OAuth compliant IdPs (such as SAP SSO)

Run XSJS code in NodeJS

Benefits

Applications developed for SAP HANA XS Advanced run on-premise and in the cloud

Natively integrates established standard development tools and processes (e.g. Git, Eclipse)

Runtimes and Containers, (CF) Buildpacks

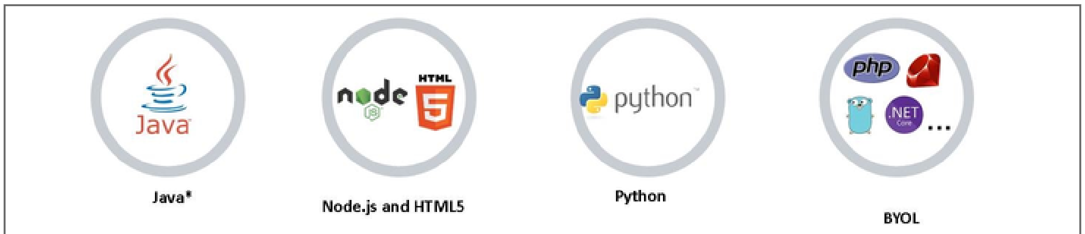


Figure 61: Runtimes and Containers, (CF) Buildpacks

With Cloud Foundry environment, SAP Cloud Platform enables the usage of additional programming languages, runtimes and frameworks with SAP Enterprise Support for Java.

*Java and the Java logo are registered trademarks of Oracle and/or its affiliates.

The following are properties of the mentioned objects above:

Java

Develop, deploy and operate Java applications,

E.g. complex integration projects.

Node.js and HTML5

Develop and run node.jsand HTML5 applications.

Benefit from multiple Node.jspackages available and SAPUI5 code templates.

Python

Develop and run python applications in the cloud.

Use multiple libraries available for data manipulation, machine learning and analysis.

BYOL

Bring your own language and runtime of choice.

Choose from existing buildpacks or develop your own.

The following are benefits of using the Cloud foundry environment:

Develop and run lightweight HTML5/SAPUI5 applications in a cloud environment.

Benefit from SAPUI5 code templates and SAP best practices to rapidly build applications,

E.g. mash-up connections to existing data sources or backend systems.

LESSON SUMMARY
You should now be able to:

- Use runtime and containers

Unit 2

Learning Assessment

1. Which of the following is an infrastructure and database service offered on SAP Cloud Platform?

   Choose the correct answer.

   - [ ] **A** PostgreSQL
   - [ ] **B** BusinessObjects Cloud
   - [ ] **C** Smart Business Service
   - [ ] **D** Data Quality Management
   - [ ] **E** Localizatiion Hub, Tax Service

2. Which of the following is not a supported size for SAP Cloud Platform SAP ASE Service?

   Choose the correct answer.

   - [ ] **A** 32 GB
   - [ ] **B** 80 GB
   - [ ] **C** 160 GB
   - [ ] **D** 256 GB
   - [ ] **E** 640 GB

3. The SAP Cloud Platform Cockpit is the consolidated destination for all operational needs from configuration to deployment and monitoring on SAP Cloud Platform.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

4. Which of the following is not a runtime \ container in the NEO environment on SAP Cloud Platform?

   Choose the correct answer.

   - [ ] **A** HTML5
   - [ ] **B** HANA XS
   - [ ] **C** JAVA
   - [ ] **D** C++

5. A browser based IDE for development of SAP HANA objects is the SAP HANA Web-based Development Workbench.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

Learning Assessment - Answers

1. Which of the following is an infrastructure and database service offered on SAP Cloud Platform?

   Choose the correct answer.

   - [x] **A** PostgreSQL
   - [ ] **B** BusinessObjects Cloud
   - [ ] **C** Smart Business Service
   - [ ] **D** Data Quality Management
   - [ ] **E** Localizatiion Hub, Tax Service

   This is correct. See Unit 2, Lesson 1 of CLD100, Col 18.

2. Which of the following is not a supported size for SAP Cloud Platform SAP ASE Service?

   Choose the correct answer.

   - [ ] **A** 32 GB
   - [ ] **B** 80 GB
   - [ ] **C** 160 GB
   - [x] **D** 256 GB
   - [ ] **E** 640 GB

   This is correct. See Unit 2, Lesson 2 of CLD100, Col 18.

3. The SAP Cloud Platform Cockpit is the consolidated destination for all operational needs from configuration to deployment and monitoring on SAP Cloud Platform.

   Determine whether this statement is true or false.

   - [x] True
   - [ ] False

   This is correct. See Unit 2, Lesson 2 of CLD100, Col 18.

4. Which of the following is not a runtime \ container in the NEO environment on SAP Cloud Platform?

   Choose the correct answer.

   - [ ] **A** HTML5
   - [ ] **B** HANA XS
   - [ ] **C** JAVA
   - [x] **D** C++

   This is correct. See Unit 2, Lesson 3 of CLD100, Col 18.

5. A browser based IDE for development of SAP HANA objects is the SAP HANA Web-based Development Workbench.

   Determine whether this statement is true or false.

   - [x] True
   - [ ] False

   This is correct. See Unit 2, Lesson 3 of CLD100, Col 18.

UNIT 3

Application Development Services in PaaS

UNIT OBJECTIVES

- Introduce SaaS extensions
- Explain analytics in PaaS
- Explain business services in PaaS
- Explain collaboration in PaaS

Explain integration services

Explain multi cloud

Explain security in PaaS

Explain user experience

Unit 3
Lesson 1

Introducing SaaS Extensions on PaaS

LESSON OBJECTIVES
After completing this lesson, you will be able to:

- Introduce SaaS extensions

SaaS Extensions

These are SaaS extensions on the SAP Cloud Platform:

SAP SuccessFactors

- Automatic provisioning of SAP Cloud Platform extension accounts for SAP SuccessFactors customers and partners
- Solution deployment, subscription, & monitoring

SAP Ariba

Leverage existing SAP Ariba Web Service (cXML) APIs to access data and interact with SAP Ariba workflows

SAP S/4HANA

Using SAP Web IDE to develop key user screens for SAP Cloud.

SAP Hybris

- OData API integration with ID propagation
- Leveraging SAP Cloud Identity for SSO and User provisioning
- UI mashups

Concur

- REST APIs access with ID propagation from extension apps
- Available showcases for both Travel and Expense module extensions

LESSON SUMMARY
You should now be able to:

- Introduce SaaS extensions

Unit 3
Lesson 2

Explaining Analytics in PaaS

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain analytics in PaaS

Analytics in PaaS

SAP Cloud Platform, Analytics

With SAP Cloud Platform embed advanced analytics into your solutions, allowing you to identify, combine, and manage multiple sources of data and build advanced analytics models within your business applications:

SAP BusinessObjects Cloud:

- Software-as-a-Service solution built on SAP Cloud Platform.
- Prepare and model cloud and on-premise data.
- Design, visualize, and create your stories.
- Create plans with forecasts and allocations.

SAP HANA advanced analytics: Spatial/GIS

- Allow for visualization, interaction, and exploration of spatial data in SAP HANA via maps.
- Uncover patterns user Spatial clustering algo: Grid, K-Means and DBScan.

SAP HANA advanced analytics: Predictive Analytics

- Native In-database predictive algorithms for fast results.
- Quicker implementations and ability to embed predictive into applications, solutions, and BI environments to deliver business impact.

SAP Smart Business service

- Framework to visualize analytic content in the form of charts and tiles.
- Visualize KPIs and OPIs as SAP Fiori applications.
- Supports Embedded analytics and Predictive analytics.

SAP HANA for Advanced Analytics: Search, Text Analysis, & Mining

Sophisticated text processing capabilities for unstructured content (e.g. sentiment analysis).

Native full-text and fuzzy search.

Graphical modeling of search models.

SAP HANA advanced analytics: Series Data

Comparison of data across multiple series tables.

Support very high volumes of data using effective compression techniques.

SAP Cloud Platform Predictive service(s)

allows any application on SAP Cloud Platform to easily embed automated predictive capabilities without needing skills of a data scientist.

SAP Cloud Platform Streaming Analytics

Analyzes and transforms high-velocity, high-volume event streams in real time.

Generates alerts or notify applications to respond.

Can be used with SAP Cloud Platform IoT Services to analyze incoming data in real-time.

Analytics, SAP BusinessObjects Cloud

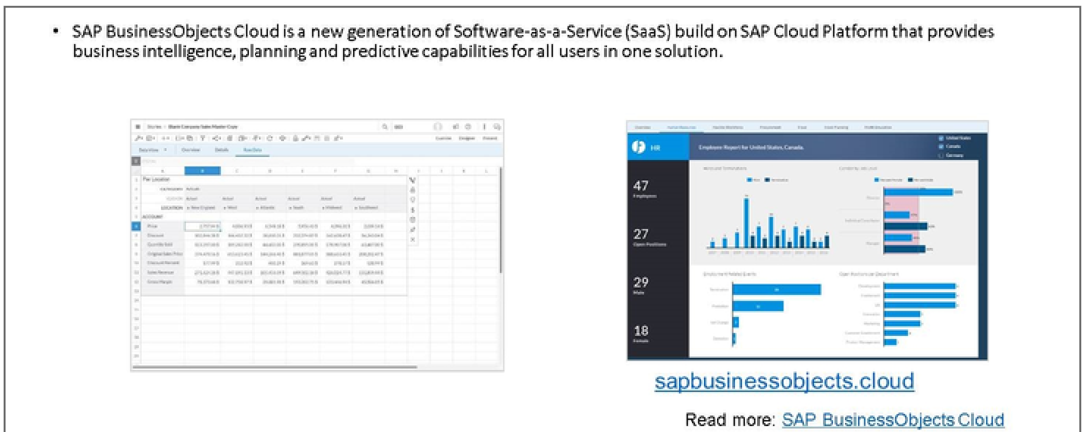


Figure 62: Analytics, SAP BusinessObjects Cloud

Key capabilities and Benefits:

Key capabilities

Prepare and model cloud and on-premise data.

Design, visualize, and create your stories.

Create plans with forecasts and allocations.

Collaborate with your team and receive notifications.

SAP Digital Boardroom for a consolidated view of the organization’s performance.

Live connection to SAP BW 7.5 (SP6+ notes applied).

Acquire data from SAP Fieldglass.

Acquire data from Progress OpenEdge JDBC database.

Save a Story as a Template.

Stories can now be saved and shared as a template for new stories. You can select specific pages to be included, and choose to replace images with empty placeholders or keep them as part of the template.

Benefits

Cloud based - No desktop required and scale without worry!

Save time by not having to jump between products to do your job.

Reduce errors by not having to integrate multiple point solutions.

Serve the needs of all and break down silos between functions.

Analytics, SAP HANA advanced analytics: Spatial/GIS

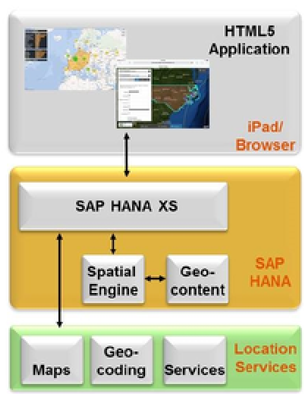


Figure 63: Analytics, SAP HANA advanced analytics: Spatial/GIS

The SAP HANA spatial/GIS options provide functions to analyze and process geospatial information in SAP HANA with high performance.

Key capabilities and Benefits:

Key capabilities

Store, process, manipulate, share and retrieve spatial data directly in the database.

50+ geospatial functions and algorithms.

Allow for visualization, interaction, and exploration of spatial data.

Performance enhancements with spatial partitioning.

SQLScript support for easy access and manipulation spatial data.

Spatial clustering algorithms: Grid, K-Means and DBScan.

Open standards (OGC, 1999 SQL/MM).

SDK for custom geospatial algorithms.

"50+ geospatial functions and algorithms".

Open standards and SDK.

Benefits

High performing platform for both business (operational, finance, marketing, sales, engineering) and spatial data.

Re-imagine digital intelligence.

Connect your enterprise to the digital economy.

Analytics, SAP HANA advanced analytics: Predictive Analytics

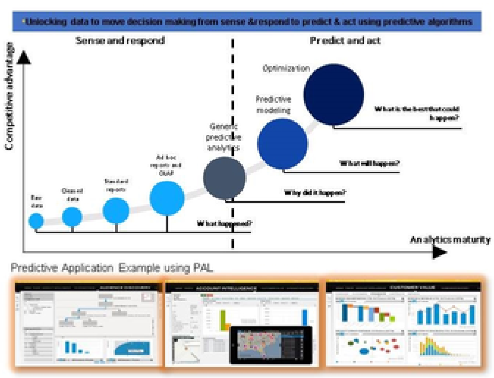


Figure 64: Analytics, SAP HANA advanced analytics: Predictive Analytics

The SAP HANA predictive analysis library (PAL) is a built-in and native SAP HANA library to perform in-database data mining, machine learning, and statistical calculations within SQLScript procedures.

Key capabilities and Benefits:

Key capabilities

Support over 90+ algorithms in the area of:

- Association (Apriori, FP-Growth, KORD – Top K Rule Discovery).
- Classification (C4.5/CART/CHAID Tree, Logistic Reg. Elastic Net, SVM, Random Forest, Naïve Bayes, K-NN, Neural Network, etc.).
- Clustering (K-Means, DBSCAN, Self Organized Maps, Affinity Propagation, Latent Dirichlet Allocation, Gaussian Mixture Model, Agglomerate Hierarchical, etc.).
- Statistical Functions (Principal Component Analysis, Covariance Matrix, etc).
- Regression (Liner/Polynomial/Exponential/Geometric/Logarithmic regression).

- Link Prediction (Common Neighbors, Jaccard's Coefficient, etc).
- Time series (Exponential Smoothing, ARIMA, etc.).

Integration with SAP Predictive Analytics.

Graphical Machine Learning pipeline modelling support.

Benefits

Native In-database predictive algorithms for fast results.

Quicker implementations and ability to embed predictive into applications, solutions, and BI environments to deliver business impact.

SAP HANA for Advanced Analytics: Search, Text Analysis and Mining

Figure 65: Search, Text Analysis and Mining

Key capabilities and Benefits:

Key capabilities

Native full-text and fuzzy search.

Graphical modeling of search models.

Search API to facilitate the development of search UIs.

In-database text analysis and mining.

Support for 33 languages.

XSA integration.

Benefits

Less data duplication and movement – leverage one infrastructure for analytical and search workloads.

Extract salient information from unstructured textual data.

Easy modeling & customization tools – Web IDE for SAP HANA.

Analytics, SAP HANA Advanced Analytics: Series Data

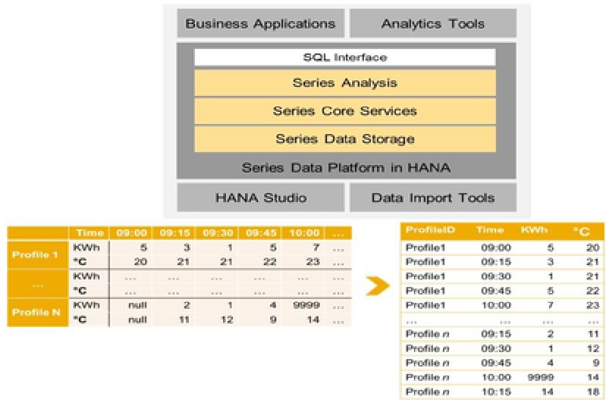


Figure 66: Series Data

SAP HANA provides series data capability that allows the measuring of data over time where time is commonly equidistant; it allows to detect and forecast trends in data.

Key capabilities and Benefits:

Key capabilities

Comparison of data across multiple series tables.

Support very high volumes of data using effective compression techniques.

Greater PAL integration.

Improved performance of series functions.

Benefits

Strong foundation for building industry based use cases for system of innovations.

Enable analysis of real-time operation intelligence from various device.

Analytics, SAP Cloud Platform Predictive service(s)

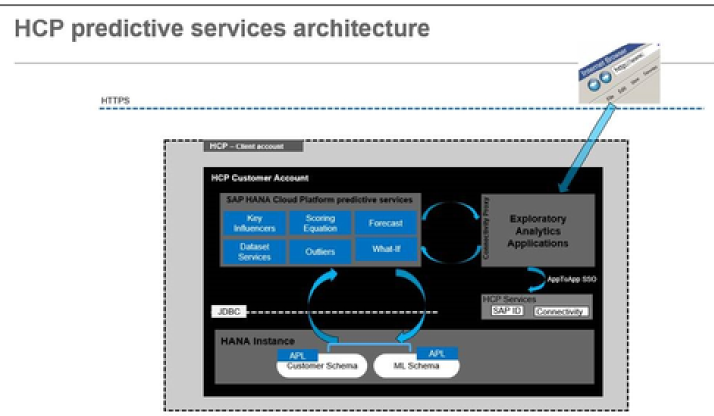


Figure 67: SAP Cloud Platform Predictive service(s)

SAP Cloud Platform Predictive service is a service that allows any application on SAP Cloud Platform to easily embed automated predictive capabilities without needing skills of a data scientist.

Key capabilities and Benefits:

Key capabilities

Strong set of predictive REST services.

No data mining skills required to understand:

- The business utility of these services.
- How to use them.
- The results they provide.

Each service returns understandable insights.

Provide synchronous or asynchronous model execution.

Reduced development cycle, data analysis possible with a single service call, you don't have to code it yourself from scratch.

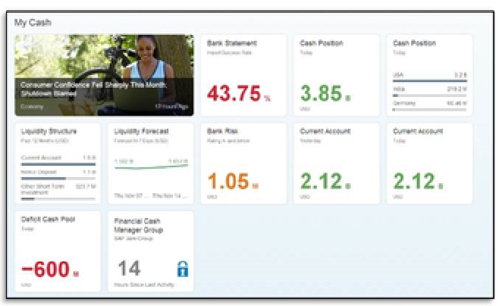
Benefits

Easy and quick integration of powerful data mining features into new or existing SAP Cloud Platform applications.

Use any supported programming language to develop your application in the cloud, for example Java, HTML5, or JavaScript for X.

Flexible – can be used for custom app development or OEM environments.

Analytics, SAP Smart Business service



This is the current state of planning and may be changed by SAP at any time.

Figure 68: SAP Smart Business service

Smart Business enables Rapid publishing of SAP Fiori Analytic applications into role specific SAP Fiori Launchpads by key users. For normal users, Smart Business supports insight to action by providing Quick overview of business performance and enabling interactive analysis.

Key capabilities and Benefits:

Key capabilities

SAP Smart Business is a framework for visualizing analytic content in the form of charts and tiles.

KPIs and OPIs can be visualized as SAP Fiori applications without writing any code.

Supports Automated analysis on behalf of business user in background jobs and Alerting whenever there is a breach of KPI thresholds.

Provides advance features like Embedded Analytics and Predictive Analysis.

Benefits

Benefit from Consumer Grade SAP Fiori Analytics powered by Smart Business in SAP Cloud Platform. Publish Analytics in Codeless environment significantly reducing your Total Cost of Development.

Helps design Charting visualizations in WYSIWYG modeler.

SAP Fiori Elements (Object Page, Analytic List Pages, Overview Pages, …) simplifies UI development dramatically.

Out-of-the-box trust configuration with the SAP Cloud Platform account.

Analytics, SAP Cloud Platform Streaming Analytics

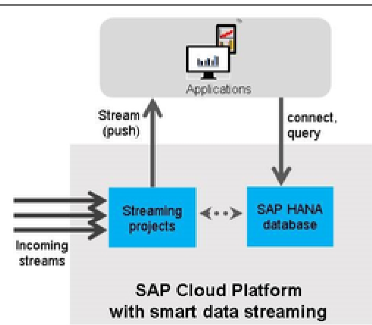


Figure 69: SAP Cloud Platform Streaming Analytics

The SAP Cloud Platform Streaming Analytics analyzes and transforms high-velocity, high-volume event streams in real time, allowing applications to detect and respond to changing conditions in real-time, and with the ability to compute and capture useful information in the SAP HANA database.

Key capabilities and Benefits:

Key capabilities

Monitor and analyze incoming data to watch for trends, correlation, patterns, missing data.

Generate alerts or notify applications to respond.

Filter, sample, aggregate and correlate incoming data.

Combine raw events into actionable information.

Capture only the data you want, in the form you need it.

Stream live updates to operational dashboards.

Can be used with SAP Cloud Platform IoT Services to analyze incoming data in real-time.

Benefits

Respond immediately to new information.

Optimize the data being stored in SAP HANA, storing only useful information in optimized data models.

LESSON SUMMARY
You should now be able to:

Explain analytics in PaaS

Unit 3
Lesson 3

Explaining Business Services in PaaS

LESSON OBJECTIVES

After completing this lesson, you will be able to:

Explain business services in PaaS

SAP Cloud Platform, Business Services in PaaS

SAP Business Services fuel the fast development of Business apps and services for the cloud, and power an open marketplace for new business apps.

SAP Hybris

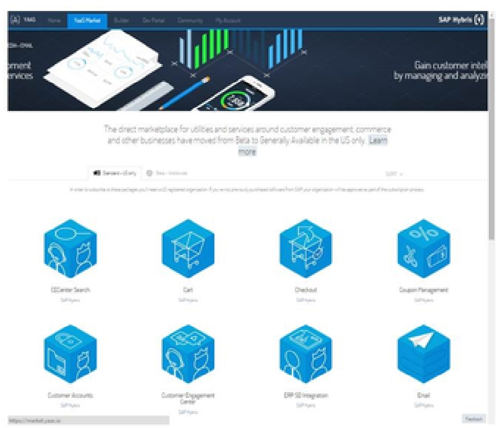


Figure 70: Business Services, SAP Hybris as a Service on SAP Cloud Platform

SAP Hybris as a Service (YaaS) on SAP Cloud Platform allows developers to create, offer and sell business service packages based on a micro-service centric programming model and customers to leverage these services for extending existing and new front-office solutions faster than ever.

Key capabilities and Benefits

Key capabilities

Offers SAP Hybris as a Service packages:

- Core Service packages.
- SAP Hybris Commerce as a Service packages.

- SAP Hybris Marketing Loyalty packages.
- SAP Hybris Service Customer Engagement Center packages.

Allows partners to offer such packages as well.

Augmenting front-office solutions with new functionality and deliver new LoB centric apps.

Benefits

Lower implementation costs and broaden development community by building on existing service packages.

Contribute to the ecosystem and offer your own service packages through the YaaS market.

SAP Hybris as a Service on SAP Cloud Platform:

Platform allows developers to create, offer and sell business service packages based on a micro-service centric programming model.

Allows partners to offer such packages as well.

Augmenting front-office solutions with new functionality and deliver new LoB centric apps.

SAP Data Quality Management, microservices for location data (beta)

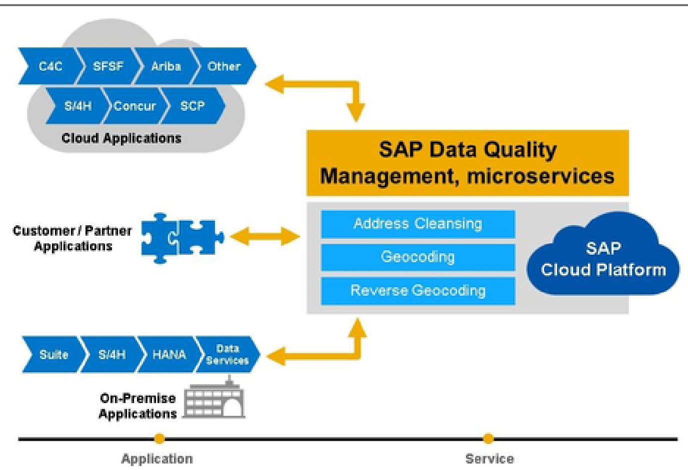


Figure 71: Business Services, SAP Data Quality Management, microservices for location data

Key Capabilities and Benefits

Key Capabilities

Address Cleansing– parsing, standardization, correction, validation, enhancement globally.

Geocoding– Address to LAT/LONG.

Reverse Geocoding– LAT/LONG to Address(es).

Benefits

Simple consumption model in the Cloud Integration into SAP applications Available Today - SAP ERP, CRM, Master Data Governance, SAP S/4HANA onPremise, Data Services, SAP Hybris Cloud for Customer (1705) SuccessFactors (1705 Beta) Consumption-based pricing model Leading solution, 30+ years experience in address cleansing and geocoding Strong partnerships with many worldwide postal authorities and data providers.

Business Services, SAP Localization Hub, tax service

The SAP Localization Hub, tax service provides one central framework to automate the determination and calculation of indirect taxes world-wide. The service can be accessed on the SAP Cloud Platform and is designed as RESTful API.

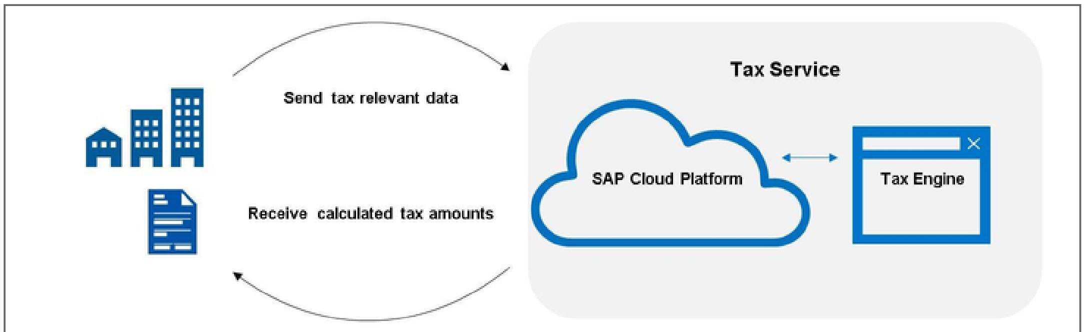


Figure 72: Business Services, SAP Localization Hub, tax service

Key Capabilities and Benefits

Key Capabilities

Global Coverage: Pre-delivered indirect tax determination and computation content for 120 countries.

Compliance: With it's global network, SAP keeps this service up to date.

Central Framework: The tax service can be used as central tax computation framework across the customers' system landscape.

Benefits

Cost Savings by reduced maintenance efforts.

Single source to determine and compute tax.

Automation and scalability of tax calculation.

High quality and consistent tax computation results.

SAP Localization Hub, tax service

Global Coverage: Pre-delivered indirect tax determination and computation content for 120 countries.

Cloud Offering: Provide consumers with access to tax computation across all products using SAP Cloud Platform.

Central Framework: The tax service can be used as central tax computation framework across the customer landscape.

Further details:

The SAP Localization Hub, tax service provides one central framework to simplify the adoption of global tax regulations. The tax service includes a computation content with pre-delivered and inbuilt rules for 120 countries to support various indirect taxes. For most of the supported countries the tax content supports tax types, tax rates, deductibility, as well as exemption.

Tax updates and respective legal changes are pushed automatically into the tax calculation micro service to enable legal compliance.

The service can be accessed on the SAP Cloud Platform and is designed as RESTful API / Webservice. The interaction with the services happens in a predefined, structured format and you will receive endpoints to connect (including authorization) and to use the services.

With the global coverage of the predelivered and perpetually updated content you are able to reduce the country specific development efforts in your organization. The maintenance cost will also decrease as legal changes are pushed automatically into your system. The legal change updates implies cost savings and in addition it enables you to keep you tax calculation legal compliant.

You achieve cost transparency in your business transaction and can adapt your international business processes accordingly.

Tax determination and calculation in one central framework helps you to simplify the taxation process in your organization. Manual errors are reduces and consistent results are ensured across the complete system landscape.

Deploying the SAP Localization Hub, tax service as a microservices creates flexibility and scalability to react on new business opportunities. Business software becomes a network of applications on an open platform which supports agile business approach and gains speed to market.

Business Services, SAP Leonardo Machine Learning

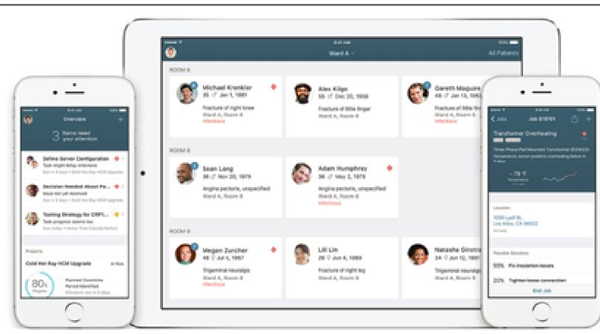


Figure 73: Business Services, SAP Leonardo Machine Learning

SAP Leonardo Machine learning portfolio, integrates learning from data to help businesses grow, optimize processes, assist employees, and make customers happy. A part of this portfolio are “Business Services” which enable the intelligent enterprise by providing machine learning services targeted for your lines of business. Integrate these Business Services with your extension applications via stand-alone APIs or direct integration with SAP’s line of business solution portfolio.

Main Properties:

SAP Machine Learning for Talent Acquisition.

Automate the screening process for recruiters by using machine learning to identify best-fit candidates with **SAP Machine Learning for Resume Matching**

Provide candidates with job recommendations based on their skills and qualifications with **SAP Machine Learning for Job Matching**

Enable recruiters to classify job descriptions against national occupational definition databases (e.g. O*NET) with **SAP Machine Learning for Job Standardization**

Integrate each of these Business Services with extension applications for SAP SuccessFactors Recruiting and SAP Fieldglass via the SAP API Business Hub.

SAP Machine Learning for Customer Experience.

Automatically process customer enquiries and scale digital interactions with **SAP Machine Learning for Service Ticket Intelligence**

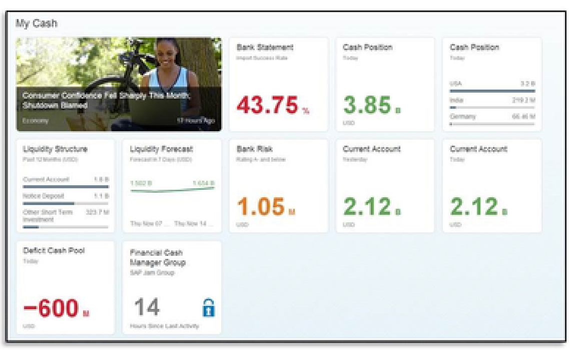
Integrate each of these Business Services with extension applications for SAP Hybris Cloud for Customer via the SAP API Business Hub.

SAP Leonardo Machine Learning

Integrates learning from data to help businesses grow, optimize processes, assist employees

Provides machine learning services targeted for your lines of business

Business Services, SAP Smart Business service



This is the current state of planning and may be changed by SAP at any time.

Figure 74: Business Services, SAP Smart Business service

Smart Business enables Rapid publishing of SAP Fiori Analytic applications into role specific SAP Fiori Launchpads by key users. For normal users, Smart Business supports insight to action by providing Quick overview of business performance and enabling interactive analysis.

Key Capabilities and Benefits

Key Capabilities

SAP Smart Business is a framework for visualizing analytic content in the form of charts and tiles.

KPIs and OPIs can be visualized as SAP Fiori applications without writing any code.

Supports Automated analysis on behalf of business user in background jobs and Alerting whenever there is a breach of KPI thresholds.

Provides advance features like Embedded Analytics and Predictive Analysis.

Benefits

Benefit from Consumer Grade SAP Fiori Analytics powered by Smart Business in SAP Cloud Platform.

Publish Analytics in Codeless environment significantly reducing your Total Cost of Development.

Helps design Charting visualizations in WYSIWYG modeler.

SAP Fiori Elements (Object Page, Analytic List Pages, Overview Pages, …) simplifies UI development dramatically.

Out-of-the-box trust configuration with the SAP Cloud Platform account.

SAP Smart Business service:

SAP Smart Business is a framework for visualizing analytic content in the form of charts and tiles.

KPIs and OPIs can be visualized as SAP Fiori applications without writing any code.

Supports Automated analysis on behalf of business user in background jobs and Alerting whenever there is a breach of KPI thresholds.

LESSON SUMMARY

You should now be able to:

Explain business services in PaaS

Unit 3
Lesson 4

Explaining Collaboration in PaaS

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain collaboration in PaaS

Collaboration

With SAP Cloud Platform you can bring people together with secure access to shared business content, information, applications and processes - to drive results and increase team productivity.

These are the provided services:

SAP Jam

- Build collaborative application experiences.
- Render SAP Jam Collaboration feeds and forums directly in your extensions or SAP / 3rd party apps.
- Embed apps built on SAP Cloud Platform in your SAP Jam work patterns and intranet solution via OpenSocial.

SAP Cloud Platform Gamification

- Implementation of complex game mechanics with web-based gamification workbench.
- Improved usability of gamification workbench (new design, entity update, mass change, etc.).
- Easy and secure integration of gamification via Web API.
- Basic analytics on runtime gamification data.

SAP Document Center

- Securely share content with business partners.
- Enable & mobilize existing on-premise business content (SAP Business Suite or 3rd party such as Microsoft SharePoint).
- Leverage content in SAP Jam, SAP Fiori launchpad, or SAP S/4HANA and SAP Business Suite.

Collaboration, SAP Jam

Figure 75: SAP Jam

SAP Jam Collaboration on the SAP Cloud Platform enables developers to build collaborative application experiences.

Key capabilities and Benefits

Key capabilities

Integration with SAP Web IDE for easy integrations of collaborative elements like feeds, forums and groups into Cloud Platform applications.

Expose data contextually from SAP or 3rd party apps via OData.

Embed apps built on SAP Cloud Platform in your SAP Jam work patterns and intranet solution via OpenSocial.

Benefits

Fast time to value with pre-built work patterns and easy customization.

Ensure all participants have the latest relevant business updates.

Flexibility to modify collaborative processes to meet your unique needs with LoB or industry specific work patterns.

Collaboration, SAP Cloud Platform Gamification

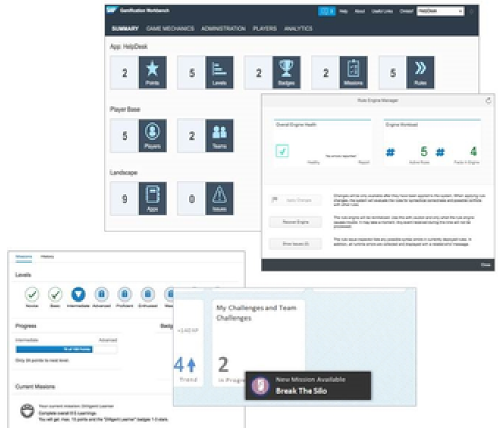


Figure 76: SAP Cloud Platform Gamification

SAP Cloud Platform Gamification can be used to engage and motivate people with rewarding experiences.

Key capabilities and Benefits

Key capabilities

Implementation of complex game mechanics with web-based gamification workbench Improved usability of gamification workbench (new design, entity update, mass change, etc.) Easy and secure integration of gamification via Web API Basic analytics on runtime gamification data.

Benefits

Easy integration of advanced gamification into new or existing applications via a technical service Long term user engagement with real-time feedback on achievements Adoption of gamification design during runtime and continuously improvement via gamification analytics.

Collaboration, SAP Document Center

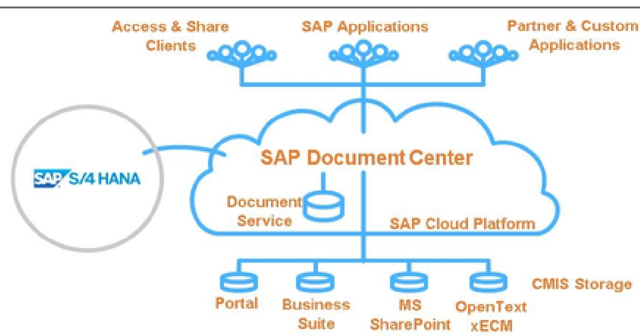


Figure 77: SAP Document Center

SAP Document Center is designed for enterprises where collaboration, security and control of business content is critical. Users have unified and easy access to all business documents including collaboration and sharing on any device. Running on SAP Cloud Platform, SAP Document Center is the standard extension for SAP S/4HANA and SAP Business Suite for file access with a strong focus on application integration and the cloud.

Key capabilities and Benefits

Key capabilities

- Ready to use
- File Share & Sync Solution
- Native clients
- Productivity features
- Tailored security for cloud and mobile usage

Benefits

- Integrate
- Harmonized API and SPI
- SDKs for multiple programming languages
- Ready to use SAP Fiori Library
- Business applications and content repositories

LESSON SUMMARY
You should now be able to:

- Explain collaboration in PaaS

Unit 3
Lesson 5

Explaining Integration Services

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain integration services

Integration Services

SAP Cloud Platform, Integration Services, Overview

SAP Cloud Platform Integration Services, improve business agility and prevent data and application silos by seamlessly and securely integrating your cloud applications into your business landscape.

It consist of the following parts:

SAP Cloud Platform Integration

- Pre-packaged integration content hub in the cloud - "Discover, Configure, Manage"
- Cloud-based technology with subscription-based usage
- Compatible with SAP Process Integration content (e.g. mappings)

SAP Cloud Platform Connectivity

Allows a secure integration with on-premise systems.

SAP HANA Smart Data Integration

Real-time replication, physical bulk/batch movement, and federation in a unified framework.

SAP Cloud Platform API Management

Simple, scalable and secure access to digital assets through application programming interfaces (APIs) and enables developer communities to consume these.

SAP Cloud Platform Business Rules (beta)

Encapsulates business logic from application logic, enabling business key users and application developers to change the decision logic without re-writing the application.

SAP API Business Hub

- Discover APIs of a range of different SAP solutions in a central catalog
- Access to API's in sandbox systems

SAP Cloud Platform Enterprise Messaging (beta)

Cloud-based messaging framework that enables you to connect applications, services, and devices across different technologies.

RabbitMQ on SAP Cloud Platform (beta)

Open source message broker

Robust messaging between applications

SAP Cloud Platform OData Provisioning

Expose business data and business logic from on-premises SAP systems as standards-based services

Re-uses and leverages SAP investment in on-premise SAP systems

SAP Cloud Platform Workflow

Orchestrates a sequence of tasks across different people and organizations.

SAP Cloud Platform Integration

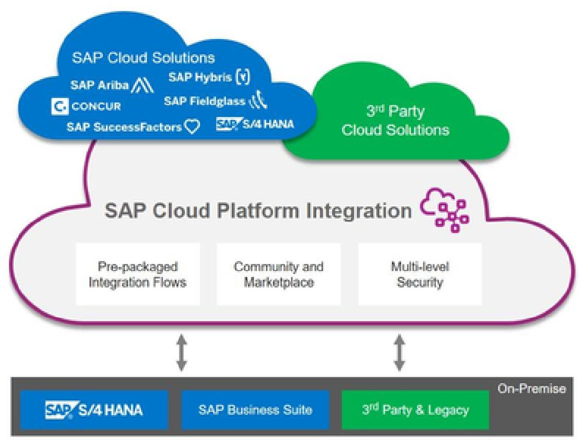


Figure 78: SAP Cloud Platform Integration

SAP Cloud Platform Integration seamlessly connects cloud applications with other cloud and on-premises apps, both from SAP and third-party providers.

Key Capabilities and Benefits

Key Capabilities

Additional pre-built integration flows:

- Concur Expense with SAP S/4HANA and SAP ERP.
- SAP Fieldglass with SAP S/4HANA.
- SAP Hybris Cloud for Customer with SAP SuccessFactors Employee Central and SAP Hybris Cloud for Customer for Utilities B2C.
- SAP Access Control with SAP SuccessFactors for user provisioning.
- SAP Health Engagement with digital health platform Validic.
- Open OData APIs for monitoring

Partner-supplied adapters at no additional cost.

ISO27001, SOC1, SOC2 certifications

Integration content lifecycle management incl. automated standard content update

Benefits

Reduced effort to integrate with popular applications.

SAP Cloud Platform: Connectivity

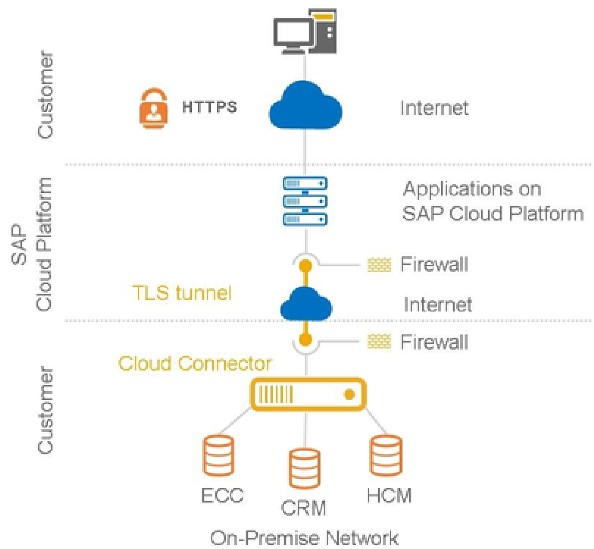


Figure 79: SAP Cloud Platform Connectivity

SAP Cloud Platform Connectivity using the Cloud Connector allows a secure integration with on-premise systems. This enables easy integration of on-premise data with SAP Cloud Platform.

Key Capabilities and Benefits

Key Capabilities

On-premise agent establishes secure TLS tunnel connection between the SAP Cloud Platform and on-premise systems

Supports pre-configured “Destination API” and certificate inspection to safeguard against forgeries

Supports multiple protocols and standardized APIs (HTTP, RFC, JDBC), high availability and principal propagation

On-premise connectivity for Cloud Foundry based services of the SAP Cloud Platform.

On-Premise Connectivity for SAP Cloud Platform.

Integration of the Cloud Connector in the SAP Solution Manager.

Disaster Recovery support.

RFC service channels for SAP Cloud Platform.

Benefits

Complementary to SAP Gateway, SAP Cloud Platform Integration and 3rd party integration suites both on-premise and in the cloud.

SAP Cloud Platform: Smart Data Integration

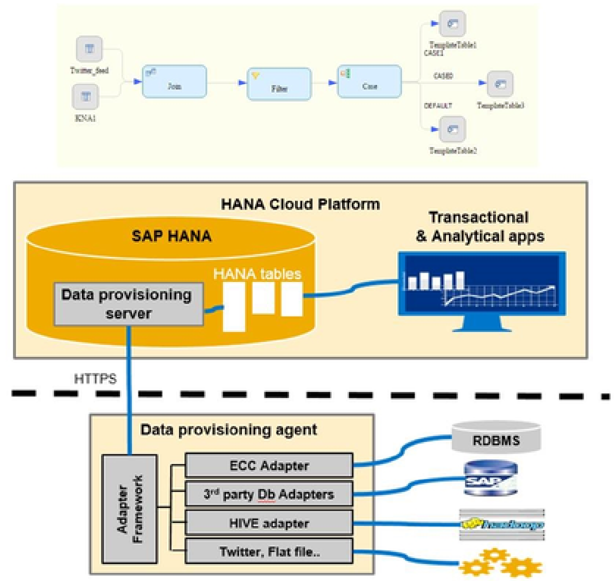


Figure 80: Smart Data Integration

SAP Cloud Platform Smart Data Integration provides the capability to access, provision, replicate, and transform data from various sources in SAP HANA on SAP Cloud Platform.

Key Capabilities and Benefits

Key Capabilities

Native component in the SAP HANA database on SAP Cloud Platform.

Real-time replication, physical bulk/batch movement, and federation in a unified framework.

Many ETL-type transformations available in graphical UI (Flowgraph).

Light on-premise agent to securely access all on-premise data sources from the cloud.

Wide range of built-in adapters for common sources (databases, files, social media, OData, …), open and extensible SDK for creating custom adapters.

Benefits

Simplified – one common modelling environment to provision and consume in SAP Web IDE (or SAP HANA Studio).

Real-time – lower latency due to real time replication and in-memory performance.

Open & extensible – supports data of any shape and size; open framework for new data sources.

Integration, SAP Cloud Platform API management

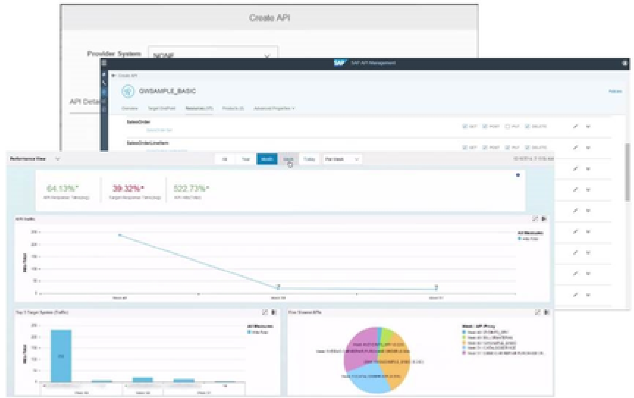


Figure 81: SAP Cloud Platform API management

SAP Cloud Platform API management provides simple, scalable and secure access to digital assets through application programming interfaces (APIs) and enables developer communities to consume these.

Key Capabilities and Benefits

Key Capabilities

Unified standards-based API access of REST/OData or SOAP services.

Enterprise Grade Security for the APIs against attacks like DoS, CSRF, XSS etc. and robust traffic management.

Real-time insights & analytics on the APIs traffic, usage, error reporting and monitoring.

Developer services to enable developers to try, subscribe, use and manage API consumption.

Benefits

Platform for engaging with and enabling employees and developers - internal and external.

The use case the API management solves is that if you have a developer who is creating services through our platform and wants to expose that, there are several issues that it solves.

Publishing, promoting, administer (access/availability) and support end user resources like documentation made available to consumers of that API.

SAP API Business Hub

Figure 82: SAP API Business Hub

SAP API Business Hub provides a central catalog of SAP and selected partner APIs allowing developers to search, discover, test and consume the services they need to build new applications, app extensions or process integrations, using the SAP Cloud Platform.

Key Capabilities and Benefits

Key Capabilities

Growing list of SAP and selected partner APIs for application developers to discover, learn and try out.

Sandbox test environment for easily discovering APIs.

Integration with SAP Web IDE to easily consume APIs and develop SAP Fiori extension apps.

Comprehensive API documentation in a vendor-neutral OpenAPI standard to accelerate learning.

Benefits

Convenient access to critical APIs information in one place to learn faster.

Comprehensive learning and testing enviroment to improve development productivity.

SAP Cloud Platform: Enterprise Messaging (beta)

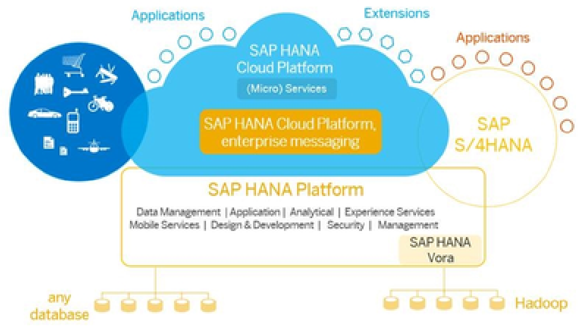


Figure 83: Enterprise Messaging (beta)

SAP Cloud Platform enterprise messaging is a cloud-based messaging framework that enables you to connect applications, services, and devices across different technologies, platforms, and clouds. Scaling to millions of messages per second in real-time, you can send and receive messages reliably using open standards and protocols such as MQTT.

Key Capabilities and Benefits

Key Capabilities

Connect all your applications, services, and devices seamlessly.

Communicate reliably at scale.

Send and receive millions of messages per second.

Gain insights with SAP HANA.

Benefits

Simple implementation.

High throughput, low latency.

High availability clusters and elastic scalability.

Multiple message protocols (e.g. AMQP, MQTT, REST), APIs (Java JMS, Node.js) and exchange patterns (e.g. pub/sub, point-to-point, request/reply).

RabbitMQ on SAP Cloud Platform

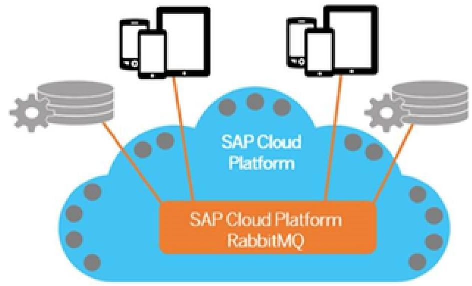


Figure 84: RabbitMQ on SAP Cloud Platform

SAP Cloud Platform provides support for RabbitMQ, an open-source message broker software. Use RabbitMQ as messaging-oriented middleware to implement message queues for application-to-application messaging.

Key Capabilities and Benefits

Key Capabilities

Provides robust messaging between applications.

Supports High Availability (HA) via clustering.

Enables straightforward connectivity between applications written in different programming languages.

Benefits

High Availability & Reliability.

Implements the Advanced Message Queuing Protocol (AMQP).

Business case

Asynchronous communication between services.

Deliver notifications.

Asynchronous operations like billing.

SAP Cloud Platform: OData Provisioning

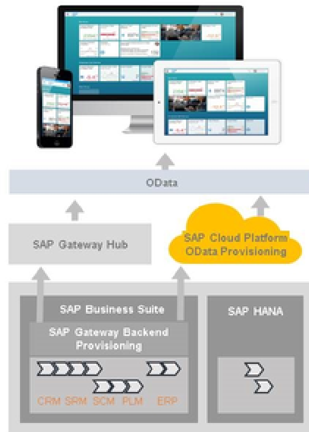


Figure 85: OData Provisioning

OData Provisioning exposes business data and business logic from on-premises SAP systems as standards-based services on SAP Cloud Platform, thereby enabling customers to develop user-centric applications faster.

Key Capabilities and Benefits

Key Capabilities

Publish (register, activate, maintain) OData services from SAP Business suite on cloud.

Provide security features on services such as Authentication, Authorizations, and Cross-Site Request Forgery Protection.

Connect to multiple backends and address multi-origin use cases from cloud.

Maintain metadata cache for performance.

Troubleshoot for error analysis and access performance statistics.

Benefits

Eliminate the need to maintain central SAP Gateway Server thereby reducing TCO.

Access SAP Business Suite Data in a secure manner.

Build cloud apps on top of backend data without disrupting the same.

Rolling software updates and subscription-based usage.

SAP Cloud Platform: Business Rules (beta)

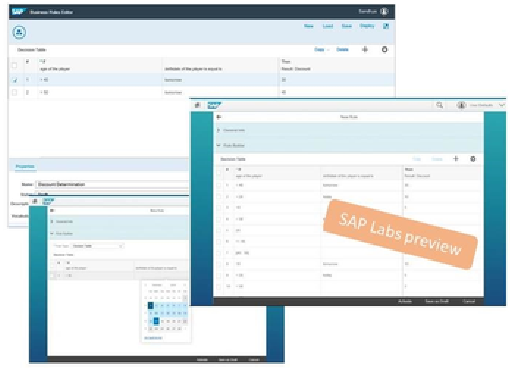


Figure 86: Business Rules (beta)

The SAP Cloud Platform Business Rules service can be used to encapsulate business logic from application logic, enabling business key users and application developers to change the decision logic without re-writing the application.

Key Capabilities and Benefits

Key Capabilities

Web-based Business Rules editor for managing a business rules project.

Embeddable UI5 Controls for decision table business rules.

API based Rule Engine invocation REST API to build, author and manage decision logic.

Benefits

Rapidly build and author decision logic for workflow-based applications and extension applications on SAP Cloud Platform.

Support IoT 4.0 based use cases(beta mode) for rule push down into the IoT 4.0 Edge or Gateway.

SAP Cloud Platform: Workflow

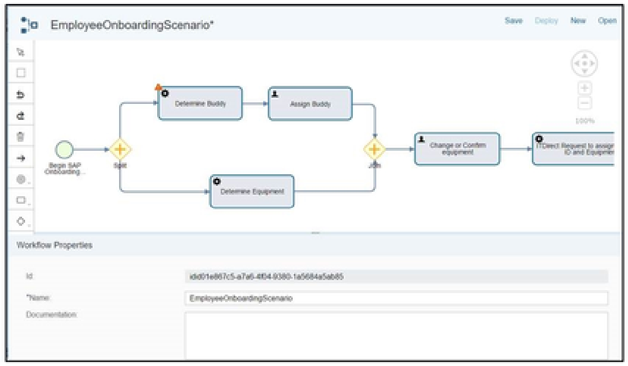


Figure 87: Workflow

The SAP Cloud Platform Workflow service can be used to orchestrate a sequence of tasks across different people and organizations.

The workflow service enables you to build and run workflows on the SAP Cloud Platform.

Key Capabilities and Benefits

Key Capabilities

Web-based BPMN editor for defining the workflow.

My Inbox application for end users to manage their tasks.

Monitoring and administration of running workflows.

HTTP-based connectivity to other applications via service tasks.

REST API to manage workflows and tasks.

Benefits

Rapidly build new workflow-based applications on SAP Cloud Platform.

Easily extend standard cloud application workflows.

LESSON SUMMARY

You should now be able to:

Explain integration services

Unit 3
Lesson 6

Explaining Multi Cloud

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain multi cloud

Multi Cloud

Multi Cloud, expanded cloud options

The following are advantages of multi clouds:

Build cutting-edge open innovations

- Bespoke cloud-native applications readily leveraging community innovations and emerging technologies
- Accelerated time to market at lower costs due to openness of the platform.
- Scalable, fast, simple and robust platform to build cloud architecture to suit your business needs
- Leverage multitude of PaaS development and business services.

Transform to a Digital Enterprise

- Build new business models leveraging enterprise grade services for emerging technologies such as Machine learning (ML) empowering new age SaaS solutions.
- Reduce the cost of digital transformation leveraging open platform, with application development options to support XSA, Node.js, Python, and BYOL - all powered by Cloud Foundry based services.
- Flexible deployment option with reduced vendor lock-in and broader coverage

Expand development options in Cloud

- Make best of both worlds with hybrid cloud -Develop XSA applications on-premise on SAP S/4HANA. Readily deploy the same app in cloud on SAP Cloud Platform. Letting you maximize your ROI from existing on-premise investments
- Natively integrate established standard development tools and processes (e.g. Git, Eclipse)

LESSON SUMMARY

You should now be able to:

- Explain multi cloud

Unit 3
Lesson 7

Explaining Security in PaaS

LESSON OBJECTIVES
After completing this lesson, you will be able to:

- Explain security in PaaS

Security in PaaS

SAP Cloud Platform, Security

SAP Cloud Platform's closely integrated security services include authentication, single sign-on, on-premises integration and self-services such as registration and password reset for employees, customers, partners and consumers.

SAP Cloud Platform, Security, topics:

SAP Cloud Platform Security

- Customizable roles for account members.
- Secure storage for confidential data.
- Securely integrate with customer's corporate user directory.

SAP Cloud Platform Identity Authentication

- central user authentication, web single sign-on and user management as Software-as-a-Service. It's based on open industry-standards - SAML 2.0, SCIM.
- User self-services (e.g. registration, password reset, user profile).
- Out-of-the-box trust configuration with the SAP Cloud Platform account.

SAP Cloud Platform Authorization Management

Offers a REST API to manage the authorizations of users of applications running in SAP Cloud Platform.

SAP Cloud Platform Identity Provisioning

- Automatically create and manage user accounts.
- Policy based authorizations management.
- Re-using existing corporate user stores.
- Centralized end-to-end life-cycle management of corporate identities in the cloud.
- Automated provisioning of existing on-premise identities to cloud applications.

SAP Cloud Platform Password Storage and Keystore service

Provides a repository for cryptographic keys and certificates for applications running on SAP Cloud Platform.

SAP Cloud Platform OAuth 2.0 service

Provides authorization and authentication for a wide range of application scenarios, such as Web, Mobile and IoT.

Security, SAP Cloud Platform Security

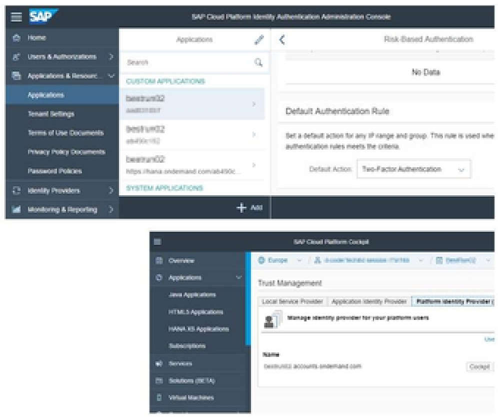


Figure 88: Security, Trust Management

Key Capabilities and Benefits

Key Capabilities

Leverage single sign-on.

Protect customer data.

Secure mobile and IoT scenarios.

Securely integrate with customer's corporate user directory.

Propagate the logged on user.

Secure storage of confidential data.

Additional OAuth 2.0 Authorization Grant Flows (SAML2 Bearer Assertion).

Principal Propagation from Java/HTML5 to HANA XS with AppToAppSSO.

Default trust with SAP ID Service can be enhanced with additonal SAML Identity Providers.

Added Groups Management to Authorization Platform API.

Benefits

Security – Re-use of functions and features for secure development of applications and secure runtime environment.

Security - Secure authentication with one strong password (optionally with additional factors), supporting strong password & authentication policies.

Cost saving – Functions to efficiently set up and manage server-side security capabilities.

Simplicity - Supports secure transition of business processes into the cloud.

Security, SAP Cloud Platform Identity Authentication

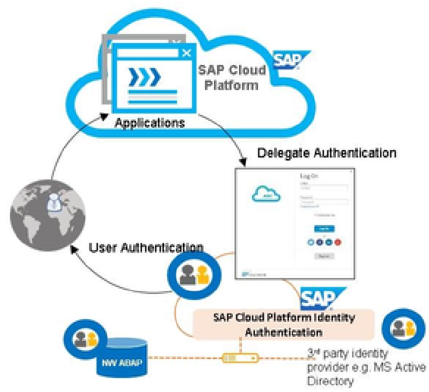


Figure 89: SAP Cloud Platform Identity Authentication

SAP Cloud Platform Identity Authentication provides central user authentication, web single sign-on and user management as Software-as-a-Service. It's based on open industry-standards - SAML 2.0, SCIM.

Key Capabilities and Benefits

Key Capabilities

Authentication against cloud or on-premise user store for SAP and non-SAP cloud applications.

A variety of authentication methods: username and password, two-factor-authentication, social login, SPNEGO, risk-based authentication rules.

User self-services (e.g. registration, password reset, user profile).

Look & feel and policies customization.

Programmatic integration via SCIM standard.

Benefits

Uniform log-on and web SSO across applications.

Central user management or reuse of existing identity infrastructure.

Out-of-the-box trust configuration with the SAP Cloud Platform account.

Security, SAP Cloud Platform Identity Provisioning

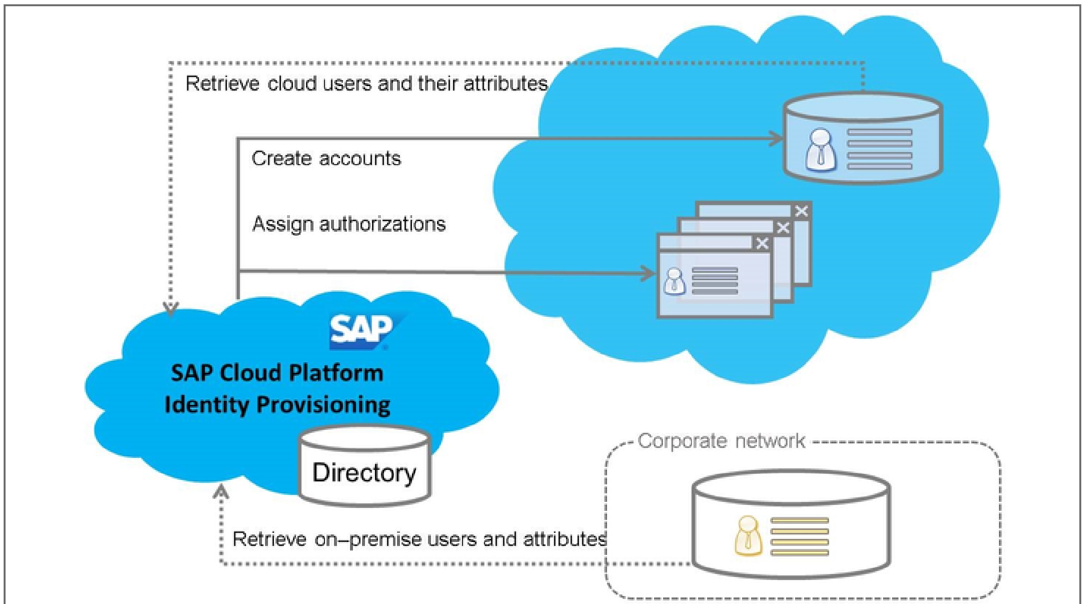


Figure 90: SAP Cloud Platform Identity Provisioning

Identity Provisioning offers a comprehensive, low cost approach to identity lifecycle management in the cloud.

Key Capabilities and Benefits

Key Capabilities

Automatically create and manage user accounts.

Policy based authorizations management.

Re-using existing corporate user stores.

Optimized for SAP cloud applications and SCIM-enabled solutions.

Integrated with Identity Authentication and SAP Cloud Platform, integration service.

Integration option with SAP Identity Management.

Benefits

Fast and efficient administration of user on-boarding.

Centralized end-to-end life-cycle management of corporate identities in the cloud.

Automated provisioning of existing on-premise identities to cloud applications.

SAP Cloud Platform Authorization Management

SAP Cloud Platform Authorization Management is a service offering a REST API to manage the authorizations of users of applications running in SAP Cloud Platform.

Key Capabilities and Benefits

Key Capabilities

Provide functionality to manage roles and their assignments to users and groups in the specified account and application.

Manage the rights that your application users have dynamically and easily during runtime through automation.

Benefits

Roles can be provided within the web.xml or web-fragment.xml.

Roles can either be deployed with the application or created via the API.

SAP Cloud Platform Password Storage and Keystore service

SAP Cloud Platform Password Storage and Keystore service provides a repository for cryptographic keys and certificates for applications running on SAP Cloud Platform.

Key Capabilities and Benefits

Key Capabilities

Enabling applications to retrieve keystores easily.

Signing and verifying of digital signatures.

Encrypting of and decrypting of messages.

Performing SSL communication.

Benefits

Configuring keystores on a local server or via the SAP Cloud Platform console client.

Searching for a keystore on the different levels (Subscription level, Application level, Account level).

Keystore console commands allow users to list, upload, download and delete keystores.

SAP Cloud Platform OAuth 2.0 service

SAP Cloud Platform OAuth 2.0 service provides authorization and authentication for a wide range of application scenarios, such as Web, Mobile and IoT.

Key Capabilities and Benefits

Key Capabilities

Protect applications and APIs with OAuth 2.0.

Avoid storing credentials at the third-party location.

Limit the access permissions granted to third parties.

Enable easy access right revocation without the need to change credentials.

Benefits

Support the OAuth 2.0 protocol based on the IETF RFC 6749.

Granting access without explicit credentials sharing.

LESSON SUMMARY
You should now be able to:

Explain security in PaaS

Unit 3
Lesson 8

Explaining User Experience

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain user experience

User Experience

User Experience

SAP Cloud Platform empowers organizations to build and scale simple, personalized and responsive user experience.

The following are the building blocks of the user experience:

SAP Cloud Platform Portal

- Easily build branded business sites using intuitive site designer.
- Integrate SAP Fiori apps, custom web apps, unstructured content and third party content (cloud or on-premise).
- Extend SAP solutions via predefined templates and accelerators.

SAP Fiori Cloud

- Enable a modern and intuitive user experience for SAP users.
- Customize, extend and develop and mobilize SAP Fiori apps in the cloud.

SAP Cloud Platform Build

- Create interactive prototypes without the need for development.
- Integrated end-user feedback.

SAP Cloud Platform Forms by Adobe

- Online and offline interactive scenarios.
- Print forms.

UI Theme Designer

Browser-based tool which allows you to create your own themes to adapt the visual appearance of applications. It is the single tool for theming and branding of key SAP UI technologies.

User Experience, SAP Cloud Platform Portal

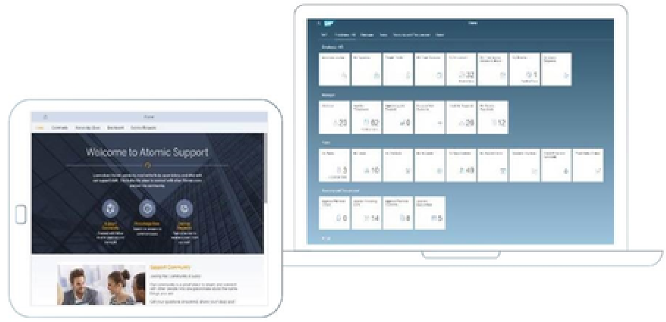


Figure 91: SAP Cloud Platform Portal

SAP Cloud Platform Portal provides an easy way for creating appealing business sites to engage with customers, partners and employees. Users benefit from central, role-based point of access to information and services.

Key capabilities and Benefits

Key capabilities

Build branded business sites or SAP Fiori 2.0 launchpad sites.

Manage and integrate SAP Fiori apps, custom web apps, unstructured content and third party content (cloud or on-premise).

Leverage SAP Fiori content for SAP Business Suite & SAP S/4HANA.

Extend SAP solutions via predefined templates and accelerators.

Develop an appealing, custom site experience.

SAP Fiori Launchpad sites:

SAP Fiori 2.0 design (SAPUI5 1.44, new navigation, Me Area, Notifications Area*, etc.) *requires Gateway (on-premise) as notification hub.

Improved SAP Fiori Configuration Cockpit for managing tiles, roles, catalogs (e.g. assignments; group, filter, sort elements).

Enhanced integration with SAP Easy Access menu to launch classic SAPUIs from the launchpad

Integration between PFCG roles and app catalogs to automatically determine required authorizations.

Freestyle sites:

- Add multiple widgets to sections (in page layout).
- Improved branding & theming process.
- Additional widgets and content types (e.g. video, image carousel, Jam widget, etc).

Benefits

Drive UX innovations and agility via cloud-based portal services.

Improve user satisfaction via intuitive user interface.

Enhance admin efficiency trough prebuild SAP business content (SAP Fiori apps) and open integration framework.

SAP Cloud Platform Portal service is a service that allows enterprise/customers/partners to easily create a business sites for their audience:

Audience may be internal employees or external customers and partners.

Allows them to bring together both content and applications all in one central location.

Allows them to theme and create a site that has the look and feel of the company's individual brand.

SAP delivers standardize templates and accelerators to quickly create and adapt sites.

User Experience, SAP Fiori Cloud

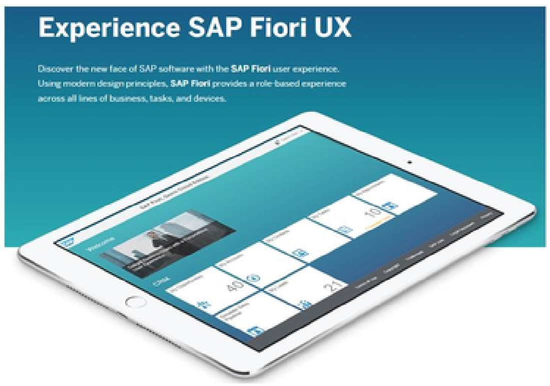


Figure 92: SAP Fiori Cloud

SAP Fiori Cloud provides **SAP S/4HANA** and **SAP Business Suite** customers a simple approach to adopt the SAP Fiori user experience (UX) in the cloud by leveraging SAP Fiori pre-packaged content and services.

Key capabilities and Benefits

Key capabilities

Enable a modern and intuitive user experience for SAP users.

Design, develop, extend, and mobilize SAP Fiori apps in the cloud.

Connect securely to SAP Business Suite and SAP S/4HANA.

Experience and explore SAP Fiori UX on www.sap.com/fiori-demo.

Ability to launch classic UIs from launchpad: Web Dynpro ABAP, Screen Personas and Web GUI.

Design new apps with Build and develop/extend with Web IDE.

Benefits

Simplify SAP Fiori implementation and accelerate time-to-value while leveraging existing investments in SAP backend systems.

Easily connect and run selected SAP Fiori Cloud apps for various lines of business: human resources, finance, sales, supply chain and procurement.

Enable fast UX innovation cycles.

User Experience, SAP Cloud Platform Build

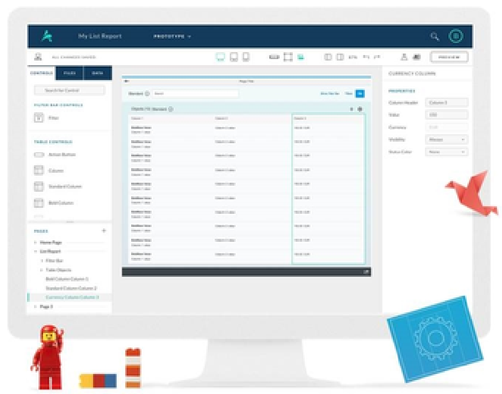


Figure 93: SAP Cloud Platform Build

SAP Cloud Platform Build is an open source, collaborative design tool that makes it easy for anyone to learn and apply design thinking principles to any IT project.

Key capabilities and Benefits

Key capabilities

Create interactive prototypes without the need for development.

Integrated end-user feedback.

Realistic sample data: add data in-line, import data from Excel, or bind to real services.

Design with real SAP Fiori controls, floorplans and smart templates.

Jumpstart development through the generation of UI5 code and integration with SAP Web IDE.

Prototyping:

- Support for import of custom themes.
- New UI Logic including stacked actions.
- Transitions between pages.
- New floorplans, controls and additional control simplification.

Benefits

Incorporate user feedback prior to development, reducing frequency of costly change requests.

Lower UI development costs.

Increased end-user satisfaction.

User Experience, SAP Cloud Platform Forms by Adobe

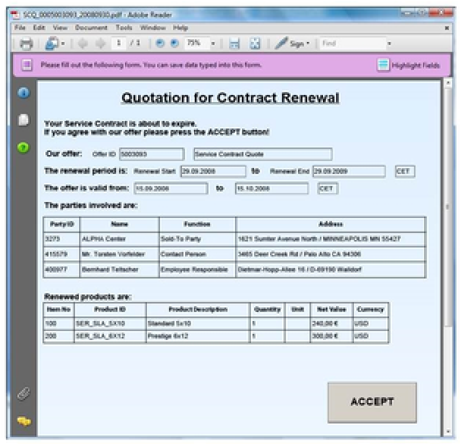


Figure 94: SAP Cloud Platform Forms by Adobe

SAP Cloud Platform Forms by Adobe is SAP's cloud solution for print forms and interactive forms.

Key capabilities and Benefits

Key capabilities

Online and offline interactive scenarios.

Print forms.

Optimal integration with SAP technology including BPM.

Large number of predefined forms.

Digital signatures.

Bar codes.

Pre-filled application values, list boxes…

Automatic data extraction and integration in SAP applications.

Printing, saving locally and archiving.

Standard Adobe Reader functionality.

Distribution by e-mail / portal.

Adherence to SAP product standards.

Benefits

Self-explanatory (paper-like).

High number of predefined forms for SAP Business Suite, SAP Business ByDesign and SAP S/4HANA (currently 2400+).

Business case (when applicable a brief description on how this feature will help solve what/ which business scenario) e.g. This feature will help x person (role) do y faster/better.

SAP Cloud Platform Forms by Adobe allows developers to generate print forms and interactive forms, online- and offline, without installing the ADS locally.

It provides a similar range of functionality to the pure on-premise version.

For now, this is a hybrid setup where the ADS runs on SAP CP and the rest of the solution on-premise on the NetWeaver stack.

User Experience, UI Theme Designer

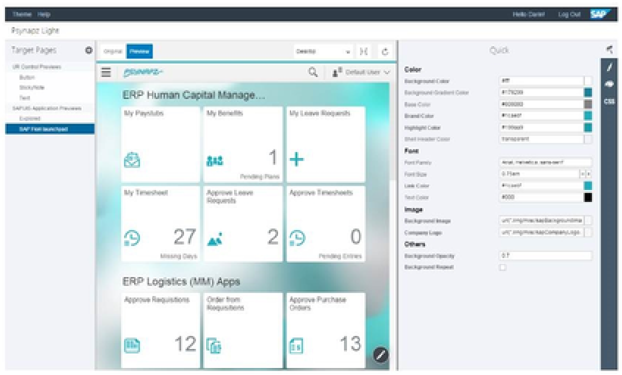


Figure 95: User Experience, UI Theme Designer

SAP's UI theme designer is a browser-based tool which allows you to create your own themes to adapt the visual appearance of applications. It is the single tool for theming and branding of key SAP UI technologies.

Key capabilities and Benefits

Key capabilities

Graphical web-based WYSIWYG editor for controls and application previews.

Different skill levels of theming: quick, expert and CSS.

Custom CSS3 and LESS support.

Great support of SAPUI5.

Benefits

Reduce visual breaks for end users.

Increased theming/branding efficiency for applying your corporate branding and look to SAP-based applications.

Zero footprint and low TCO.

LESSON SUMMARY

You should now be able to:

- Explain user experience

Unit 3

Learning Assessment

1. Which of the following is not an application development service in SAP Cloud Platform?

   Choose the correct answer.

   - [ ] **A** BusinessObjects Cloud
   - [ ] **B** SAP S/4HANA Advanced Analytics
   - [ ] **C** Smart Business Service
   - [ ] **D** Streaming Analytics Service
   - [ ] **E** Internet of Things

2. The SAP Cloud Platform can be used to build extensions for SAP cloud products such as SuccessFactors, Ariba, etc.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

3. Which service would allow you to design IFlow's to implement integration between systems?

   Choose the correct answer.

   - [ ] **A** Integration Service
   - [ ] **B** OData Provisioning
   - [ ] **C** RabbitMQ
   - [ ] **D** Workflow
   - [ ] **E** API Business Hub

4. One of the key capabilities of the Cloud Connector is to establish TLS, RFC, and SMTP secure tunnel connections between the SAP Cloud Platform and on-premise systems.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

5. Which of the following is not a collaboration service in SAP Cloud Platform?

   Choose the correct answer.

   - [ ] **A** Jam
   - [ ] **B** Gamification
   - [ ] **C** Document Center
   - [ ] **D** Enterprise Messaging

Learning Assessment - Answers

1. Which of the following is not an application development service in SAP Cloud Platform?

   Choose the correct answer.

   - [ ] **A** BusinessObjects Cloud
   - [ ] **B** SAP S/4HANA Advanced Analytics
   - [ ] **C** Smart Business Service
   - [ ] **D** Streaming Analytics Service
   - [X] **E** Internet of Things

   This is correct. See unit 3, lesson 1 of CLD100, Col18.

2. The SAP Cloud Platform can be used to build extensions for SAP cloud products such as SuccessFactors, Ariba, etc.

   Determine whether this statement is true or false.

   - [X] True
   - [ ] False

   This is correct. See unit 3, lesson 1 of CLD100, Col18.

3. Which service would allow you to design IFlow's to implement integration between systems?

   Choose the correct answer.

   - [X] **A** Integration Service
   - [ ] **B** OData Provisioning
   - [ ] **C** RabbitMQ
   - [ ] **D** Workflow
   - [ ] **E** API Business Hub

   This is correct. See unit 3, lesson 3 of CLD100, Col18.

4. One of the key capabilities of the Cloud Connector is to establish TLS, RFC, and SMTP secure tunnel connections between the SAP Cloud Platform and on-premise systems.

   Determine whether this statement is true or false.

   - [ ] True
   - [x] False

   This is correct. See unit 3, lesson 3 of CLD100, Col18.

5. Which of the following is not a collaboration service in SAP Cloud Platform?

   Choose the correct answer.

   - [ ] **A** Jam
   - [ ] **B** Gamification
   - [ ] **C** Document Center
   - [x] **D** Enterprise Messaging

   This is correct. See unit 3, lesson 4 of CLD100, Col18.

UNIT 4 Software as a Service (SaaS)

UNIT OBJECTIVES

- Get an introduction about SaaS
- Explore core business applications
- Explain business services on SAP Cloud Platform
- Use API business hub
- Explaining SaaS extensions on SAP Cloud Platform

Unit 4
Lesson 1

Introducing Software as a Service

LESSON OBJECTIVES

After completing this lesson, you will be able to:

Get an introduction about SaaS

Software as a Service, Introduction

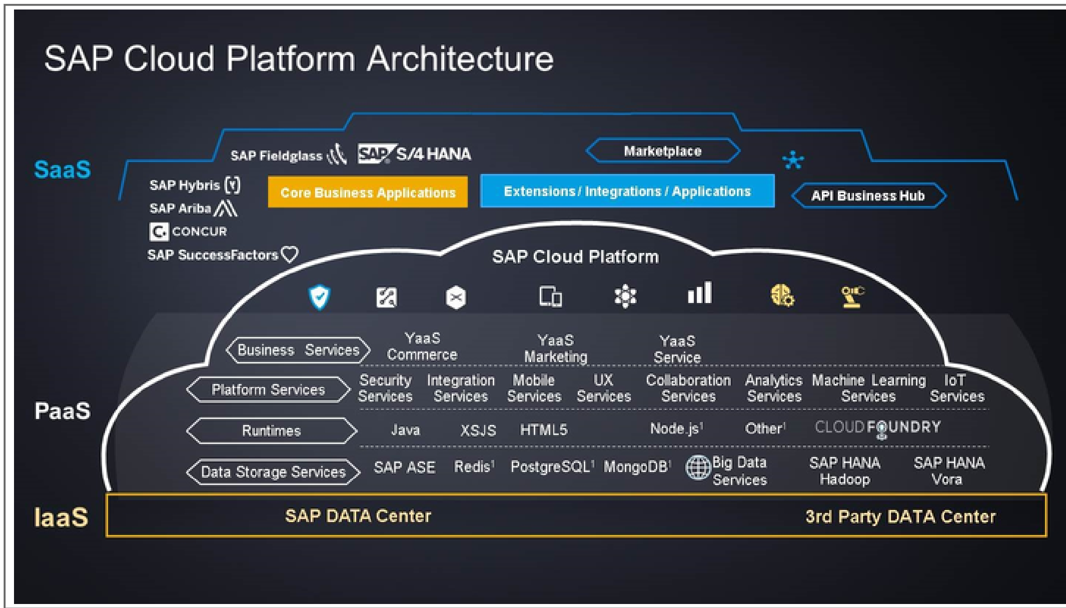


Figure 96: Software as a Service, Introduction

The figure gives a a closer look at the Architecture of SAP CP. You see, that Software as a Serice (SaaS) is located (using) the SAP Cloud Platform, especially the Services Platform as a Service (PaaS) and Infrastructure as a Service (IaaS).

SaaS, Overview

What is SaaS?

SaaS - Software-as-a-Service - is a way of delivering applications over the Internet.

For customers:

- the advantage is, that they can access SaaS applications right from a Web browser, which means there is no hardware or

- software to buy, install, maintain, or update. The SaaS provider takes care of everything - and
- the customer always has the latest version of the application.

Explore our SaaS products:

SAP SuccessFactors HR Solutions

SAP S/4HANA Finance

SAP Hybris Cloud for Customer

SAP Hybris Cloud for Sales

SAP Hybris Cloud for Service

SAP Hybris Marketing

SAP Hybris Commerce

SAP Business ByDesign (ERP for Midsize Business)

SAP Business One Cloud (ERP for Small Business)

SAP BusinessObjects Cloud for Analytics

SAP Jam Collaboration

SAP Jam Communities

Ariba Network: Sourcing, Procurement, & Finance

Concur: Travel & Expense Management

Fieldglass: Flexible Labor Management

And **SAP Leonardo**

SaaS Roadmaps, Introduction

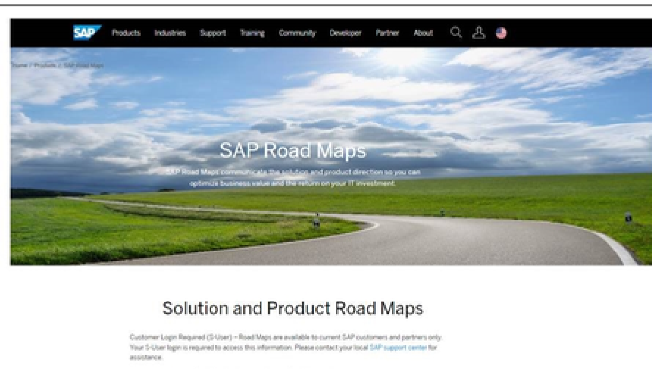


Figure 97: Introducing the roadmaps

Learn more about the core Services at **https://www.sap.com/products/roadmaps.html** .

SAP Leonardo, Introduction

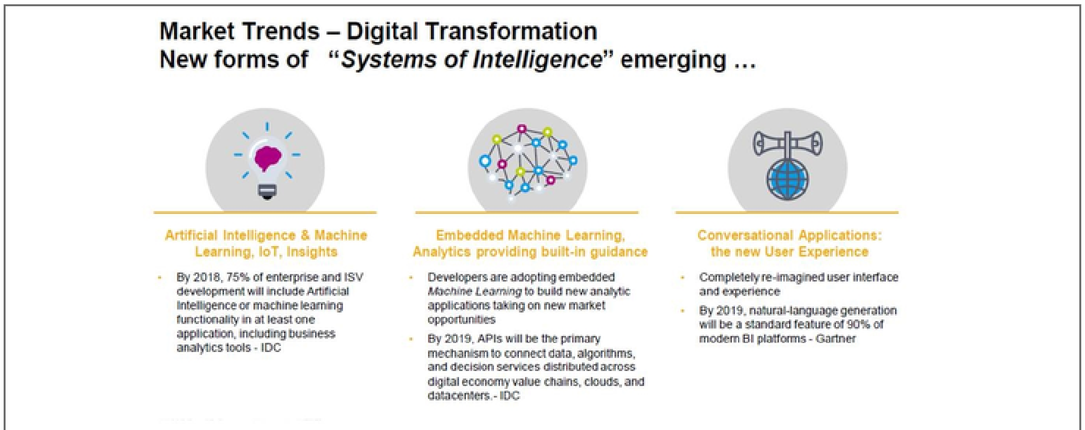


Figure 98: SAP Leonardo, Introduction

The figure shows the reasons for establishing SAP Leonardo.

SAP Leonardo, Benefits

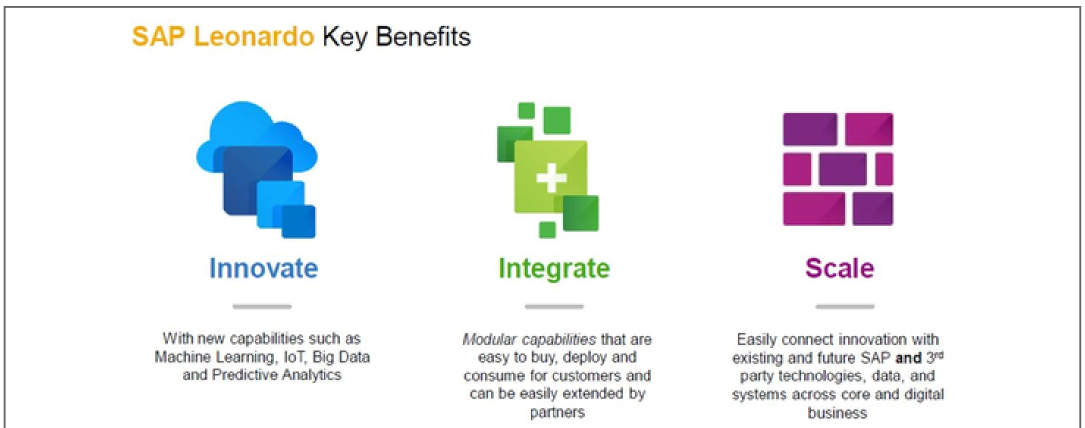


Figure 99: SAP Leonardo, Benefits

The figure shows the 3 main Benefits of SAP Leonardo.

SAP Leonardo, Context

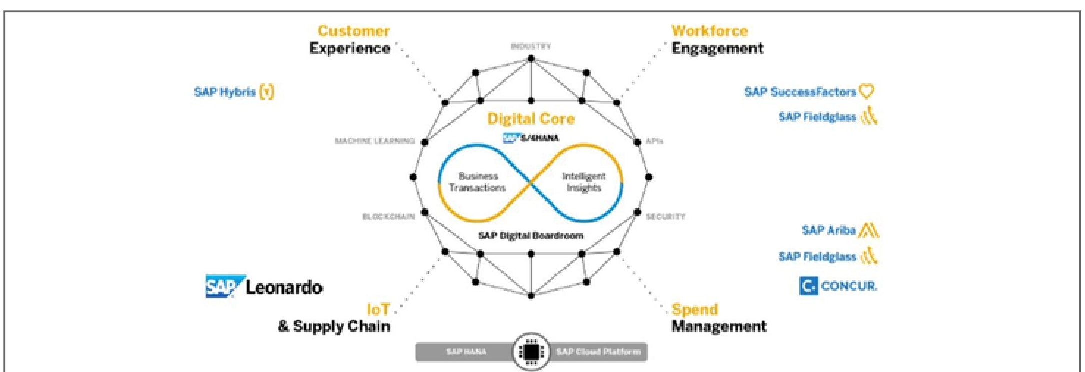


Figure 100: SAP Leonardo, Context

The figure shows the context around SAP Leonardo.

SAP Leonardo, Vision

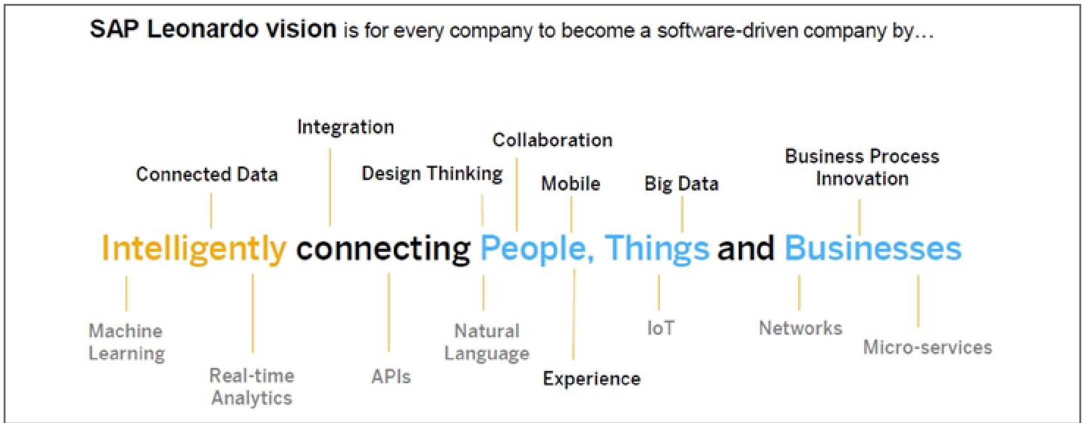


Figure 101: SAP Leonardo, Vision

The figure shows the vision of SAP Leonardo,

SAP Leonardo, Technologies

Figure 102: SAP Leonardo, Technologies

The figure shows the technologies, used by SAP Leonardo.

SAP Leonardo Internet of Things

Figure 103: SAP Leonardo Internet of Things

The figure shows how the topics of the Internet of Things are handled by SAP Leonardo.

SAP Leonardo Internet of Things, SAP Leonardo Bridge

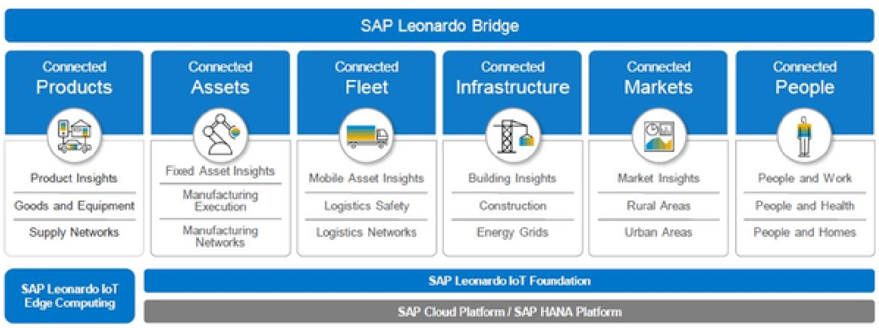


Figure 104: SAP Leonardo Bridge

The figure shows the span of SAP Leonardo Bridge.

SAP Leonardo and Digitizing Business, the big picture

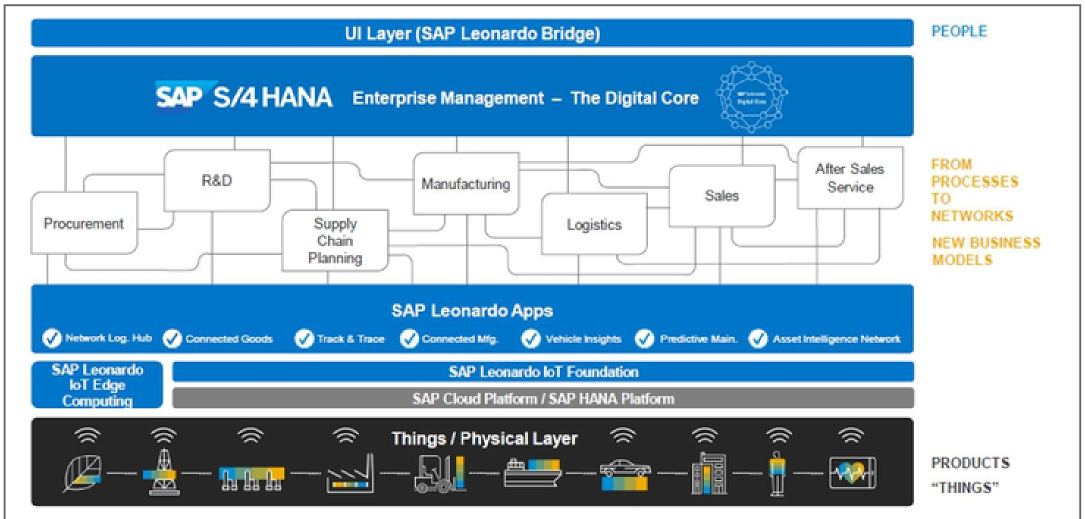


Figure 105: SAP Leonardo and Digitizing Business, the big picture

The figure shows the big picture of the SAP Leonardo and Digitizing Business.

SAP Leonardo, Machine Learning

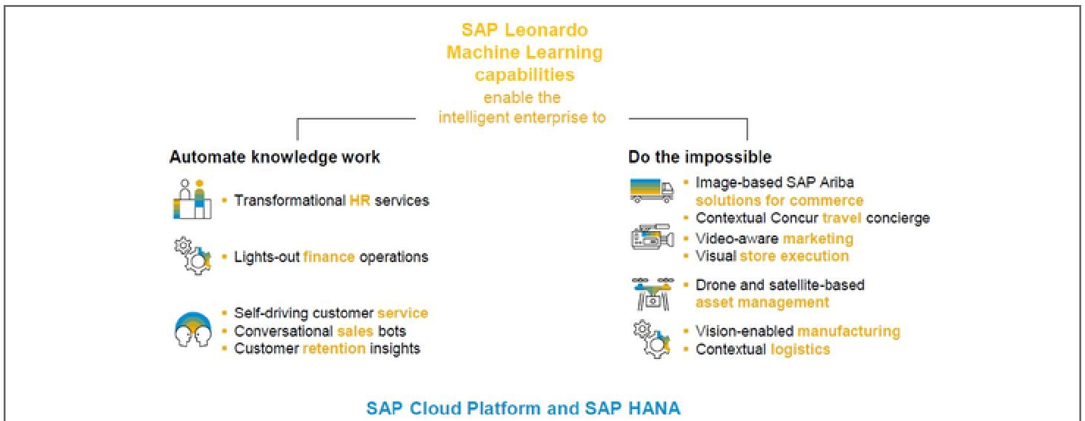


Figure 106: SAP Leonardo, Machine Learning

The figure shows the approach of SAP Leonardo about machine Learning

SAP Leonardo Machine Learning, processing steps

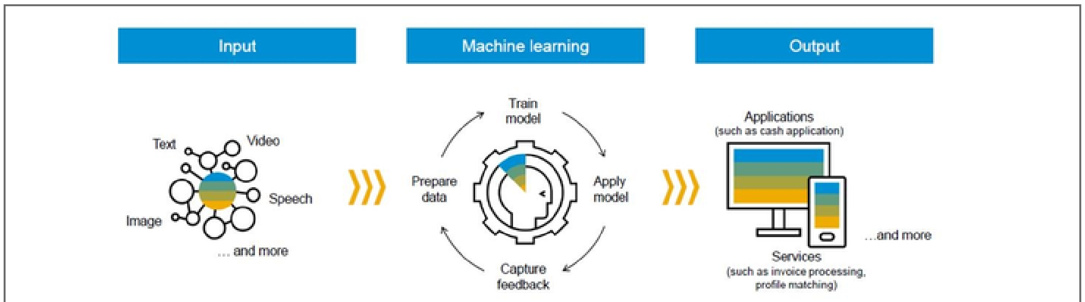


Figure 107: SAP Leonardo Machine Learning, processing steps

The figure shows the different steps of machine learning in SAP Leonardo.

SAP Leonardo Machine Learning, business placement

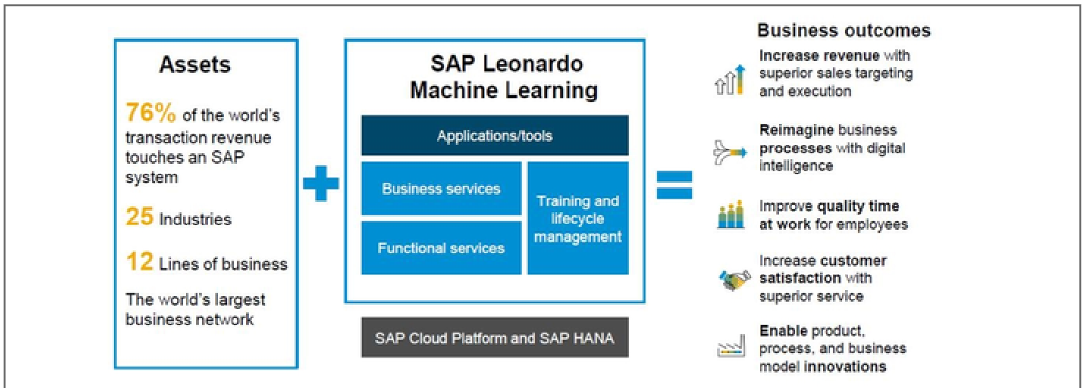


Figure 108: SAP Leonardo Machine Learning, business placement

The figure shows the business placement in SAP Leonardo.

LESSON SUMMARY

You should now be able to:

- Get an introduction about SaaS

Unit 4
Lesson 2

Exploring Core Business Applications

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explore core business applications

Core Business Applications

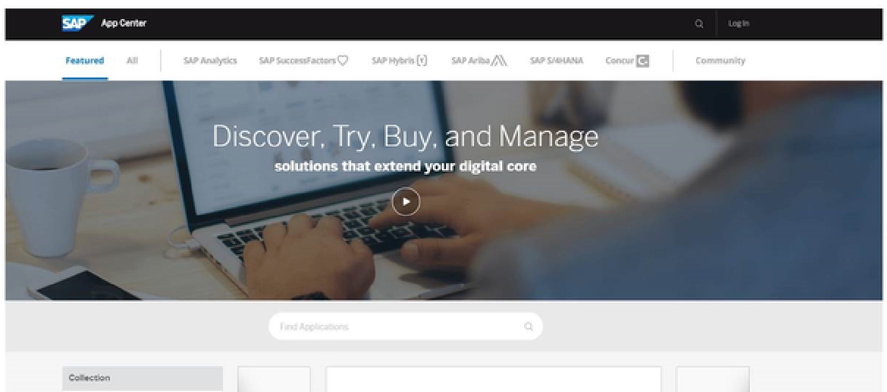


Figure 109: Core Business Applications

You will find a complete overview in the App Center https://www.sapappcenter.com/home

LESSON SUMMARY

You should now be able to:

- Explore core business applications

Unit 4
Lesson 3

Explaining Business Services on SAP Cloud Platform

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain business services on SAP Cloud Platform

Business Services on SAP Cloud Platform

SAP Business Services fuel the fast development of Business apps and services for the cloud, and power an open marketplace for new business apps.

SAP Hybris as a Service on SAP Cloud Platform:

- Platform allows developers to create, offer and sell business service packages based on a micro-service centric programming model.
- Allows partners to offer such packages as well.
- Augmenting front-office solutions with new functionality and deliver new LoB centric apps.

SAP Data Quality Management, microservices for location data (beta):

- Address Cleansing
- Geocoding
- Reverse Geocoding

SAP Leonardo Machine Learning:

- Integrates learning from data to help businesses grow, optimize processes, assist employees.
- Provides machine learning services targeted for your lines of business.

SAP Localization Hub, tax service (beta):

- Global Coverage: Pre-delivered indirect tax determination and computation content for 120 countries.
- Cloud Offering: Provide consumers with access to tax computation across all products using SAP Cloud Platform.
- Central Framework: The tax service can be used as central tax computation framework across the customer landscape.

SAP Smart Business service:

SAP Smart Business is a framework for visualizing analytic content in the form of charts and tiles.

KPIs and OPIs can be visualized as SAP Fiori applications without writing any code.

Supports Automated analysis on behalf of business user in background jobs and Alerting whenever there is a breach of KPI thresholds.

SAP Cloud Platform, Business Services landing page

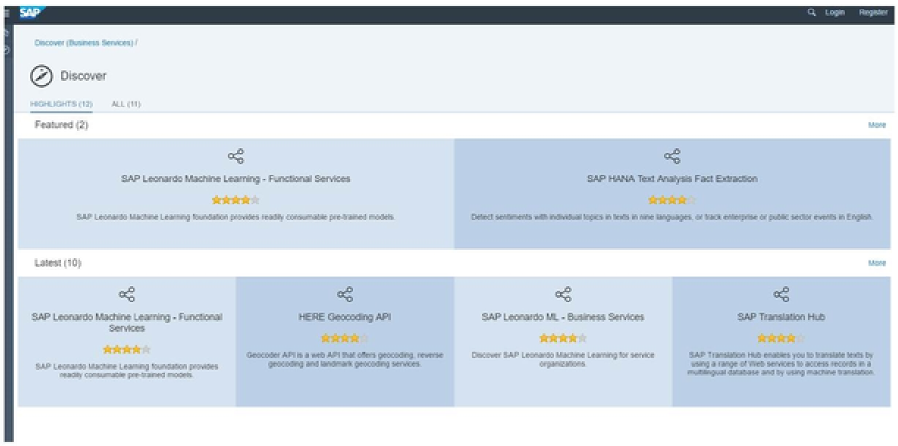


Figure 110: Business Services landing page

The figure shows the Business Services landing page: **https://api.sap.com/#/shell/businessservice**

Business Services: SAP Hybris as a Service on SAP Cloud Platform

Figure 111: SAP Hybris as a Service on SAP Cloud Platform

SAP Hybris as a Service (YaaS) on SAP Cloud Platform allows developers to create, offer and sell business service packages based on a micro-service centric programming model and customers to leverage these services for extending existing and new front-office solutions faster than ever.

Key capabilities

Offers SAP Hybris as a Service packages:

- Core Service packages.
- SAP Hybris Commerce as a Service packages.
- SAP Hybris Marketing Loyalty packages.
- SAP Hybris Service Customer Engagement Center packages.

Allows partners to offer such packages as well.

Augmenting front-office solutions with new functionality and deliver new LoB centric apps.

Benefits

Lower implementation costs and broaden development community by building on existing service packages.

Contribute to the ecosystem and offer your own service packages through the YaaS market.

SAP Data Quality Management, microservices for location data

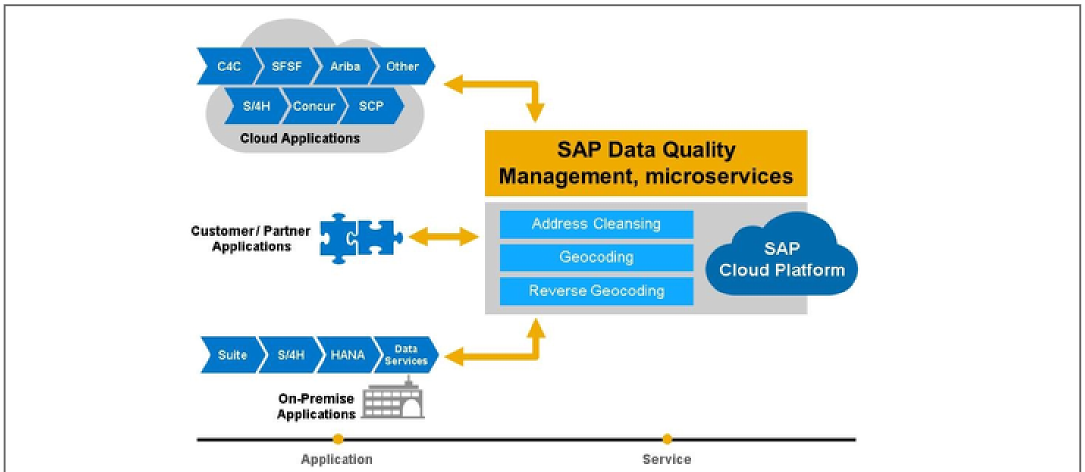


Figure 112: Microservices for location data

Key Capabilities and Benefits

Key Capabilities

Address Cleansing– parsing, standardization, correction, validation, enhancement globally

Geocoding– Address to LAT/LONG

Reverse Geocoding— LAT/LONG to Address(es)

Benefits

Simple consumption model in the Cloud

Integration into SAP applications

- Available Today - SAP ERP, CRM, Master Data Governance, SAP S/4HANA onPremise, Data Services, SAP Hybris Cloud for Customer (1705)
- SuccessFactors (1705 Beta)

Consumption-based pricing model

Leading solution, 30+ years experience in address cleansing and geocoding

Strong partnerships with many worldwide postal authorities and data providers

Description of the microservice for location data

This is a microservice offering providing services for application developers to integrate these services into their applications.

When integrated these services can provide a way for consuming applications to offer address validation, geocoding, and reverse geocoding capabilities to its end users.

Value:

1. Address validation - ensure high quality address data is entered into your applications. If you're shipping products, invoices, etc. to customers. Ensure accurate address so products are delivered and not returned thus reducing returned package costs and frustrated customers who did not receive their deliveries on time. Provide more accurate location analytics by standadizing attributes such as Country, Region, City, Postcode, etc..
2. Geocoding - want to provide a way to visual locations on a map given an address. You need a geocode and this service can provide a geocode given an address for further display or geospatial analysis. E.g. A customer is ordering a product on your website, determine the closes pickup location and display on a map using our geocoding service to retrieve the LAT/LONG information to perform this calculation.
3. Reverse Geocoding - you have an entities current location (LAT/LONG) and wish to provide a list of nearby addresses. E.g. Account manager is visiting a new potential client and entering contact information into their CRM system via their mobile device. We want to enhance the CRM system to automatically enter the address for the sales rep based on the currently location (LAT/LONG) of their mobile device. The reverse geocoding service can provide the ability o retrieve the closes addresses for the CRM application.

What is the key differentiator of this service?

1. Get it all from SAP - Most other Cloud PaaS Vendors do not offer address cleansing/ validation. Some, like Google and Microsoft have some geocoding/reverse geocoding capabilities as part of their Map API's. However they do not offer a combined address validation service w/ their geocoding. So many of them will interpolate some address locations to an address that doesn't really exist. With our address validation we can provide more accurate results in many cases.
2. Runs on SAP Cloud Platform - leverages existing security infrastructure, easy to integrate into other SAP Cloud Platform applications
3. Simpler Consumption model (than our onPremise offerings) - we provide a simple microservice API in the Cloud (on SCP) to make it easy to consume these features in your applications.

SAP provides address cleansing, geocoding and reverse geocoding in some of our onPremise applications today. However, this requires the customer to have additional software licenses onPremise, requires them to individually license address cleansing and geocoding content from SAP under an annual subscription model enable SAP Data Services and SAP S/4HANA, smart data quality to perform these actions, and requires customers to download and deploy this content on a monthly/quarterly basis to keep the solution up to date wtih the latest data from our 3rd party providers. With SAP Data Quality Management, microservices SAP takes are of all of this in the Cloud for our customers. SAP Customers can work entirely with SAP vs. working with many 3rd party vendors to get these services for their applications and get the features in an onPremise or now a Cloud/SaaS based model.

- SAP Data Services (provides onPremise address cleansing, geocoding, & reverse geocoding natively within SAP S/4HANA using the same engines as this Cloud service)

  Address Cleansing - https://wiki.scn.sap.com/wiki/display/EIM/Global+Address+Cleanse+Transform

  Geocoding/Reverse Geocoding - https://wiki.scn.sap.com/wiki/display/EIM/Geocoder+Transform

- SAP S/4HANA smart data quality (provides onPremise address cleansing, geocoding, & reverse geocoding natively within SAP S/4HANA using the same engines as this Cloud service)

  Address Cleansing - https://www.youtube.com/watch?v=uFFnXeg5BEU

  Geocoding (old based on Studio now available in Web IDE and integrated with Geocode Index) - https://www.youtube.com/watch?v=3NJyfNAbPJA&index=31&list=PLkzo92owKnVwQ_preA3cxlQjn_v3W0Eh5

  Reverse Geocoding - https://www.youtube.com/watch?v=QXOsH0heLDw&list=PLtTA9OzhFl86fLIz_S-coeDU-NGWWQepD&index=7

- Embedded in the business suite: Data Quality Management, version for SAP - ABAP add-on that provides address cleaning/validation in SAP ECC, CRM, Master Data Governance and SAP S/4HANA onPremise today based on Data Services or SAP S/4HANA smart data quality. Now supports the DQM, microservices. See example of this working here: http://www.sdn.sap.com/irj/scn/index?rid=/library/uuid/d00c3204-f2ad-2e10-579d-a5bb940281cd&overridelayout=true

Integration into SAP Applications - our key value proposition we feel is to offer these services pre-integrated into other SAP applications. So we are working on the following integrations which will be available in the future:

- SAP Business Suite - our existing ABAP Add-on is being enhanced to support the SAP DQM, microservices API's to support address cleansing and geocoding within SAP ECC, CRM, and Master Data Governance.

- SAP S/4HANA - we're working on an integration of these features into SAP S/4HANA, smart data quality to provide a Cloud based model via these API's and also integrating the Geocoding feature with SAP S/4HANA spatial geocoding index.

- SAP Data Services - SAP's ETL tool, which supports these features today but completely onPremise will also be adding a new transformation to make it easy to consume these microservices from within the Data Services platform.
- SuccessFactors Employee Central - 1705 beta, 1708/1711 GA (limited countries). Address cleansing only.
- Hybris Cloud for Customer - GA in 1705 release. Address cleansing and geocoding.
- Other integrations and POC's at infant stages are ongoing with - Concur, Ariba, Hybris Commerce.

Leading solution w/ 25+ years experience

- Technology purchased by BusinessObjects from leading data quality vendor Firstlogic (http://www.informationweek.com/software/information-management/business-objects-to-acquire-firstlogic-inc/d/d-id/1040437?)
- Became part of SAP in 2008 with BusinessObjects acquisition by SAP.

Strong Partnerships

- 20+ years partnering directly with the worlds leading postal authorities, data providers, and geospatial providers (e.g. TomTom, HERE) to provide one of the most robust solutions in the industry.

Distributed or Centralized Configuration

- All capabilities can be controlled directly through the API and managed in the consuming application OR we also provide a UI on SAP Cloud Platform which can be used to store information about mapping from an applications data model to the service and controlling the address cleansing and geocoding options. The second option allows a consuming application to reuse our UI so they don't have to provide their own persistency and UI for mapping and configuration the to the service.

SAP Localization Hub, tax service

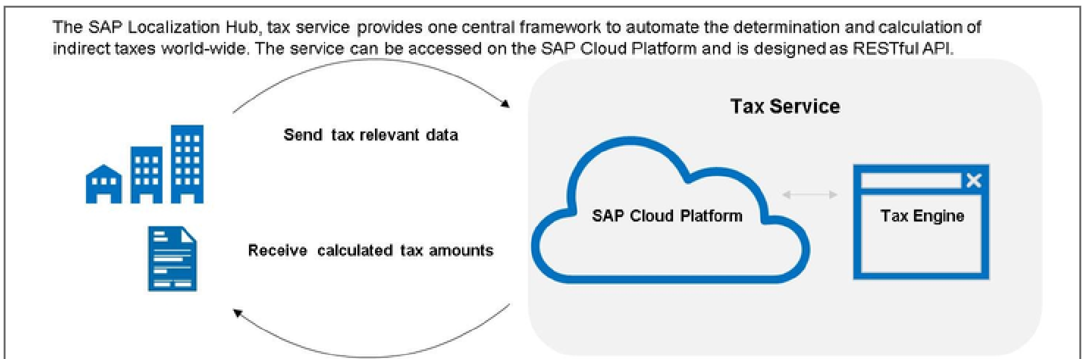


Figure 113: SAP Localization Hub, tax service

Key capabilities

Global Coverage: Pre-delivered indirect tax determination and computation content for 120 countries

Compliance: With it’s global network, SAP keeps this service up to date

Central Framework: The tax service can be used as central tax computation framework across the customers’ sytem landscape

Benefits

Cost Savings by reduced maintenance efforts

Single source to determine and compute tax

Automation and scalability of tax calculation

High quality and consistent tax computation results

SAP Leonardo Machine Learning

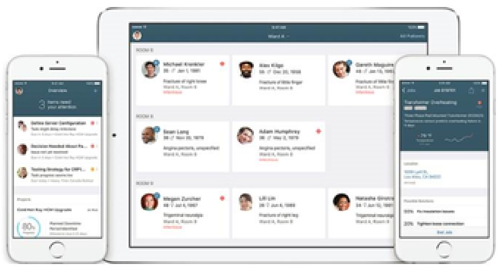


Figure 114: SAP Leonardo Machine Learning

SAP Leonardo Machine learning portfolio, integrates learning from data to help businesses grow, optimize processes, assist employees, and make customers happy. A part of this portfolio are "Business Services" which enable the intelligent enterprise by providing machine learning services targeted for your lines of business. Integrate these Business Services with your extension applications via stand-alone APIs or direct integration with SAP's line of business solution portfolio.

SAP Machine Learning for Talent Acquisition:

Automate the screening process for recruiters by using machine learning to identify best-fit candidates with SAP Machine Learning for Resume Matching

Provide candidates with job recommendations based on their skills and qualifications with SAP Machine Learning for Job Matching

Enable recruiters to classify job descriptions against national occupational definition databases (e.g. O*NET) with SAP Machine Learning for Job Standardization

Integrate each of these Business Services with extension applications for SAP SuccessFactors Recruiting and SAP Fieldglass via the SAP API Business Hub

SAP Machine Learning for Customer Experience:

Automatically process customer enquiries and scale digital interactions with SAP Machine Learning for Service Ticket Intelligence

Integrate each of these Business Services with extension applications for SAP Hybris Cloud for Customer via the SAP API Business Hub

SAP Smart Business service

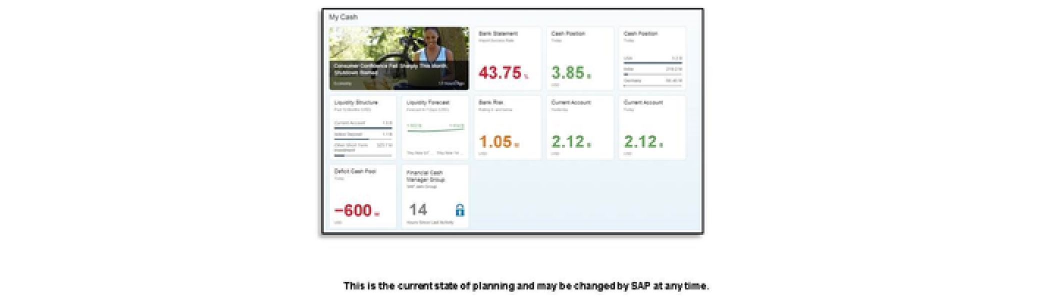


Figure 115: SAP Smart Business service

Smart Business enables Rapid publishing of SAP Fiori Analytic applications into role specific SAP Fiori Launchpads by key users. For normal users, Smart Business supports insight to action by providing Quick overview of business performance and enabling interactive analysis

Key capabilities

- SAP Smart Business is a framework for visualizing analytic content in the form of charts and tiles
- KPIs and OPIs can be visualized as SAP Fiori applications without writing any code
- Supports Automated analysis on behalf of business user in background jobs and Alerting whenever there is a breach of KPI thresholds
- Provides advance features like Embedded Analytics and Predictive Analysis

Benefits

- Benefit from Consumer Grade SAP Fiori Analytics powered by Smart Business in SAP Cloud Platform.
- Publish Analytics in Codeless environment significantly reducing your Total Cost of Development.
- Helps design Charting visualizations in WYSIWYG modeler.
- SAP Fiori Elements (Object Page, Analytic List Pages, Overview Pages, …) simplifies UI development dramatically.
- Out-of-the-box trust configuration with the SAP Cloud Platform account.

LESSON SUMMARY

You should now be able to:

- Explain business services on SAP Cloud Platform

Unit 4
Lesson 4

Using API Business Hub

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Use API business hub

API Business Hub

Figure 116: SAP API Business Hub

SAP API Business Hub provides a central catalog of SAP and selected partner APIs allowing developers to search, discover, test and consume the services they need to build new applications, app extensions or process integrations, using the SAP Cloud Platform.

Key Capabilities

- Growing list of SAP and selected partner APIs for application developers to discover, learn and try out.
- Sandbox test environment for easily discovering APIs.
- Integration with SAP Web IDE to easily consume APIs and develop SAP Fiori extension apps.
- Comprehensive API documentation in a vendor-neutral OpenAPI standard to accelerate learning.

Benefits

- Convenient access to critical APIs information in one place to learn faster.
- Comprehensive learning and testing enviroment to improve development productivity.

Integration Services, Landing page

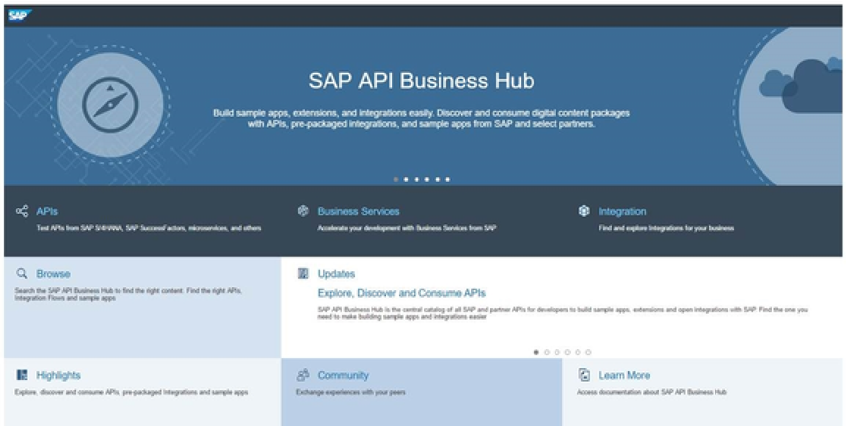


Figure 117: Integration Services, Landing page

The figure shows the Landing Page of the Integration Services.

SAP API Business Hub landing page **https://api.sap.com/landingPage.html** .

LESSON SUMMARY

You should now be able to:

- Use API business hub

Unit 4
Lesson 5

Explainng SaaS Extensions on SAP Cloud Platform

LESSON OBJECTIVES

After completing this lesson, you will be able to:

Explaining SaaS extensions on SAP Cloud Platform

SaaS Extensions on SAP Cloud Platform

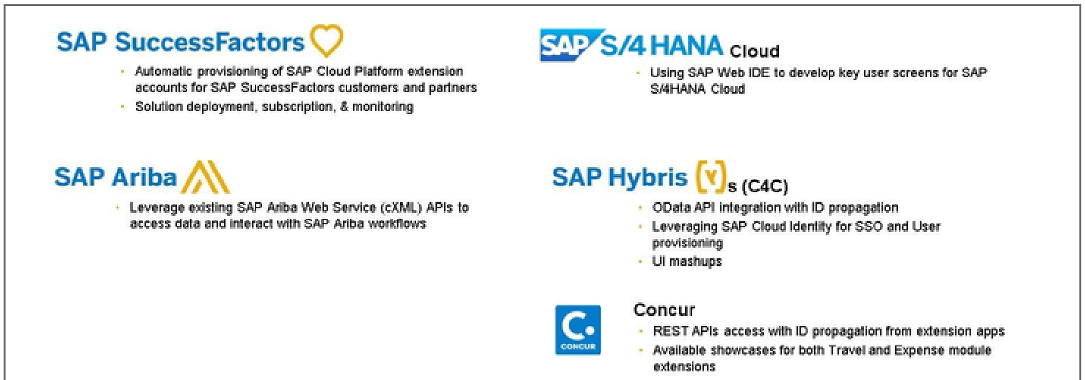


Figure 118: SAP Cloud Platform, SaaS Extensions

The figure shows the available SaaS extensions for 5 SAP Products.

LESSON SUMMARY

You should now be able to:

Explaining SaaS extensions on SAP Cloud Platform

Unit 4

Learning Assessment

1. Which of the following is not an SaaS product ?

   Choose the correct answer.

   - [ ] **A** SuccessFactors
   - [ ] **B** Ariba Network
   - [ ] **C** Fieldglass
   - [ ] **D** Hybris Marketing
   - [ ] **E** SAP S/4HANA

2. SAP Hybris as a Service runs on SAP Cloud Platform.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

3. Which of the following should be used to find SAP and selected partner API's?

   Choose the correct answer.

   - [ ] **A** Integration Service
   - [ ] **B** SuccessFactors
   - [ ] **C** Machine Learning
   - [ ] **D** API Business Hub
   - [ ] **E** Internet of Things

4. SAP Machine Learning for Customer Experience processes customer enquiries and scale digital interactions.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

5. SAP Smart Business is a framework for visualizing analytic content in the form of charts and tiles.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

Learning Assessment - Answers

1. Which of the following is not an SaaS product ?

   Choose the correct answer.

   - [ ] **A** SuccessFactors
   - [ ] **B** Ariba Network
   - [ ] **C** Fieldglass
   - [ ] **D** Hybris Marketing
   - [x] **E** SAP S/4HANA

   This is correct. See unit 4, lesson 1 of CLD100, Col18.

2. SAP Hybris as a Service runs on SAP Cloud Platform.

   Determine whether this statement is true or false.

   - [x] True
   - [ ] False

   This is correct. See unit 4, lesson 1 of CLD100, Col18.

3. Which of the following should be used to find SAP and selected partner API's?

   Choose the correct answer.

   - [ ] **A** Integration Service
   - [ ] **B** SuccessFactors
   - [ ] **C** Machine Learning
   - [x] **D** API Business Hub
   - [ ] **E** Internet of Things

   This is correct. See unit 4, lesson 1 of CLD100, Col18.

4. SAP Machine Learning for Customer Experience processes customer enquiries and scale digital interactions.

   Determine whether this statement is true or false.

   [X] True

   [ ] False

   This is correct. See unit 4, lesson 3 of CLD100, Col18.

5. SAP Smart Business is a framework for visualizing analytic content in the form of charts and tiles.

   Determine whether this statement is true or false.

   [X] True

   [ ] False

   This is correct. See unit 4, lesson 5 of CLD100, Col18.

UNIT 5

SAP HANA Enterprise Cloud (HEC)

UNIT OBJECTIVES

- Explain SAP HANA Enterprise Cloud
- Join SAP HANA Enterprise Cloud and SAP Cloud Platform together
- Unlock the digital business

Unit 5
Lesson 1

Explaining SAP HANA Enterprise Cloud

LESSON OBJECTIVES

After completing this lesson, you will be able to:

Explain SAP HANA Enterprise Cloud

SAP HANA Enterprise Cloud

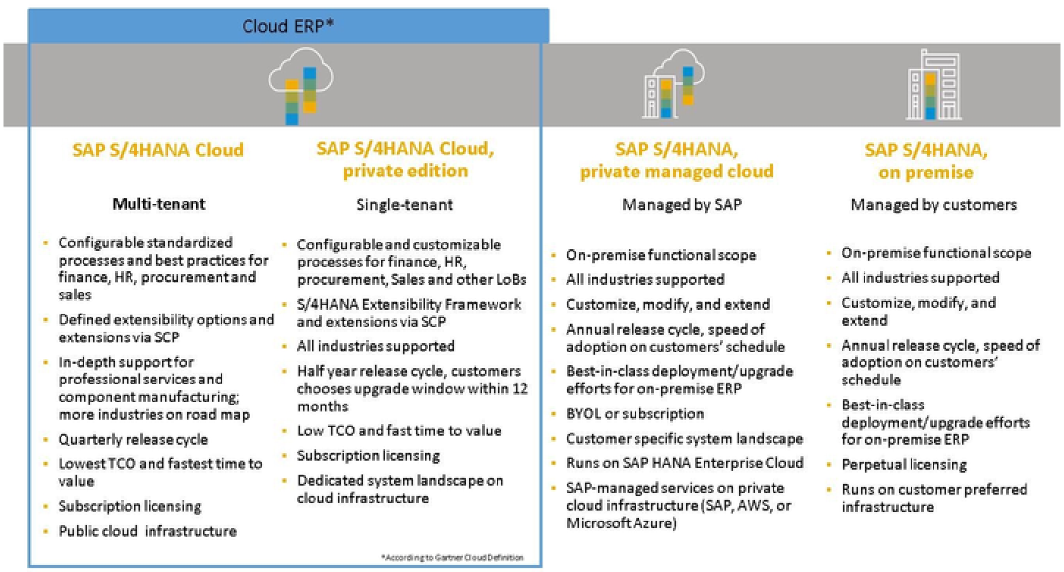


Figure 119: The right Solution for each Customer Situation

The Figure, The right Solution for each Customer Situation, shows the possibilities for customers as they transition their IT to the cloud. While ultimately the goal is to use the public cloud, a mix of hybrid options may be considered for a length of time. For example a currently onPremise customer can utilize a private managed cloud (HEC) for their SAP S/4HANA suite initially and then after a few years transition to SAP S/4HANA public cloud.

SAP S/4HANA Deployment Options

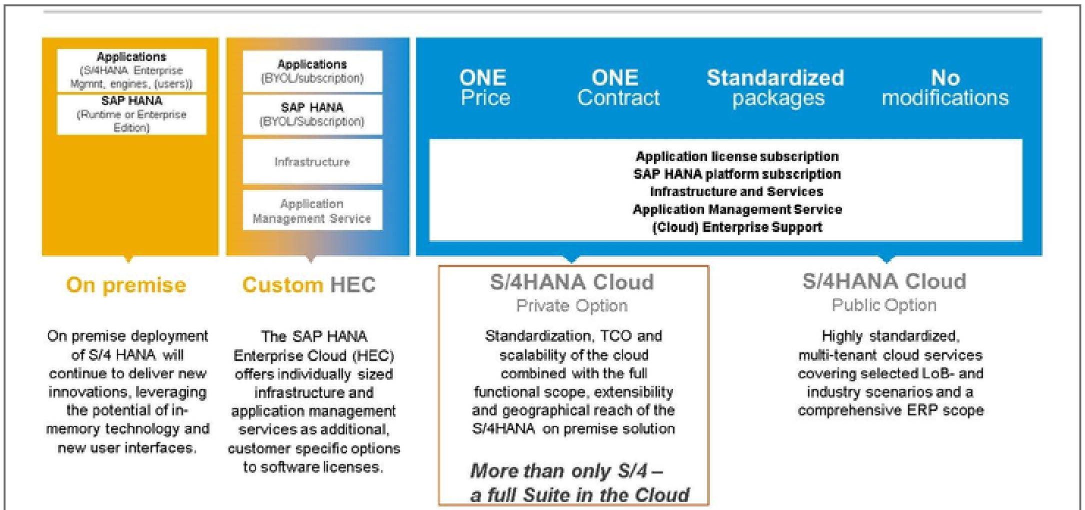


Figure 120: SAP S/4HANA Deployment Options

With an onPremise deployment of SAP S/4HANA suite the customer will continue to have a perpetual license along with total governance responsibilities. Also the customer will control which release is operational and when to apply support packages, upgrade, etc.

With Custom HEC the Infrastructure and application management services are provided by SAP while the governance remains with the customer. Licensing is either “Bring Your Own License” (BYOL) or by subscription. As with onPremise the customer will control which release is operational and when to apply support packages, upgrade, etc.

With both SAP S/4HANA Cloud (Private Edition) and SAP S/4HANA Cloud (Public Option), the licensing model changes to a single subscription contract and governance is done by SAP. The releases are the same for all customers with quarterly updates. The main difference is that the private edition offers more extensibility options as opposed to the highly standardized public edition.

LESSON SUMMARY

You should now be able to:

- Explain SAP HANA Enterprise Cloud

Unit 5
Lesson 2

Joining SAP HANA Enterprise Cloud and SAP Cloud Platform Together

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Join SAP HANA Enterprise Cloud and SAP Cloud Platform together

SAP HANA Enterprise Cloud and SAP Cloud Platform Together

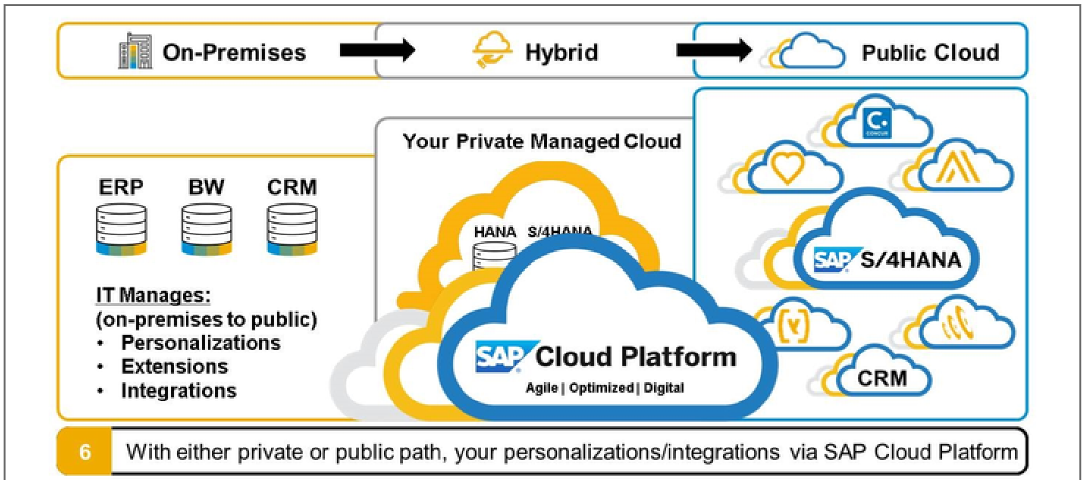


Figure 121: Path to Public Starts with SAP HANA Enterprise Cloud

A key requirement of our customers is the ability to create personalizations along with extensions and integrations. The SAP Cloud Platform with its PaaS capabilities allows them to do just that. Whether it's building new cloud apps, or integrating onPremise private cloud and public cloud based apps with each other.

SAP S/4HANA In-App and Side-by-Side Extensibility

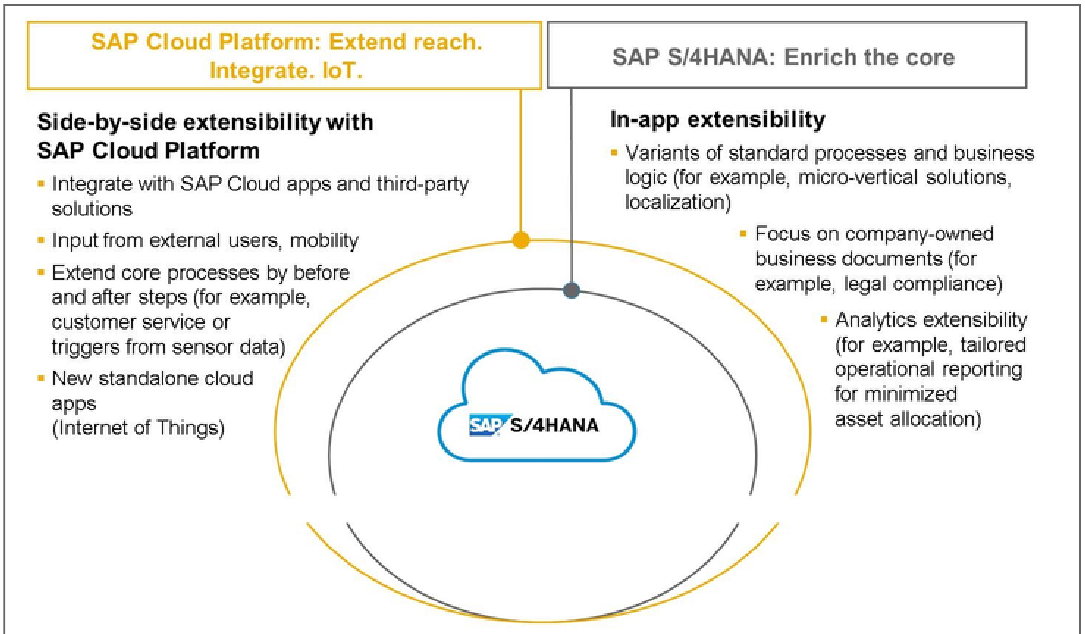


Figure 122: SAP S/4HANA In-App and Side-by-Side Extensibility

In-app extensibility allows the customer to extend their existing SAP S/4HANA applications in the SAP S/4HANA system itself (ABAP extensions, SAP Fiori front end extensions via JavaScript, etc.). The target group is the business expert in the company (the “key user”) and the tool set consists of web based apps for end to end creation of extensions.

Side-by-side extensibility allows the customer to extend their SAP S/4HANA applications by creating new applications running on SAP Cloud Platform. This will be done by developers. Their tools will vary from the SAP Web IDE (for frontend and backend JavaScript as well as HANA development) to Eclipse (for JAVA Development).

LESSON SUMMARY

You should now be able to:

- Join SAP HANA Enterprise Cloud and SAP Cloud Platform together

Unit 5
Lesson 3

Unlocking the Digital Business

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Unlock the digital business

Digital Business

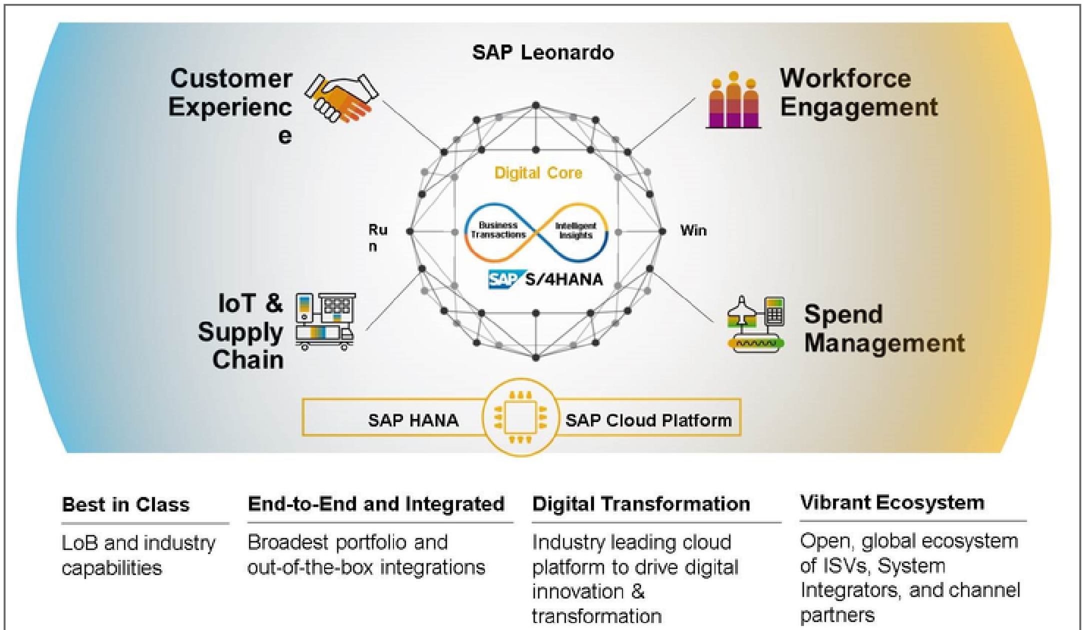


Figure 123: SAP Cloud is THE Digital Business Platform

Through organic innovations and strategic acquisitions, SAP has the best solution portfolio and expertise required to enable our customer's digital strategy.

The leading solutions in each category - procurement, manufacturing, supply chain, financials, human capital, travel and expense - so you can drive digital transformation where ever it happens within your organization.

We are the only partner that can help customers with every aspect of transforming into a digital business:

- We digitize the Core of business processes and the "back office" systems with SAP S/4HANA.
- We digitize the "front office", allowing businesses to seamlessly serve their customers and offer them the omni-channel experience they expect with Hybris and C4C.

We manage and optimize their supply chain, enabling them to connect with their suppliers on our Business Networks with Ariba, Concur and Fieldglass .

We engage their total workforce across employees, contractors, and partners with Successfactors, Fieldglass and SAP Fiori.

And we optimize their assets and innovate through the Internet of Things with the SAP Cloud Platform.

The SAP Cloud strategy has led to number of design paradigms we are committed to deliver to enable the digital journey of an enterprise.

Foundations for Digital Business Transformation

Figure 124: Foundations for Digital Business Transformation

Foundations for Digital Business Transformation are:

Simplify IT Landscape

Reduced cost to go-live

Deployment Choices and Lower TCO

Move mission critical business applications to the cloud

Shift from a CapEx to an OpEx model

Adaptive and scalable operating model

Make strategic decisions on the scope and direction of your business

Modernize IT and increase their capabilities and service levels

Innovation at your own pace

Flexibility to Innovate

Confidently focus on your business rather than on keeping the lights on

Complexity Hampers Innovation

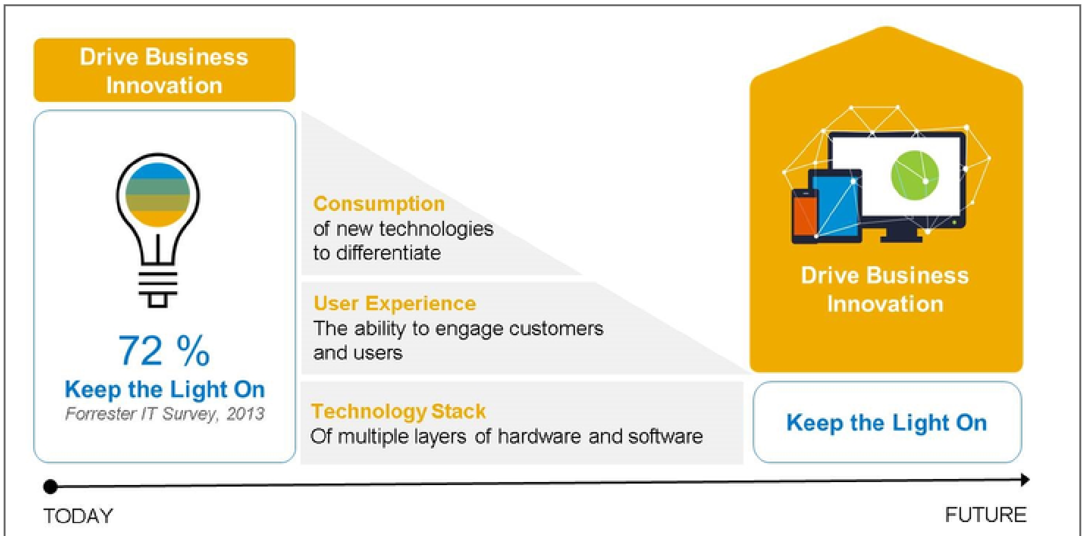


Figure 125: Complexity Hampers Innovation

The figure shows:

Simplify IT Landscape

Reduced cost to go-live

Deployment Choices and Lower TCO

Move mission critical business applications to the cloud

Shift from a CapEx to an OpEx model

Adaptive and scalable operating model

Make strategic decisions on the scope and direction of your business

Modernize IT and increase their capabilities and service levels

Innovation at your own pace

Flexibility to Innovate

Confidently focus on your business rather than on keeping the lights on

Cloud Transformation Options

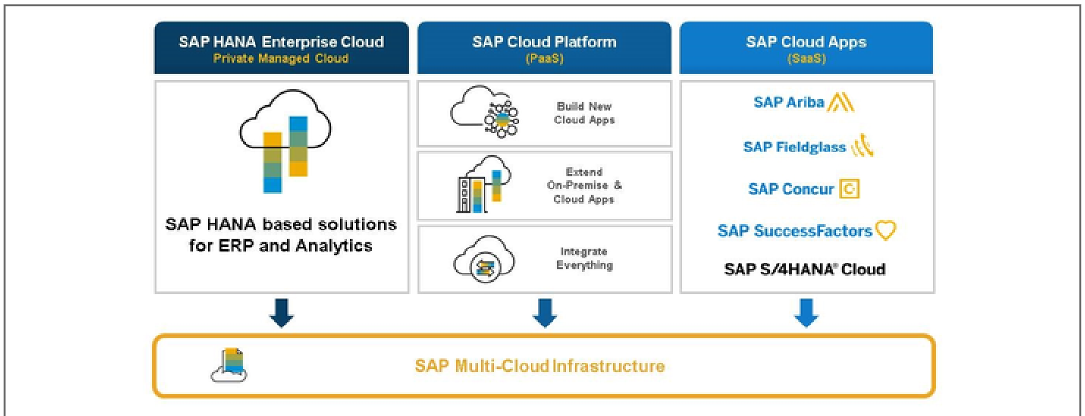


Figure 126: Cloud Transformation Options

To enable Cloud transformation at the speed of each individual business SAP offers the most complete stack of options:

- Native cloud apps for fast, fully standardized value creation in the LOB
- Private Managed Cloud environments for companies wanting to move their legacy onPremise investments into the cloud without losing the historic set up customizations or willingness / need to standardize
- Hybrid deployments of onPremise / private managed & public cloud stacks
- Supported by an innovation platform, integration platform and extensibility framework

Cloud Transformation Journeys

Every cloud journey is different. To enable your cloud transformation, SAP offers the most complete stack of flexible paths to support the pace of each individual business.

In the center, SAP Cloud Platform, an open and agile Platform-as-a-Service (PaaS) with comprehensive application development services for new cloud apps to extend existing SAP and non-SAP solutions. You can start here even if you're not using any SAP solution.

On the left, SAP HANA Enterprise Cloud delivers a managed cloud service that delivers unmatched SAP HANA expertise with functional application capabilities, industry processes and fully personalized with no limitations. This is ideal for customers who want to move their legacy onPremise investments into the cloud without losing the historic set up customizations or willingness / need to standardize'. Common HANA-based solutions include SAP S/4HANA (ERP), BW/4HANA (Data Warehousing) and BusinessObjects on HANA (Analytics).

And on the right, these are SAP's [native] cloud apps delivered via Software-as-a-Service (SaaS) to streamline several aspects of your operation. Functionality for human capital management (HCM) from SAP SuccessFactors, procurement (Ariba) and travel & expense management (Concur). This is where the latest innovations are delivered to you in a standardized and accelerated manner for each LoB. But don't forget, those SaaS offerings will benefit from SAP Cloud Platform to provide the integration 'glue' back to your ERP and any other applications (onPremise or in a managed cloud).

Path to Public Cloud Starts with SAP HANA Enterprise Cloud

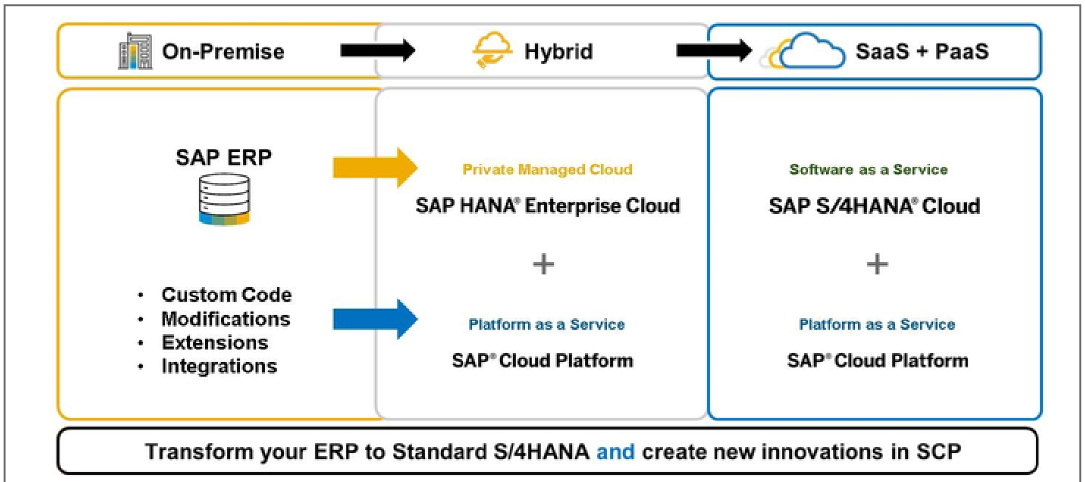


Figure 127: Path to Public Cloud Starts with SAP HANA Enterprise Cloud

To summarize, your path to innovation and unlocking the potential of your digital business strategy will rely on a cloud strategy that is comprehensive, adaptive and innovative. SAP has invested billions of dollars to modernize your SAP landscape and give you that safe passage to a public cloud future. And SAP does this in fashion where you have as much control as you want. You're always in the drivers' seat; you always control the pace.

Steps to the Public Cloud:

Step 1. Historically, all workloads have operated in a blend of onPremises with some deployed in an early hybrid model. This is common knowledge to anyone in IT.

Step 2. Customers started consuming new business-critical applications in a SaaS (Software-as-a-Service) model in the public cloud to take advantage of the latest innovations. Functionality for human capital management (HCM) from SAP SuccessFactors, procurement (Ariba) and travel & expense management (Concur) has now streamlined several aspects of your operation. But, it still required IT to manage the all the personalizations, extensions and integrations back into your core ERP application.

Step 3. Customers now have a bit more of a comfort-level with the public cloud (SaaS) and look towards certain onPremises applications that can be consumed via the public cloud, such as CRM. But, IT still has to maintain those personalizations, extensions and integrations back into your core ERP application.

Step 4. You now hit a certain threshold or compelling event where an aging platform (such as your Business Warehouse) is impacting your business and maybe you're considering SAP HANA because a simpler data model will lead to new business capabilities. But you don't have the skills/resources for a SAP HANA deployment. Not to worry, with SAP HANA Enterprise Cloud, we can help you deploy an SAP HANA environment in a matter of weeks. Let SAP provide you that SAP expertise, as-a-Service.

Step 5. You now have a better sense for how SAP HANA can impact one aspect of your business and you evaluate SAP S/4HANA in the hybrid model. For this cloud decision, you probably need a cloud solution that emulates the control and stability of your onPremises ERP - and that's where HANA Enterprise Cloud fits.

Step 6. Regardless of whether your path starts with a private managed cloud (hybrid cloud) with HANA Enterprise Cloud or SAP S/4HANA in the Public Cloud (SaaS), let SAP own all operations around SAP HANA lifecycle governance and provide you all the SAP expertise you need as-a-Service so you can focus on your business. And remember all those previous personalizations/integrations/extensions? SAP Cloud Platform delivers an open and agile Platform as a Service (PaaS)with comprehensive application development services and capabilities - enabling partners and customers to rapidly build new cloud apps, integrate, or extend existing SAP and non-SAP solutions. This is your development platform for onPremises-to-public, private-to-public, public-to-public. You can leverage SAP Cloud Platform today. Future proof your investments today.

For Example. If you want full functional scope of SAP S/4HANA along with the ability to customize, modify, and extend it, then use SAP S/4HANA, private managed cloud.

SAP S/4HANA in the Cloud - Our Flexibility, Your Choice

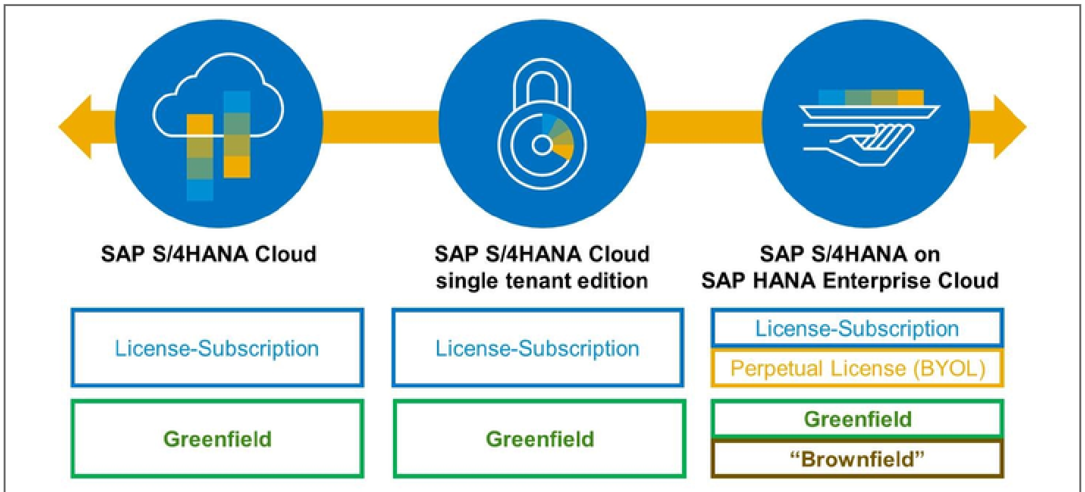


Figure 128: S/4HANA in the Cloud - Our Flexibility, Your Choice

SAP S/4HANA transforms traditional value chains into a smart value-added network. Take advantage of the opportunities created by this network to connect all internal and external aspects of your value chain. As a result, you get agile and transparent real-time information based on which you proactively make decisions to improve your business performance.

Whether you choose a greenfield approach, the brownfield approach that uses landscape transformation, SAP offers you the right SAP S/4HANA license option.

Perpetual license

A perpetual license is an entitlement for an unlimited period of time. With perpetual licenses, the bulk of the investment is made when purchasing the software. At that moment, the fee to acquire the license is paid and the software can be installed on an onPremise basis. Just to be clear: the software license does not include the right to use future releases of the licensed software. This because the perpetual license does not contain access to maintenance and support services. You'll need a separate maintenance agreement to organize this.

Subscription-based Model

With this model, you only have the entitlement to use the software for a specific period of time. During this period, payment is done on a monthly basis. This payment allows you to use the software which is hosted at a remote location. Access to maintenance and support is included in the fee. You don't need an additional maintenance and support agreement. Of course, since the entitlement is only valid for a specific period, licenses have to be renewed on a regular basis.

Cloud-Born Business Outcomes Require the Right Digital Foundation

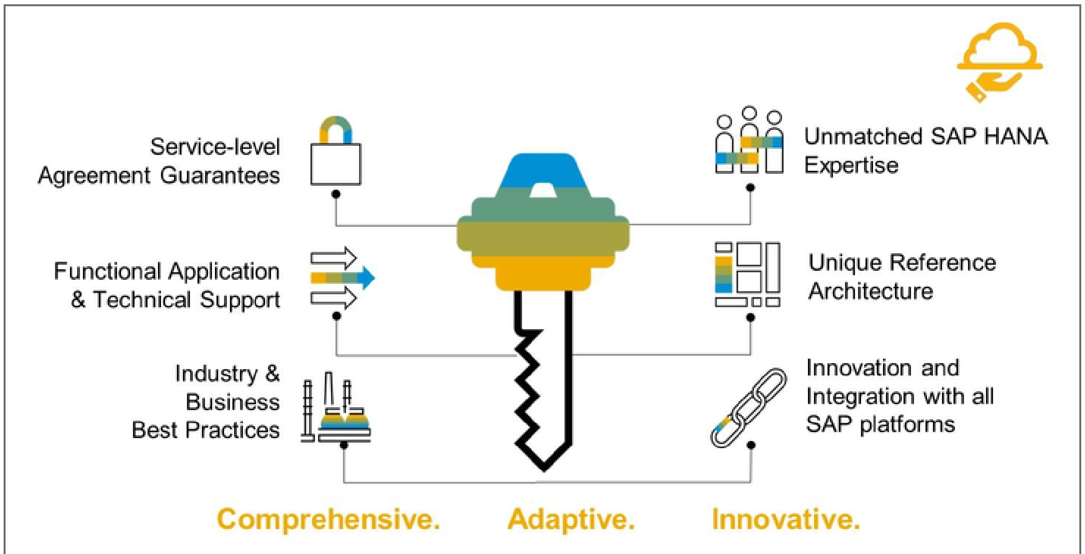


Figure 129: Cloud-Born Business Outcomes Require the Right Digital Foundation

The figure gives a visual summary of the capabilities of the SAP HANA Enterprise Cloud. The key shall visualize the unlock capabilities and the full value of SAP HANA.

Additional Value

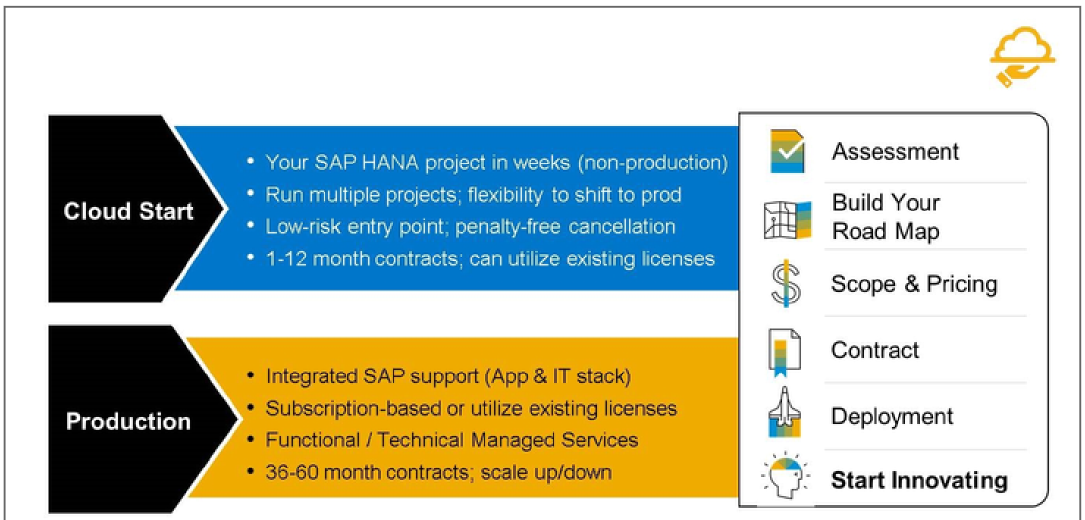


Figure 130: Digital Business at Your Pace - SAP Makes it Simple

At SAP, we want you to Run Simple. And your first SAP private cloud experience should be that simple.

We offer a simple entry point with our CloudStart program where your SAP S/4HANA project can be deployed in a matter of weeks and SAP gives you the bandwidth to run multiple projects simultaneously and shift to production whenever your business is ready. It's a low-risk entry-point where you can even bring your own licenses. Maybe you've purchased SAP S/4HANA already and your team hasn't been trained yet or you haven't acquired the infrastructure components? Either way, SAP has you covered, and we can even provide a menu of managed services (a reduced version from HEC Production) to support certain functional application services.

If you're ready to explore an even shorter path to production, we can provide you with the control and stability of your onPremise deployment, as-a-monthly service. This includes integrated support from the SAP S/4HANA software to the entire managed private cloud environment. You can leverage our complete menu of functional application and technical managed services. You can also provide existing licenses or begin with a pure subscription model. SAP HANA Enterprise Cloud is completely adaptable to your business needs.

Getting started is easy. SAP has well proven models for engagement that help you on the path from scoping to deployment, including rapid assessment of needs, standing up your own unique roadmap, delivering initial pricing indications, through to contracting, deployment and innovation, the ultimate goal.

Let us know how we can engage to help you start innovating.

SAP HANA Enterprise Cloud gives you flexibility

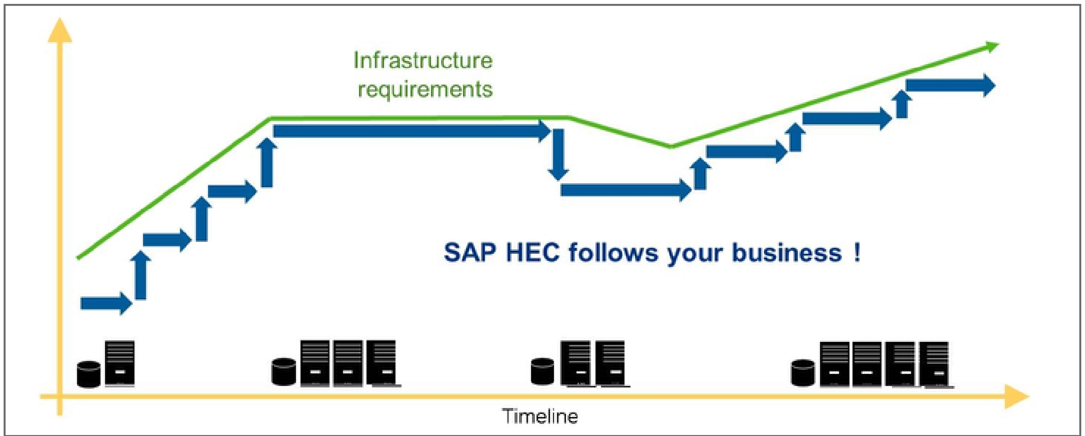


Figure 131: SAP HANA Enterprise Cloud gives you flexibility

The following are ways to flexibility:

Simplify IT Landscape

Reduced cost to go-live

Deployment Choices and Lower TCO

Move mission critical business applications to the cloud

Shift from a CapEx to an OpEx model

Adaptive and scalable operating model

Make strategic decisions on the scope and direction of your business

Modernize IT and increase their capabilities and service levels

Innovation at your own pace

Flexibility to Innovate

Confidently focus on your business rather than on keeping the lights on

SAP HEC - Roles & Responsibilities

The following are SAP HEC - Roles & Responsibilities:

A – Service Management

Account Management

Service Request Management - Technical Support

OSS / Support Portal Management

B – Infrastructure

Data Center Management

Network Management

Hardware Operations

Storage Management

Operating System

System Startup/Shutdown

Backup/Restore

Infrastructure integration

File transfer capabilities: CIFS shares

Management of Wide Area Network

C1 – Database Management SAP HANA

SAP HANA (general database operations)

SAP HANA XS

SAP HANA: Enterprise Information Management (EIM)

SAP HANA: Dynamic Tiering (DT)

SAP HANA: Smart Data Streaming (SDS)

SAP HANA: Multiple Database Containers (MDC)

Database operations

D – Core Technical Operations

System Installation

Incident Management

Event detection and notification ("monitoring")

General Operations

SAP Security Management

- Homogeneous system copy
- Release Management
- Proactive services
- Certificate handling
- Disaster Recovery

E – NetWeaver Operations

- General NetWeaver Operations
- SAP Client Operations
- Interface Administration
- Job Scheduling
- Transport Management
- Output Management
- Certificates handling

F – Server Provisioning

- Security Planning
- Hardware Operations
- Server Management
- Storage Management
- Application Management
- Disaster Recovery

See: **http://go.sap.com/about/agreements/hec.html**

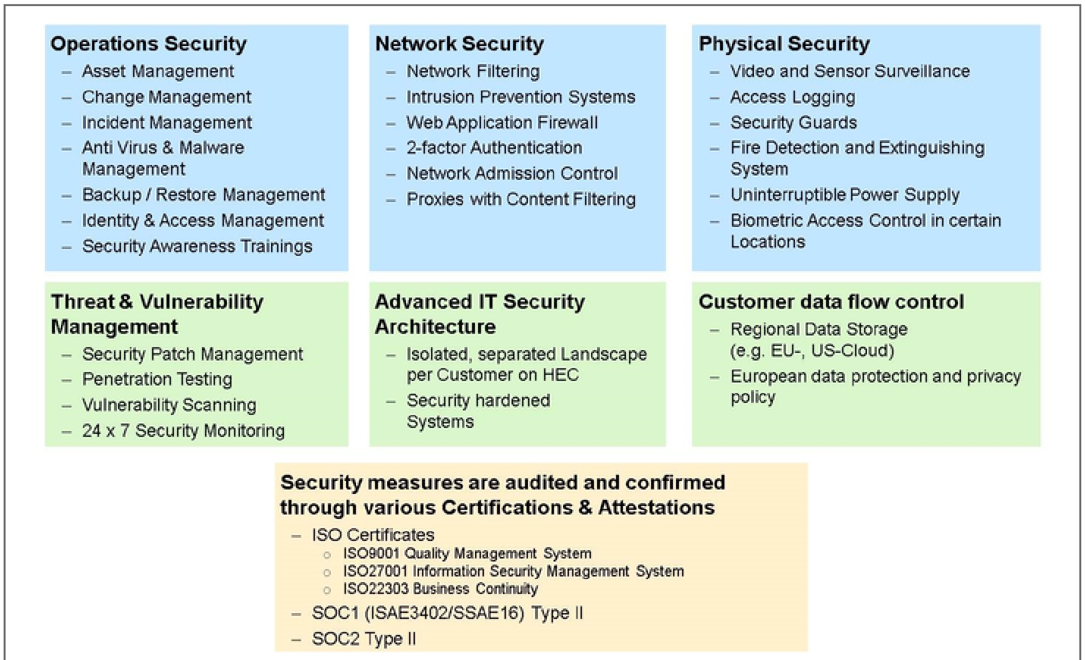


Figure 132: SAP HANA Enterprise Cloud - Security

The figure shows the different areas of security in cloud environments and the technologies used.

Roles and Responsibilities in HEC

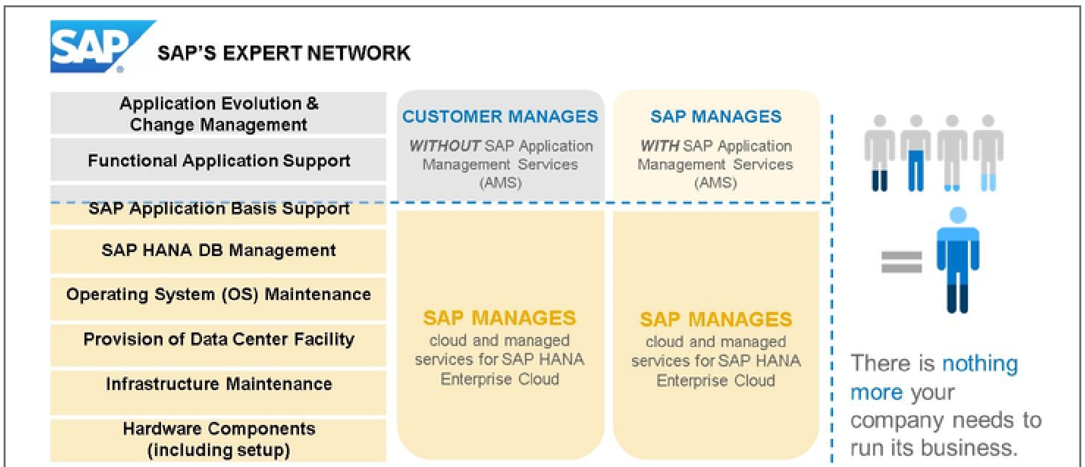


Figure 133: Roles and Responsibilities in HEC

In a world where customers do not have the time or budget to develop or acquire new skills, Application Management Services from SAP are a flexible, cost effective solution. AMS is an SLA based run time subscription service providing support and monitoring for the functional application layer of the solution. AMS complements the HEC productive infrastructure hosting and makes HEC to become an end to end Managed Service from SAP. SAP takes care of all monitoring, break fix, continuous evolution and innovation of the customer solution and lets customers refocus on their strategic business priorities while they can rely on a stable software with agility to support innovation.

Customers also get assistance with the adoption of innovative technologies, and accelerate time-to-value of cloud solutions and technology from SAP.

The service typically includes:

- Analysis, resolution, and root cause analysis of application incidents.
- Change management, including structure optimization and implementation of new functions.
- Execution of service packages and continuous operations requests.
- Monitoring, analysis, and optimization of application performance.
- Want to position with every production deal.

Configuring SAP HEC with a Powerful, Scalable Stack of Managed Services

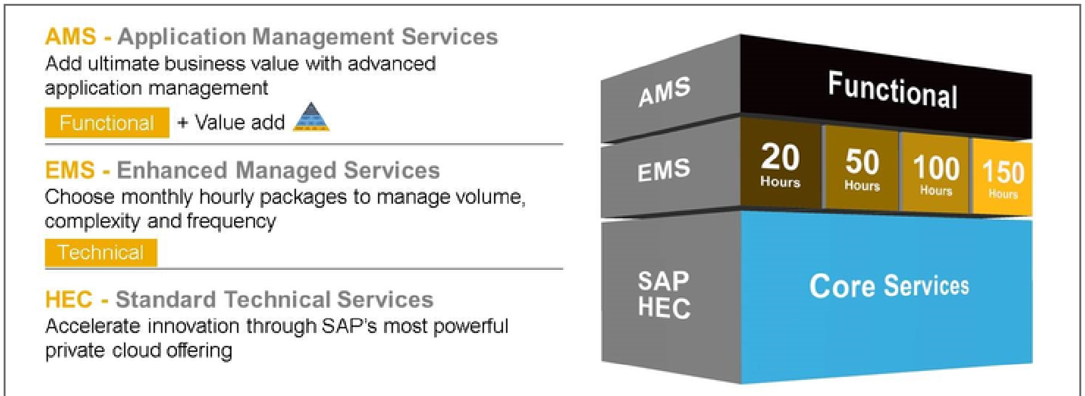


Figure 134: Configuring SAP HEC with a Powerful, Scalable Stack of Managed Services

SAP HANA Enterprise Cloud's range of Managed Services coordinates and manages all three levels of your cloud solution landscape-infrastructure, systems/technical, and applications-making it a one-stop solution and a single point of accountability for delivering SAP HANA value.

Unlocking the full value of cloud business applications requires expert management across all three layers of your solution stack:

- from infrastructure, allowing you to accelerate innovation by leveraging the cloud
- to full technical service - reducing complexity and unforeseen risk
- to functional application management in order to re-direct resources to focus on business outcomes

Reference Architecture: 100% best practice

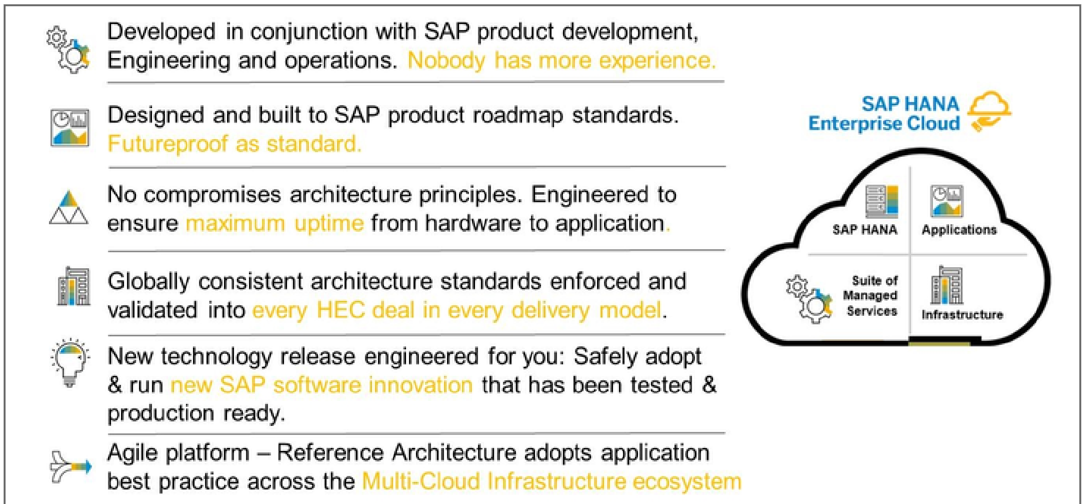


Figure 135: Reference Architecture: 100% best practice

LESSON SUMMARY

You should now be able to:

- Unlock the digital business

Unit 5

Learning Assessment

1. What does HEC provide for a customer?

   Choose the correct answers.

   - [ ] **A** Infrastructure and managed services.
   - [ ] **B** A range of enterprise applications.
   - [ ] **C** A secure private cloud environment.
   - [ ] **D** A multitenant environment.

2. HEC applications and services are available on a subscription basis.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

3. Which of the following is not provided by Enhanced Management Services?

   Choose the correct answer.

   - [ ] **A** Analysis, resolution, and root cause analysis of application incidents.
   - [ ] **B** Monitoring, analysis, and optimization of application performance.
   - [ ] **C** Enterprise grade security.

Learning Assessment - Answers

1. What does HEC provide for a customer?

   Choose the correct answers.

   - [x] **A** Infrastructure and managed services.
   - [x] **B** A range of enterprise applications.
   - [x] **C** A secure private cloud environment.
   - [ ] **D** A multitenant environment.

   This is correct. See unit 5, lesson 1 of CLD100, Col18.

2. HEC applications and services are available on a subscription basis.

   Determine whether this statement is true or false.

   - [x] True
   - [ ] False

   This is correct. See unit 5, lesson 1 of CLD100, Col18.

3. Which of the following is not provided by Enhanced Management Services?

   Choose the correct answer.

   - [ ] **A** Analysis, resolution, and root cause analysis of application incidents.
   - [ ] **B** Monitoring, analysis, and optimization of application performance.
   - [x] **C** Enterprise grade security.

   This is correct. See unit 5, lesson 3 of CLD100, Col18.

UNIT 6 Interaction of all Components

Lesson 1

UNIT OBJECTIVES

- Explain the interaction of all components

Unit 6
Lesson 1

Interacting of all Components

LESSON OBJECTIVES

After completing this lesson, you will be able to:

Explain the interaction of all components

Interaction of all Components

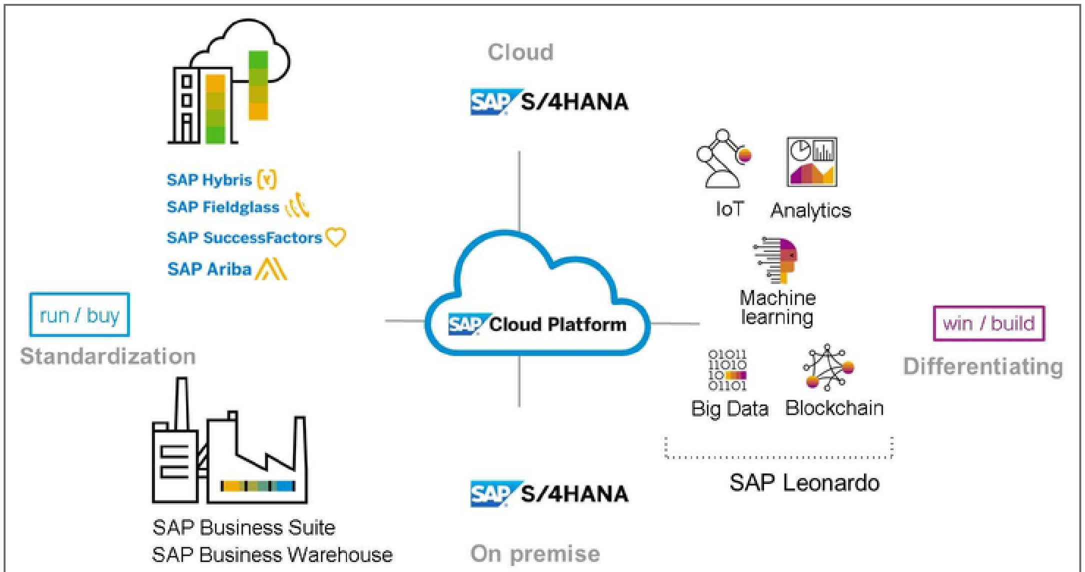


Figure 136: Putting al Things Together

The figure, Putting all Things Together, gives an overview of the general architecture of the SAP Cloud Platform and how other solutions are assigned to it.

SAP's cloud portfolio is both broad and deep. Companies have several SAAS solutions covering both core and extended business processes to choose from, available in several delivery models and in addition the SAP Cloud Platform.

Taken together this entire breadth of solutions provides companies with their Digital business roadmap. Digital business is built on new computing infrastructure – the pillars of mobile, cloud, Big Data, and analytics – accelerated by the Internet of Things (IoT), advances in machine learning, and innovations like blockchain. These disruptive technologies are giving companies the ability to radically change business models, and create new products and services.

Putting all Things Together, SAP Cloud Platform, Key Use Cases

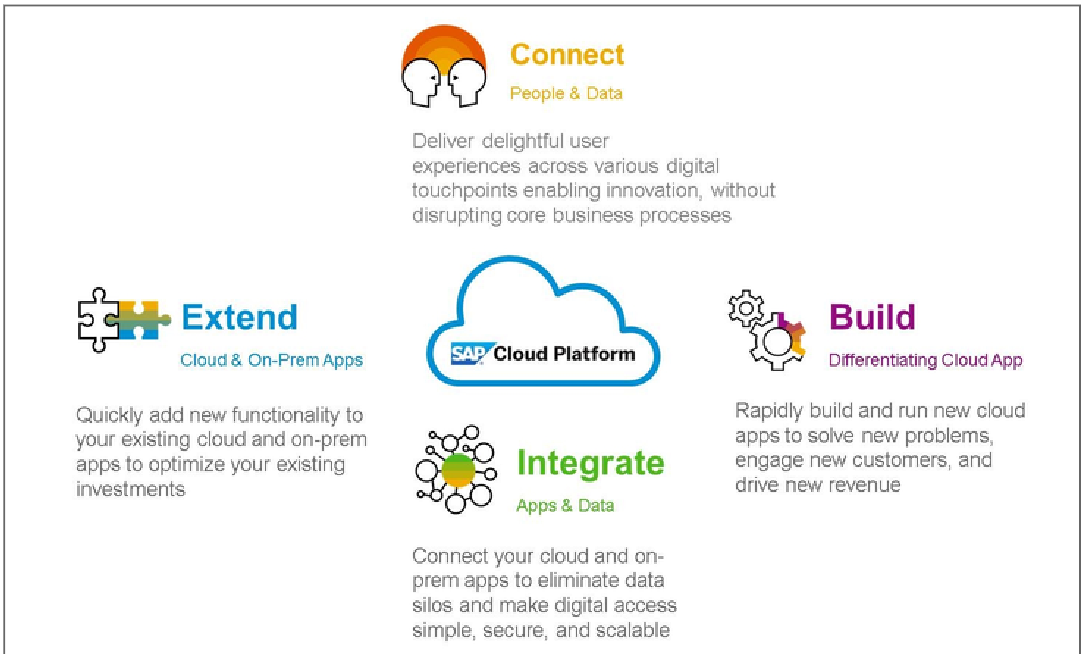


Figure 137: Putting all Things Together, SAP Cloud Platform, Key Use Cases

The figure, Putting all Things Together, SAP Cloud Platform, Key Use Cases, gives an overview of the possible use cases of SAP Cloud Platform.

Seven years ago Mark Andressen wrote in his famous Wall Street Journal Editorial “Why Software Is Eating The World” that every company must be come a “software company” to thrive in the new digital economy. No matter your industry, you’re expected to be reimagining your business to make sure you’re not the next local taxi company or hotel chain caught completely off guard by your equivalent of Uber or Airbnb. As the article states:

“Six decades into the computer revolution, four decades since the invention of the microprocessor, and two decades into the rise of the modern Internet, all of the technology required to transform industries through software finally works and can be widely delivered at global scale.”

SAP Cloud platform delivers all the necessary pieces to reimagine your business through “Building”, “Connecting”, “Integrating”, and “Extending”.

LESSON SUMMARY

You should now be able to:

- Explain the interaction of all components

Learning Assessment

1. Reimaging a business is characterized by:

   Choose the correct answers.

   - [ ] **A** Building
   - [ ] **B** Connecting
   - [ ] **C** Outsourcing
   - [ ] **D** Breaking
   - [ ] **E** Integrating

Learning Assessment - Answers

1. Reimaging a business is characterized by:

   Choose the correct answers.

   - [x] **A** Building
   - [x] **B** Connecting
   - [ ] **C** Outsourcing
   - [ ] **D** Breaking
   - [x] **E** Integrating

   This is correct. See Unit 6, Lesson 2 of CLD100, Col18.